The TFP–Welfare Disconnect: Evidence from Southern Europe

Luis Pérez*

Southern Methodist University

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Abstract

This paper studies the joint evolution of aggregate productivity and welfare in Spain and Italy between 1970 and 2010. Using a growth-accounting framework for open and distorted economies with input-output linkages, I construct aggregate TFP measures that correct biases in conventional Solow residuals. Once distortions are taken into account, the productivity decline in Spain begins seven years later and is roughly half as large as the traditional Solow residual suggests. To identify the sources of this decline, I develop a novel decomposition of aggregate TFP growth that separates the contributions of technical efficiency, factor reallocation, and changes in domestic and foreign intermediate linkages. I find that Spain's TFP decline is driven entirely by the reallocation of capital and labor toward low-productivity service sectors, whereas in Italy the decline is driven mainly by deteriorating technical efficiency. Aggregate welfare nevertheless continued to rise in both countries, sustained by factor accumulation and technological progress at home and abroad, revealing a systematic TFP–welfare disconnect.

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1 Introduction

Measured total factor productivity (TFP) in Southern European countries has stagnated or declined over the past several decades according to widely used statistical sources such as EU KLEMS and the Penn World Tables.[1] Conventional estimates based on the Solow residual suggest that aggregate productivity peaked in the late 1980s in Spain and the mid-1990s in Italy, and has followed a sustained downward trajectory since then. Figure 1 illustrates this pattern, showing that aggregate TFP in both countries reached its maximum decades ago and has steadily declined thereafter.

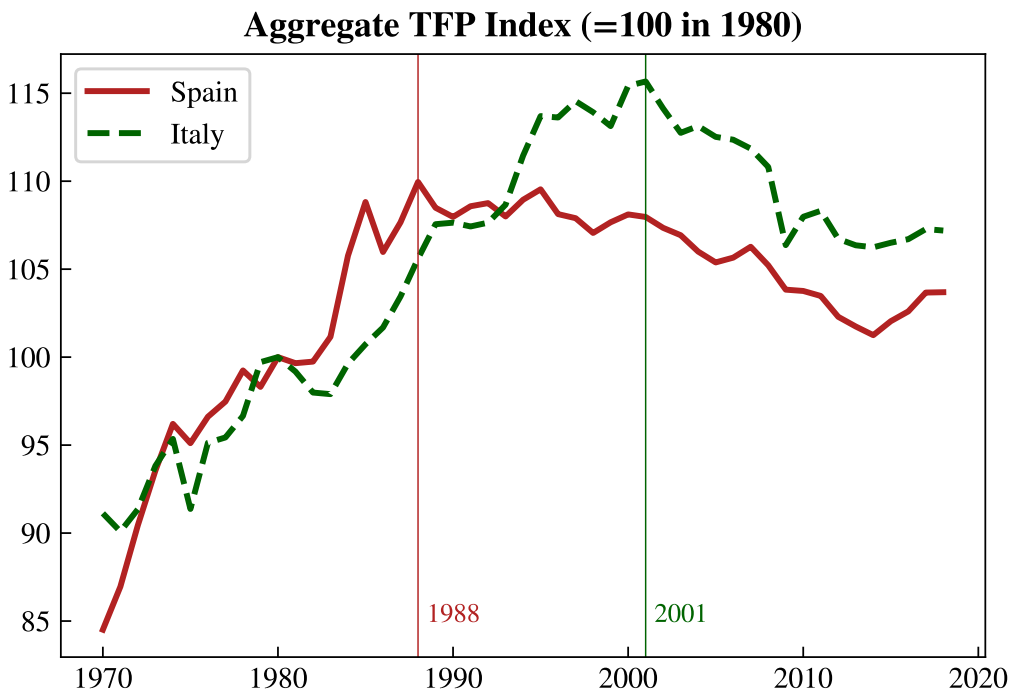


Figure 1: Evolution of Aggregate TFP in Spain and Italy.

Figure Notes. Aggregate TFP indices are constructed using two releases (2007/08 and 2021) of the *EU KLEMS Growth and Productivity Accounts*. Each index is based on the value-added concept of TFP—implicitly assuming that technical change operates only through capital and labor—and follows a top-down approach in which TFP is computed as the traditional Solow residual. The resulting series is normalized to 100 in 1980. TFP is reported at the level of total industries ("`TOT`"), which includes both market and non-market sectors. Vertical lines indicate the year in which each TFP series reaches its respective peak.

This apparent deterioration in efficiency has puzzled economists and raised important questions about the drivers of long-run growth and competitiveness in these economies. A large literature has sought to explain these trends through various mechanisms, including structural transformation, capital misallocation, financial frictions, institutional decline, and technological stagnation. Yet no unified explanation has

[1] Data from the *Penn World Tables 10* show that aggregate TFP ("`rtfpna`") has stagnated or declined in Spain, Portugal, Italy, and Greece since the 1970s–1980s. The *EU KLEMS Growth and Productivity Accounts* report similar declines ("`TFP_vaI`") for Spain and Italy, although well-known discrepancies in timing exist between datasets—particularly for Italy—while data for Portugal and Greece are unavailable. As is common in the literature, I rely on KLEMS data, which are generally considered more reliable due to their greater sectoral detail and methodological rigor.

emerged. Existing studies often focus on specific sectors, rely on closed-economy frameworks, or examine time periods that start long after productivity started to decline. Moreover, many analyses consider only one or a small subset of these mechanisms in isolation, limiting our understanding of how different structural forces interact.

In addition, much of the literature takes measured productivity based on Solow's residual as a sufficient summary statistic of underlying efficiency. As first emphasized by Hall (1988), the traditional Solow residual—the foundation for most empirical TFP series—is generally biased in the presence of markups, taxes, and other frictions. When such distortions are present, measured productivity may either overstate or understate changes in production efficiency. This observation motivates a reassessment of conventional TFP measures in environments where markups, taxes, and other frictions drive a wedge between observed factor payments and true marginal products, potentially distorting the mapping between measured TFP and underlying productive efficiency.

In this paper, I construct cost-based measures of aggregate TFP that account for distortions and use them to study the joint evolution of aggregate productivity and welfare in Spain and Italy. The analysis builds on the general-equilibrium accounting framework developed in Baqaee and Farhi (2024), which accommodates heterogeneous producers and consumers, arbitrary input-output linkages, imperfect competition modeled as price wedges, and openness to international trade in both intermediate inputs and final consumption. Within this framework, I decompose changes in aggregate productivity and welfare into contributions from sectoral technical efficiency, factor reallocation, and domestic and foreign intermediate linkages.

Main Findings. I document that aggregate productivity in Spain between 1970 and 2010 evolved more favorably than conventional estimates based on the Solow residual suggest. When productivity is measured using cost-based factor shares that incorporate distortions generated by markups, taxes, and other frictions, the decline in aggregate productivity appears both later and milder than under the standard measure. Under the traditional Solow residual, aggregate TFP in Spain peaks in 1988 and falls by roughly six percentage points through 2010. Under the distortion-adjusted measure, it continues to rise until 1995 and declines by only about three and a half percentage points. Correcting for distortions therefore delays the onset of the decline by seven years and roughly halves its magnitude. The reason is that distortions rose secularly: the income share accruing to wedges increased from 4% in 1970 to 12% in 2010, progressively biasing the traditional Solow residual downward.

Motivated by these findings, I then examine the sources of the productivity decline and its welfare implications. The analysis reveals that, in Spain, technical efficiency was the main driver of TFP growth over the full 1970–2010 period, explaining essentially all

of the cumulative gain, while factor reallocation exerted a modest drag. By contrast, the post-1995 TFP decline is fully explained by factor reallocation, as labor and capital moved from high-productivity tradable sectors toward lower-productivity non-tradable activities, especially construction, real estate, and hospitality.

Turning to welfare, I show that the decline in aggregate TFP did not translate into lower living standards—the so-called TFP–welfare disconnect. In Spain, welfare continued to rise after 1995 because the adverse effects of domestic reallocation were more than offset by factor accumulation and technological progress, both at home and abroad. Foreign technological progress contributed positively to welfare by lowering the prices of imported final and intermediate goods.

The analysis of Italy reveals a different pattern. The distortion correction has only a level effect on measured TFP because the aggregate wedge remained roughly constant over time. Italy's TFP losses are driven mainly by deteriorating technical efficiency rather than factor reallocation, yet the same TFP–welfare disconnect emerges: welfare continued to rise despite falling TFP. The contrast between the two economies—reallocation in Spain, technology in Italy—illustrates the value of the decomposition: the same aggregate outcome has fundamentally different sources.

Contributions. This paper makes two contributions: one methodological and one empirical. On the methodological side, I derive a novel discrete-time decomposition of aggregate TFP growth within the general-equilibrium accounting framework of Baqaee and Farhi (2024). The decomposition separates TFP growth into four channels: technical efficiency, factor reallocation, and domestic and foreign intermediate linkages, with factor reallocation further split into a distortion-driven component and a composition effect. The central difference relative to the Baqaee–Farhi first-order decomposition is that factor reallocation appears here as a separate, economically interpretable channel, rather than being distributed across changes in distortions and income shares together with approximation error. This distinction matters quantitatively in long-run growth accounting: in the Spanish data, the Baqaee–Farhi approximation leaves approximately 14% of cumulative TFP growth unexplained. Because both decompositions are applied to the same discrete annual data, the residual primarily reflects the economic content excluded by the first-order formula, especially factor reallocation.[2] This is not a minor omission: factor reallocation turns out to explain all of Spain's post-1995 TFP decline. By contrast, my decomposition identifies that channel explicitly and leaves an empirically negligible residual.

This contribution builds on a long tradition in growth accounting. From the foundational work of Tinbergen (1942) and Solow (1957), the standard approach has been to

[2]The residual also captures differences in how the remaining channels are weighted and defined.

decompose output growth into contributions from factor inputs and a residual, often interpreted as technical change. Building on this foundation, a large literature has relaxed Solow's original assumptions, one at a time, to examine their implications for productivity measurement.[3] Key contributions include Domar (1961), Jorgenson and Griliches (1967), Hulten (1978), Hall (1988), and Basu and Fernald (2002). More recently, Baqaee and Farhi have extended the traditional framework to permit the study of inefficient, disaggregated, and open economies with heterogeneous agents.[4] My paper contributes to this literature by deriving a discrete-time decomposition within the Baqaee–Farhi framework that is designed for finite changes and isolates factor reallocation as a separate component of aggregate TFP growth.

On the empirical side, my findings complement and extend a large empirical literature on the causes of productivity declines in Southern Europe. Previous studies have emphasized between-sector misallocation, in which labor and capital shift toward less productive sectors, often driven by falling interest rates and increased access to foreign capital (Benigno and Fornaro, 2014; Reis, 2013). Others have focused on within-sector misallocation, highlighting the role of financial frictions (Gopinath et al., 2017), institutional deterioration (García-Santana et al., 2020), or declining allocative efficiency (Calligaris et al., 2016; Fu and Moral Benito, 2018). A third view stresses technological stagnation, managerial inefficiencies, and policy distortions such as subsidies (Díaz and Franjo, 2016; Pellegrino and Zingales, 2019; Schivardi and Schmitz, 2020). While insightful, these studies typically abstract from the joint role of international integration, input-output linkages, and distortions in shaping aggregate productivity *and* welfare.

Relative to this literature, the paper makes three empirical advances. First, whereas much of the existing work studies misallocation in closed-economy or partial-equilibrium environments, this paper embeds misallocation in an open general-equilibrium framework with input-output linkages and trade in both intermediate inputs and final goods. Second, rather than assessing misallocation through static comparisons to efficient benchmarks, this paper provides a dynamic decomposition of changes in aggregate productivity into sectoral technical efficiency, factor reallocation, and trade in both domestic and foreign intermediate goods. Third, and most importantly, the analysis goes beyond productivity to study welfare, showing that the forces that depress aggregate TFP may simultaneously raise welfare in open economies.

This paper also connects to the literature on international trade and aggregate productivity. The emphasis on trade is motivated by a series of major trade-related developments during this period—including the creation of the Single Market, the adoption of the euro, and the WTO accessions of China and India—that profoundly

[3] Solow's framework assumes a representative firm, perfectly competitive product and factor markets, and a closed economy.

[4] See Baqaee and Farhi (2019a, 2018, 2019b, 2020, 2024).

altered the structure of European economies and their integration into the global economy.[5] These transformations contributed to the rising importance of global value chains (Antrás and Chor, 2022; Baldwin and Lopez-Gonzalez, 2015; Johnson and Noguera, 2017) and coincided with a period of increasing market power (De Loecker et al., 2020; Edmond et al., 2023; Hasenzagl and Perez, 2023). Most closely related is Kehoe and Ruhl (2008), who show that sharp deteriorations in the terms of trade coincide with declines in real GDP driven by reductions in measured TFP, and argue that explaining this relationship requires new mechanisms. In this paper, I emphasize one such mechanism: when factors of production are mobile across producers and sectors, shifts in international prices can reallocate resources domestically, affecting both real GDP and aggregate TFP. Unlike the intra-industry reallocation in Melitz (2003), which raises productivity by inducing the exit of less productive firms, inter-sector reallocation in a multi-sector general-equilibrium model need not increase aggregate productivity. Trade-induced reallocation may therefore raise welfare while lowering measured TFP.

I also document empirically a systematic divergence between aggregate productivity and welfare—the so-called TFP–welfare disconnect. This divergence underscores an important conceptual point: in open economies, aggregate welfare is not necessarily proportional to domestic TFP. A country may experience a decline in aggregate productivity while still enjoying welfare gains if changes in international prices and trade patterns expand consumption possibilities. This insight resonates with a broader trade literature showing that welfare depends on more than domestic productive efficiency alone. For example, Arkolakis et al. (2012) show that, in a broad class of trade models, the gains from trade depend primarily on import penetration and trade elasticities. More closely related to the mechanism emphasized here, Caliendo and Parro (2015) show in a quantitative analysis of NAFTA that trade can raise welfare even when measured domestic productivity declines. My contribution is to document a similar disconnect in long-run historical data for Spain and Italy, and to relate it to the joint roles of factor reallocation, distortions, and trade in intermediates.

This paper is also related to recent work revisiting the measurement of aggregate productivity in Europe. In contemporaneous work, Comin et al. (2025) construct utilization- and markup-adjusted productivity measures for several European economies, including Spain and Italy, showing that standard Solow-residual measures substantially overstate the cyclicality and volatility of TFP when utilization and markups are ignored. My analysis shares the insight that conventional TFP measures conflate technology with non-technological wedges, but differs in objective. The contribution here is not

[5] Major international and regional developments during this period include: the European Economic Community (ECC)'s grant of most favored nation (MFN) status to China in 1978, the Single Market Act of 1987, the formation of the European Union in 1992, the creation of the European Economic Area in 1994, the euro's adoption in 2002, and the WTO accessions of India in 1995 and China in 2001.

primarily a new distortion-adjusted productivity series, but a general-equilibrium TFP decomposition that attributes observed productivity changes to technical efficiency, factor reallocation, and trade in intermediates, and relates those changes to welfare. Differences between my TFP series and those in Comin et al. reflect differences in data sources and in the treatment of profits and wedges, rather than conflicting views about the underlying evolution of productivity.

Finally, the paper also speaks to the literatures on misallocation and market power. In the empirical implementation, misallocation is captured through wedges introduced in the spirit of Chari et al. (2007). These wedges represent reduced-form distortions that create a gap between efficient allocations and the allocations observed in equilibrium. Such distortions may arise from market power, financial frictions, distortionary taxation, or regulatory barriers. This feature links the paper to the large literature interpreting aggregate TFP losses through the lens of misallocation (Gopinath et al., 2017; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008).

Layout of the Paper. The rest of the paper is organized as follows. Section 2 discusses measurement issues with the traditional Solow residual in the presence of distortions, explains the economic mechanisms behind the TFP–welfare disconnect, and introduces the TFP decomposition used in the empirical analysis. Section 3 presents the general-equilibrium accounting framework. Section 4 states the formal decompositions of aggregate productivity and welfare changes. Section 5 describes the data sources and estimation strategy. Section 6 presents the main empirical results for Spain; results for Italy are reported in Appendix G. Section 7 concludes.

2 Aggregate TFP, Welfare, and Reallocation

This section clarifies the conceptual framework underlying the empirical analysis that follows. The goal is not to explain the evolution of productivity or welfare observed in Spain and Italy, nor to identify causal mechanisms, but to establish how changes in aggregate TFP and welfare should be interpreted in economies with distortions, sectoral heterogeneity, and international trade.

A central theme of the paper is that aggregate productivity and welfare need not move together. In open economies, it is possible—and empirically relevant—for the *level of aggregate TFP* to decline over time while aggregate welfare continues to rise. Understanding this divergence requires distinguishing carefully between (i) how aggregate productivity is measured, (ii) which economic mechanisms affect aggregate productivity in general equilibrium, and (iii) which mechanisms affect welfare, includ-

ing some that do not raise measured TFP. This section addresses these issues and builds intuition for the general-equilibrium accounting framework used in the paper.

2.1 Measuring Aggregate TFP in the Presence of Distortions

TFP growth is commonly measured as the Solow residual: output growth minus a revenue-weighted average of input growth. In distorted economies, however, this traditional Solow residual does not generally provide a reliable measure of TFP growth.

To see this, consider an economy with a representative producer that generates output using a constant-returns-to-scale production technology combining multiple inputs. Output is given by

$$Y = AF(L_1, \ldots, L_N), \tag{1}$$

where A denotes Hicks-neutral productivity and L_n is the quantity of input $n \in \{1, \ldots, N\}$.

Suppose the producer has market power and sets price above marginal cost, so that the markup $\mu := P/\text{MC}$ exceeds one. In this environment, the traditional Solow residual can be written as:[6]

$$\underbrace{\Delta \log Y - \sum_{n=1}^{N} \Lambda_n \Delta \log L_n}_{\text{Traditional Solow residual}} = \underbrace{\Delta \log A}_{\text{TFP growth}} + \underbrace{(\mu - 1) \sum_{n=1}^{N} \Lambda_n \Delta \log L_n}_{\text{Distortion bias}}, \tag{2}$$

where $\Lambda_n \equiv w_n L_n / (PY)$ denotes the revenue share of input n.

This expression shows that, in the presence of distortions, the traditional Solow residual generally differs from true TFP growth. As a result, movements in the traditional Solow residual may conflate changes in technology with changes in distortions and the mismeasurement of input contributions.[7]

Weighting input growth by *cost-based input shares* yields an unbiased measure of productivity growth (Hall, 1988). Defining cost shares as

$$\tilde{\Lambda}_n := \frac{w_n L_n}{\sum_k w_k L_k} = \mu \Lambda_n,$$

[6]See Appendix C.2 for a derivation. This expression is the discrete-time analogue of equation (2).

[7]In empirical implementations, the capital share is often computed residually as $1 - WL/(PY)$. When $\mu > 1$, this residual includes wedge income in addition to the true cost of capital, thereby overweighting capital relative to its cost-based share and underweighting labor. If capital grows faster than labor, the traditional Solow residual therefore assigns too much weight to the faster-growing input, overstates the contribution of measured inputs to output growth, and biases measured TFP downward. As the aggregate wedge rises over time, this mismeasurement becomes larger, and so does the downward bias.

aggregate productivity growth can be written as:[8]

$$\underbrace{\Delta \log Y - \sum_{n \in N} \tilde{\Lambda}_n \Delta \log L_n}_{\text{Distorted Solow residual}} = \underbrace{\Delta \log A.}_{\text{TFP growth}} \tag{3}$$

This expression provides the appropriate definition of aggregate TFP growth in distorted economies. It applies not only in representative-producer settings, but also in economies with heterogeneous producers and differentiated goods, where aggregate output is measured using Divisia quantity indices. Throughout the paper, aggregate TFP is measured using this cost-based definition unless indicated otherwise.

2.2 The TFP–Welfare Disconnect

Even when aggregate productivity is correctly measured, changes in the level of aggregate TFP need not track changes in welfare. Aggregate TFP summarizes how efficiently primary inputs are transformed into output, whereas welfare depends on consumption possibilities, relative prices, and access to foreign goods and technologies. In open economies, changes in international relative prices affect production and trade patterns and can therefore raise welfare even when domestic productivity stagnates or declines.

Figure 2 illustrates this distinction in a two sector-economy producing wine (W) and textiles (T). Each sector employs labor and a sector-specific fixed factor. Sectoral gross output is given by

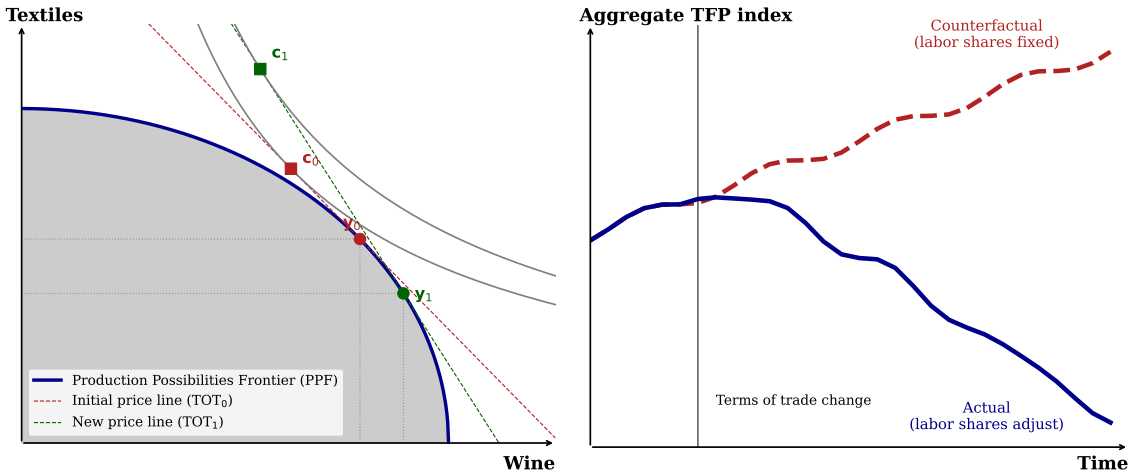
$$y_i = A_i L_{\ell,i}^{\phi}, \qquad 0 < \phi < 1, \quad i \in \{W, T\},$$

where A_i denotes sector-specific productivity and $L_{\ell,i}$ is labor employed in sector i. Aggregate labor supply is fixed.

The left panel of Figure 2 illustrates how a change in the terms of trade affects production, consumption, and welfare. The economy initially operates at a trade equilibrium with terms of trade TOT_0, producing at $\mathbf{y}_0$ on the production possibility frontier and consuming at $\mathbf{c}_0$ on the corresponding trade budget line. An increase in the relative price of wine shifts the terms of trade to TOT_1, inducing a reallocation of production along the frontier toward wine, to $\mathbf{y}_1$. Consumption moves to $\mathbf{c}_1$ on the new, steeper budget line, reaching a higher indifference curve. Although output remains on the production possibilities frontier, welfare rises because the improvement in the terms of trade expands consumption possibilities.

[8] See Appendix C.2, Proposition 2.

Figure 2: Trade–TFP Mechanism.



The right panel translates this reallocation into aggregate TFP accounting. Changes in aggregate TFP satisfy

$$\Delta \log \text{TFP} = \Delta \log Y - \sum_f \tilde{\Lambda}_f \Delta \log L_f,$$

where real output growth is measured by the Divisia quantity index

$$\Delta \log Y = \sum_{i \in \{T,W\}} \frac{p_i y_i}{PY} \Delta \log y_i.$$

Using the sectoral production function,

$$\Delta \log y_i = \Delta \log A_i + \phi \Delta \log L_{\ell,i},$$

and noting that aggregate factor supplies are constant over time, aggregate TFP growth can be written as

$$\Delta \log \text{TFP} = \underbrace{\sum_{i \in \{T,W\}} \frac{p_i y_i}{PY} \Delta \log A_i}_{\text{sectoral technical efficiency growth}} + \underbrace{\phi \sum_{i \in \{T,W\}} \frac{p_i y_i}{PY} \Delta \log L_{\ell,i}}_{\text{labor reallocation across sectors}}.$$

When sectoral productivity growth rates differ, reallocating labor toward the sector with lower productivity growth reduces aggregate TFP growth relative to a counterfactual with fixed labor shares. Persistent differences in these growth rates cumulate over time into a *lower path for the level of aggregate TFP*, as illustrated in the right panel of Figure 2. Welfare nevertheless rises because trade expands consumption possibilities.

The same logic extends beyond trade-induced reallocations. Structural change driven by shifts in demand—as in Baumol (1967)'s cost disease—as well as changes in sectoral markups, regulation, or other wedges can reallocate resources across activities with heterogeneous productivity growth. These reallocations mechanically affect the level and the growth rate of measured aggregate TFP through composition effects.

2.3 Aggregate TFP Accounting and Factor Reallocation

Standard first-order decompositions of aggregate TFP growth have two important limitations when applied to the study of long-run productivity growth.

First, efficient factor reallocations are absent by construction. In a competitive economy, the envelope theorem implies that marginal reallocations of factors have *zero* first-order effect on aggregate TFP: each factor is paid its marginal product, so the gain from moving a factor into one sector is exactly offset by the loss from withdrawing it from another. Formally, let L_{fi} denote the quantity of factor f employed in sector i. If factor markets are competitive, then $w_f = p_i \, \partial y_i / \partial L_{fi}$ for all i. Hence

$$\sum_i p_i \left(\frac{\partial y_i}{\partial L_{fi}} \right) \mathrm{d}L_{fi} = w_f \sum_i \mathrm{d}L_{fi} = w_f \mathrm{d}L_f.$$

Thus, holding the aggregate endowment L_f fixed, a pure reallocation across sectors has no first-order effect on aggregate TFP. As a result, reallocations driven by trade—such as those illustrated in Section 2.2—do not appear as a separate contribution in standard first-order decompositions. This is a fundamental feature of first-order frameworks, which assign zero to factor reallocation around an efficient allocation.

The second limitation is that inefficient factor reallocations are captured only through a local approximation and are not explicitly labeled. In distorted economies with $\mu_i > 1$, the envelope theorem no longer applies. Marginal products differ across sectors, so reallocation has a first-order effect on TFP. The decomposition in Baqaee and Farhi (2024) captures this effect, but only implicitly: it is embedded in changes in the aggregate factor income share, $\Lambda_f^{Y_c} = w_f L_f / (P_{Y_c} Y_c)$, and changes in sector-level distortions, $\Delta \log \mu_i$.[9] As a result, distortion-driven factor reallocation is not isolated as a separate and economically transparent component of TFP change.

Moreover, this local representation is accurate only when changes are small. Over the long horizons typical of growth accounting—where factor income shares, markups,

[9]For example, when rising markups in a sector cause labor to leave, aggregate TFP falls both because the markup itself worsens ($-\sum_i \tilde{\lambda}_i \Delta \log \mu_i$ records a decline) and because the induced reallocation shifts the factor share ($-\tilde{\Lambda}_f \mathrm{d} \log \Lambda_f^{Y_c}$ records a further decline). Neither term alone identifies the reallocation; together they conflate markup changes with their induced factor flows.

and sectoral composition can shift substantially—the gap between the exact change and its first-order approximation can become non-negligible. Crucially, that error does not appear as a separately identified component. Instead, it is in practice typically absorbed into the technical efficiency channel, which is calculated residually, making it difficult to determine how much of measured TFP growth reflects pure changes in technical efficiency and how much reflects composition effects that accumulate over time.

The aggregate TFP decomposition used in the empirical analysis addresses both limitations. It assigns an explicit, non-zero role to efficient factor reallocation through a discrete-time composition term, and it also isolates inefficient reallocations in a separate distortion-driven term, rather than embedding them implicitly in factor income shares. Finally, it collects the approximation error in an explicit residual, which is zero under Cobb–Douglas technologies and thus provides a diagnostic for the bias in measured technical efficiency. The decomposition accommodates arbitrary input-output linkages and international trade as it is derived from the Baqaee and Farhi (2024) framework. I state the result formally as Theorem 1 in Section 4, but preview it here to clarify how the decomposition addresses the limitations just described.

Specifically, the main aggregate TFP decomposition used in the empirical analysis is

$$\begin{aligned}\Delta \log \mathrm{TFP}_{c,t} = &\underbrace{\sum_{i\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Delta \log A_{i,t}}_{\text{(I) technical efficiency}} + \underbrace{\sum_{f\in\mathcal{F}_c}\left[\sum_{i\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}\mu_{i,t-1}\Omega_{if,t-1}\Delta\log L_{fi,t} - \tilde{\Lambda}^{Y_c}_{f,t-1}\Delta\log L_{f,t}\right]}_{\text{(II) factor reallocation}} \\ &+ \underbrace{\sum_{i\in\mathcal{I}_c}\sum_{j\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}(\mu_{i,t-1}-1)\Omega_{ij,t-1}\Delta\log x_{ij,t}}_{\text{(III) reallocation in domestic intermediates}} \\ &+ \underbrace{\sum_{i\in\mathcal{I}_c}\sum_{j\notin\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}(\mu_{i,t-1}-1)\Omega_{ij,t-1}\Delta\log x_{ij,t}}_{\text{(IV) reallocation in foreign intermediates}} + \mathcal{R}_{c,t}.\end{aligned}$$

Here, c indexes countries, t time, and i sectors. The operator $\Delta \log z_t := \log z_t - \log z_{t-1}$. The object $\lambda^{Y_c}_{i,t}$ denotes the revenue-based Domar weights in country c's GDP, $A_{i,t}$ is sector-level productivity, Ω_{if} is the revenue share of factor f in sector i's sales, L_{fi} is the quantity of factor f employed in sector i, $\tilde{\Lambda}^{Y_c}_{f,t}$ is the aggregate cost-based share of factor f, μ_i is the wedge in sector i (including its markup), x_{ij} denotes the quantity of intermediate inputs purchased by sector i from sector j, and $\mathcal{R}_{c,t} = \mathcal{O}(\|\Delta\|^2)$ is a residual that collects second-order terms.

Term (I) measures technological efficiency. It aggregates sectoral productivity growth using beginning-of-period revenue-based Domar weights. Economically, it

answers the counterfactual question: *How would aggregate TFP have evolved if the economy had experienced the observed sectoral productivity changes while keeping production weights fixed at their initial levels?* In this sense, term (I) isolates the pure technical-efficiency component of aggregate TFP growth, abstracting from changes in allocations.

Term (II) measures factor reallocation. Economically, it answers the question: *How much does aggregate TFP change because labor and capital are reallocated unevenly across sectors?* For each primary factor f, the bracket compares the growth of sector-level factor use, weighted by sectoral importance, with the growth of that factor in the aggregate. When all sectors expand factor use proportionally, the bracket is zero. It is non-zero precisely when factors are reallocated unevenly across sectors. This term contains two distinct economic forces. The first is a distortion-driven reallocation:

$$\sum_{f\in\mathcal{F}}\sum_{i\in\mathcal{I}_c}\lambda^{Y_c}_{i,t-1}(\mu_{i,t-1}-1)\Omega_{if,t-1}\Delta\log L_{fi,t}. \tag{IIa}$$

This term is non-zero only when $\boldsymbol{\mu}\neq\mathbf{1}$. It captures the first-order TFP effect of reallocating labor and capital across sectors with different wedges. In particular, reallocating factors away from high-markup sectors lowers aggregate TFP. This is the reallocation channel that the envelope theorem eliminates in competitive economies.

The second term is a discrete-time composition effect:

$$\sum_{f\in\mathcal{F}}\left[\sum_{i\in\mathcal{I}_c}\lambda^{Y_c}_{i,t-1}\Omega_{if,t-1}\Delta\log L_{fi,t}-\tilde{\Lambda}^{Y_c}_{f,t-1}\Delta\log L_{f,t}\right]. \tag{IIb}$$

This term can be non-zero even when $\boldsymbol{\mu}=\mathbf{1}$. Because it is a higher-order composition effect, it vanishes under an infinitesimal first-order approximation, but over the multi-year horizons relevant for growth accounting it can be quantitatively important.

Term (III) captures domestic intermediate reallocation. Economically, it answers the question: *How much of aggregate TFP growth is accounted for by changes in domestic input-output linkages across sectors?* It measures how changes in domestic intermediate-input use contribute to aggregate TFP when purchasing sectors are distorted. The wedge (μ_i-1) is central: in a competitive economy, an additional unit of intermediate input costs exactly what it contributes at the margin, so the first-order TFP effect is zero. With distortions, the equality breaks down, and changes in domestic input use generate non-zero aggregate TFP effects even when the underlying productivity is unchanged.

Term (IV) captures the contribution of international trade through imported intermediates. Economically, it answers the question: *How much does aggregate TFP change because sectors alter their use of foreign inputs?* Changes in imported input use affect aggregate TFP when importing sectors are distorted for the same reasons given above.

Trade can also affect aggregate TFP indirectly by reshaping the allocation of primary factors across domestic sectors, but those effects are conceptually different and are captured by term (II), not by term (IV).

Relation to Hulten (1978), Basu and Fernald (2002), and Baqaee and Farhi (2020, 2024). Hulten's theorem states that, to first order around a competitive equilibrium,

$$\Delta \log \mathrm{TFP}_c \approx \sum_{i \in \mathcal{I}_c} \lambda_i^{Y_c} \Delta \log A_i.$$

That is, aggregate TFP growth is Domar-weighted sectoral productivity growth, with no separate reallocation term. This follows from the envelope theorem: at a competitive equilibrium, factors are paid their marginal product, so the first-order TFP effect of any reallocation is zero by construction. My decomposition departs from Hulten's along two dimensions. First, even in a competitive economy ($\boldsymbol{\mu} = \mathbf{1}$), term (IIb) is non-zero over finite horizons: it is a second-order composition effect that vanishes in any first-order approximation but can be quantitatively relevant over the multi-year horizons studied here. Second, when sectors are distorted ($\boldsymbol{\mu} \neq \mathbf{1}$), the envelope theorem no longer applies and term (IIa) adds the distortion-driven reallocation channel, which is absent from Hulten by construction.

Basu and Fernald extend Hulten's framework to distorted economies and derive a reallocation term proportional to wedges and factor flows—the direct antecedent of term (IIa) here. Their result is also a first-order approximation around a steady state: higher-order composition effects are embedded in the residual rather than identified as a separate term. My decomposition extends theirs in three respects: it explicitly separates the higher-order discrete-time composition effect (term IIb), it generalizes the treatment of intermediate-input linkages to an arbitrary production network whose effects enter both directly and indirectly through Domar weights, and it incorporates imported intermediates (term IV) and open-economy trade channels.

Baqaee and Farhi extend the input-output framework to distorted open economies. Relative to their decomposition, the main contribution of the decomposition proposed above is the explicit identification of factor reallocation as a standalone contributor to aggregate TFP change. In Baqaee and Farhi, factor reallocation does not appear explicitly: distortion-driven reallocation is captured through changes in aggregate factor income shares, $\Delta \log \tilde{\Lambda}_f$, and sector-level markups, $\Delta \log \mu_i$, while efficient reallocation is absent from the first-order approximation and, in practice, is attributed to technical efficiency, which is computed residually. By contrast, the decomposition above collects both reallocation channels into a single explicit term, term (II), with two interpretable subcomponents. Term (IIa) isolates distortion-driven reallocation using revenue-based

Domar weights, λ_i, while term (IIb) isolates the discrete-time composition effect that captures mechanisms like that in Section 2.2, and thus identifies efficient reallocation that is non-zero over finite horizons.[10]

Examples. Appendix A.1 formalizes the mechanism behind the TFP–welfare disconnect shown in Figure 2. Appendix A.2 presents four stylized examples that isolate the main channels of the aggregate TFP decomposition. Example I considers efficient factor reallocation in an undistorted open economy. Example II introduces sectoral distortions in a closed economy and separates distortion-driven reallocation from the discrete-time composition effect. Example III shows how changes in domestic intermediate input use contribute to aggregate TFP when the purchasing sector is distorted. Example IV extends the analysis to imported intermediates. Together, these examples clarify the economic interpretation of each term in the decomposition.

3 General Theoretical Framework

In this section, I introduce the general-equilibrium model of Baqaee and Farhi (2024), which incorporates input-output networks, distortions, and international trade. This model serves as the foundation for the empirical analysis that follows.

Environment

The world economy consists of a set of countries $\mathcal{C} = \{1, \ldots, C\}$, a set of producers $\mathcal{I} = \{1, \ldots, I\}$, and a set of factors $\mathcal{F} = \{1, \ldots, F\}$. Factors and producers are physically located in particular countries, but can be owned by foreign households. The set of factors and producers located in country c are denoted by $\mathcal{F}_c$ and $\mathcal{I}_c$, respectively.

Households. Each country is populated by a representative household, so the terms "households" and "countries" can be used interchangeably. The preferences of household c are:

$$W_c = \mathcal{W}_c(\{c_{ci}\}_{i \in \mathcal{I}}), \tag{4}$$

where $\mathcal{W}_c$ is a homothetic aggregator that takes as an argument a sequence of consumption goods $\{c_{ci}\}$ demanded by household c and supplied by producers $i \in \mathcal{I}$.

[10] For the distortion-driven channel the two decompositions are equivalent rearrangements.

Household c demands a sequence of consumption goods to maximize its preferences, (4), subject to the budget constraint

$$\sum_{i\in\mathcal{I}} p_i c_{ci} = \sum_{f\in\mathcal{F}} \Phi_{cf} w_f L_f + \sum_{i\in\mathcal{I}} \Phi_{ci}\left(1-\frac{1}{\mu_i}\right) p_i y_i + T_c, \tag{5}$$

where p_i is the price of good i, Φ_{cf} is its ownership share of factor f, w_f is the rental price of factor f and L_f its quantity, Φ_{ci} is its ownership share on producer $i \in \mathcal{I}$, μ_i is an exogenous wedge, y_i is the quantity produced of good i, and T_c is a household-specific lump-sum tax/transfer that potentially captures trade imbalances.

Producers. Each producer $i \in \mathcal{I}$ located in $c \in \mathcal{C}$ produces output according to

$$y_i = A_i F_i\left(\{x_{ij}\}_{j\in\mathcal{I}}, \{\ell_{if}\}_{f\in\mathcal{F}_c}\right),$$

where A_i is Hicks-neutral productivity, and F_i is a producer-specific CRS technology that takes as inputs intermediate goods $\{x_{ij}\}_{j\in\mathcal{I}}$ and domestic factors $\{\ell_{if}\}_{f\in\mathcal{F}_c}$.

Each producer demands sequences of intermediate goods and factor inputs to minimize total costs, and sets price equal to marginal cost times an exogenous wedge: $p_i = \mu_i \times \mathrm{mc}_i$.

Factors. Primary factors are endowment goods which are inelastically supplied and earn Ricardian rents. These include the different types of labor and capital goods.

Ownership Shares. Ownership of firms and factors is encoded in the ownership matrix Φ, with dimension $C \times (I + F)$ and generic entries Φ_{ci} and Φ_{cf}, giving households' ownership shares of firms and factors.

Equilibrium. Given productivities $\{A_i\}$, wedges $\{\mu_i\}$, an ownership matrix Φ, and a vector of transfers $\{T_c\}$ respecting $\sum_c T_c = 0$, an equilibrium is a set of prices $\{p_i, w_f\}$, intermediate-input choices $\{x_{ij}\}$, factor-input choices $\{\ell_{if}\}$, outputs $\{y_i\}$, and final-consumption goods $\{c_{ci}\}$, such that:

(i) The price of each good equals its wedge times its marginal cost.

(ii) Producers choose intermediate goods and factor inputs to minimize total costs taking prices as given.

(iii) Households choose consumption goods to maximize preferences (4) subject to budget constraint (5) taking prices as given.

(iv) Markets clear:

$$\sum_{c\in\mathcal{C}} c_{ci} + \sum_{j\in\mathcal{I}} x_{ji} = y_i, \quad \forall i \in \mathcal{I}, \tag{Goods}$$

$$\sum_{i\in\mathcal{I}} \ell_{if} = L_f, \qquad \forall f \in \mathcal{F}. \tag{Factors}$$

Nominal Output and Expenditure. Nominal gross domestic product (GDP) is the value of domestic net output. For country c,

$$\mathrm{GDP}_c := \sum_{i\in\mathcal{I}} p_i q_{ci} = \sum_{f\in\mathcal{F}_c} w_f L_f + \sum_{i\in\mathcal{I}_c}\left(1-\frac{1}{\mu_i}\right) p_i y_i,$$

where $q_{ci} \equiv y_i \mathbf{1}_{\{i\in\mathcal{I}_c\}} - \sum_{j\in\mathcal{I}_c} x_{ji}$ is the quantity of good i produced in country c, net of the quantity of that good that is demanded for intermediate uses by domestic producers. In accordance with the System of National Accounts, GDP equals the total income generated by the factors of production employed in its production.

The nominal gross national expenditure (GNE) of a country is the total final expenditures of the residents of that country. Since there is no role for savings, GNE coincides with gross national income (GNI). Thus, for country c,

$$\mathrm{GNE}_c := \sum_{i\in\mathcal{I}} p_i c_{ci} = \sum_{f\in\mathcal{F}} \Phi_{cf} w_f L_f + \sum_{i\in\mathcal{I}} \Phi_{ci}\left(1-\frac{1}{\mu_i}\right) p_i y_i + T_c.$$

In contrast to GDP, ownership shares Φ_c are relevant for GNE. This is because income, which determines expenditures, may originate from different geographical sources.

Real Output and Expenditure. One can convert nominal variables into real ones by using Divisia indices. The change in real GDP and the corresponding GDP deflator are

$$\mathrm{d}\log Y_c = \sum_{i\in\mathcal{I}} \Omega_{Y_c,i}\,\mathrm{d}\log q_{ci}, \qquad \mathrm{d}\log P_{Y_c} = \sum_{i\in\mathcal{I}} \Omega_{Y_c,i}\,\mathrm{d}\log p_i,$$

where $\Omega_{Y_c,i} \equiv (p_i q_{ci})/\mathrm{GDP}_c$ is the share of good i in final output of country c.

Similarly, the change in real GNE for country c and the corresponding deflator are

$$\mathrm{d}\log W_c = \sum_{i\in\mathcal{I}} \Omega_{W_c,i}\,\mathrm{d}\log c_{ci}, \qquad \mathrm{d}\log P_{W_c} = \sum_{i\in\mathcal{I}} \Omega_{W_c,i}\,\mathrm{d}\log p_i,$$

where $\Omega_{W_c,i} \equiv (p_i c_{ci})/\text{GNE}_c$ is the share of good i in country c's consumption basket. As noted by Baqaee and Farhi, an implication of Shephard's lemma is that, in this economy, changes in real GNE are equal to changes in welfare.[11]

Input-Output Matrices. The world's revenue-based input-output (IO) matrix, denoted by Ω, is of dimension $(C+I+F)\times(C+I+F)$ and has generic entries Ω_{ij} specifying i's expenditures on j as a share of i's total revenues or income. Hence,

$$\Omega_{ij} = \frac{p_j c_{ij}}{\text{GNE}_i}\mathbf{1}_{\{i\in\mathcal{C}\ \wedge\ j\in\mathcal{I}\}} + \frac{p_j x_{ij}}{p_i y_i}\mathbf{1}_{\{i\in\mathcal{I}\ \wedge\ j\in\mathcal{I}\}} + \frac{w_f \ell_{if}}{p_i y_i}\mathbf{1}_{\{i\in\mathcal{I}\ \wedge\ j\in\mathcal{F}\}}.$$

Thus, whenever $i \in \mathcal{F}$ or $j \in \mathcal{C}$, $\Omega_{ij} = 0$ (factors have zero expenditures and households sell no resources). If $i \in \mathcal{C}$ and $j \in \mathcal{F}$, $\Omega_{ij} = 0$ (households demand no primary factors). The only interesting cases are: $i \in \mathcal{C}$ and $j \in \mathcal{I}$ (household's final consumptions), $i, j \in \mathcal{I}$ (intermediate input transactions), and $i \in \mathcal{I}$ and $j \in \mathcal{F}$ (producers' factor demands).

Equipped with the revenue-based IO matrix, one can construct the revenue-based Leontief inverse according to:

$$\Psi = (\boldsymbol{I} - \Omega)^{-1} = \sum_{p=0}^{\infty} \Omega^p.$$

In addition to the revenue-based IO matrix Ω, one can define the cost-based IO matrix $\tilde{\Omega}$ and the associated Leontief inverse $\tilde{\Psi}$, which capture exposures in costs. Doing so requires additional knowledge on the vector of wedges $\boldsymbol{\mu}$. These objects are:

$$\tilde{\Omega} = \text{diag}(\boldsymbol{\mu})\Omega, \qquad \tilde{\Psi} = (\boldsymbol{I} - \tilde{\Omega})^{-1}.$$

Cost-based concepts are relevant in distorted economies where $\Omega \neq \tilde{\Omega}$ and, hence, $\Psi \neq \tilde{\Psi}$. In efficient economies, these objects coincide since $\boldsymbol{\mu} = \mathbf{1}$ (there are no distortions).

Exposures and Domar Weights. Each $i \in \mathcal{C} + \mathcal{I} + \mathcal{F}$, where the symbol + here denotes set union, is exposed to each $j \in \mathcal{C} + \mathcal{I} + \mathcal{F}$ through both revenues Ψ_{ij} and costs $\tilde{\Psi}_{ij}$. While Ψ_{ij} captures *backward* linkages (i.e., how expenditures on i affect the sales of j), $\tilde{\Psi}_{ij}$ reflects *forward* linkages (i.e., how the price of j affects the marginal cost of i).

[11] In Basu and Fernald (2002)'s framework, welfare change is proportional to the change in real GDP because the economy is closed and, in that case, real GDP and real GNE coincide in the absence of savings. Moreover, factors of production are elastically supplied in Basu and Fernald's economy, in which case the relationship between welfare change and real GDP change is proportional, not exact.

A country or household is exposed in production (or GDP) and in welfare (or GNE) to each $j \in \mathcal{C} + \mathcal{I} + \mathcal{F}$, as reflected by the following equations:

$$\lambda_j^{Y_c} = \sum_{i \in \mathcal{I}} \Omega_{Y_c,i} \Psi_{ij}, \qquad \tilde{\lambda}_j^{Y_c} = \sum_{i \in \mathcal{I}} \Omega_{Y_c,i} \tilde{\Psi}_{ij}, \qquad \text{(Exposures in production)}$$

$$\lambda_j^{W_c} = \Psi_{cj} = \sum_{i \in \mathcal{I}} \Omega_{c,i} \Psi_{ij}, \qquad \tilde{\lambda}_j^{W_c} = \tilde{\Psi}_{cj} = \sum_{i \in \mathcal{I}} \tilde{\Omega}_{c,i} \tilde{\Psi}_{ij}. \qquad \text{(Exposures in welfare)}$$

Importantly, note that welfare exposures are recorded in Leontief inverses. Moreover, notice that when $j \in \mathcal{I}_c$, $\lambda_j^{Y_c}$ and $\tilde{\lambda}_j^{Y_c}$ give the revenue-based and cost-based (local) Domar weights of producer j in country c, respectively. The former is simply the sales of producer j as a fraction of country c's GDP. That is, $\lambda_j^{Y_c} = (p_j y_j)/\text{GDP}_c$, for $j \in \mathcal{I}_c$. Domar weights for foreign producers $j \in \mathcal{I} - \mathcal{I}_c$ are computed as

$$\lambda_j^{Y_c} = \sum_{k \in \mathcal{I}_c} \sum_{l \in \mathcal{I}_c} \Omega_{Y_c,k} \Psi_{kl}^c \Omega_{lj} = -\frac{p_j q_{cj}}{\text{GDP}_c}, \qquad \tilde{\lambda}_j^{Y_c} = \sum_{k \in \mathcal{I}_c} \sum_{l \in \mathcal{I}_c} \Omega_{Y_c,k} \tilde{\Psi}_{kl}^c \tilde{\Omega}_{lj},$$

where c superscripts denote local objects of country c. That is, Ψ^c and $\tilde{\Psi}^c$ are country c's revenue-based and cost-based Leontief inverses, respectively. It follows from this notation that the revenue-based Domar weight of a foreign producer is simply the ratio of this producer's sales to local producers to domestic GDP.

Since exposures of factors are prominent objects on their own, it is convenient to adopt special notation to differentiate them from others. When j is a factor, let Λ and $\tilde{\Lambda}$ denote exposures. That is, when $f \in \mathcal{F}$, $\Lambda_f^{Y_c} \equiv \lambda_f^{Y_c}$, $\Lambda_f^{W_c} \equiv \lambda_f^{W_c}$, $\tilde{\Lambda}_f^{Y_c} \equiv \tilde{\lambda}_f^{Y_c}$, and $\tilde{\Lambda}_f^{W_c} \equiv \tilde{\lambda}_f^{W_c}$. Hence, $\Lambda_f^{Y_c}$ records the income that accrues to factor f as a share of the nominal output of country c, and $\tilde{\Lambda}_f^{Y_c}$ gives the costs of factor f as a share of the total costs incurred in the production of output in country c. Importantly, $\Lambda_f^{Y_c}$ is not to be confused with the income share of factor f in country c, which is given by $\Lambda_f^c = \frac{\Phi_{cf} w_f L_f}{\text{GNI}_c} = \frac{\Phi_{cf} w_f L_f}{\text{GNE}_c}$.

4 Aggregate TFP and Welfare Decompositions

I now present the decomposition of aggregate TFP growth used in the empirical analysis, together with the corresponding first-order decompositions of aggregate TFP and welfare growth of Baqaee and Farhi (2024). Throughout, all objects are expressed in discrete time using log changes, $\Delta \log z_t := \log z_t - \log z_{t-1}$.

Definition 1 (Aggregate TFP Growth)**.** *Aggregate TFP growth of country c is defined as*

$$\Delta \log TFP_{c,t} := \Delta \log Y_{c,t} - \sum_{f \in \mathcal{F}_c} \tilde{\Lambda}_{f,t-1}^{Y_c} \Delta \log L_{f,t},$$

where $\Delta \log Y_{c,t}$ is real GDP growth, $\tilde{\Lambda}^{Y_c}_{f,t-1}$ is the cost-based factor share of domestic factor $f \in \mathcal{F}_c$ in country c's productions, and $\Delta \log L_{f,t}$ is the growth rate of that factor.

This definition generalizes the Solow residual to economies with multiple sectors, distortions, international trade, and arbitrary input-output linkages. As discussed in Section 2.1, the use of cost-based rather than revenue-based factor shares is necessary to isolate productivity changes from changes in wedges and reallocations.

Theorem 1 (Decomposition of Aggregate TFP Growth). *Aggregate TFP growth in country c can be decomposed as*

$$\begin{aligned}\Delta \log TFP_{c,t} = &\underbrace{\sum_{i\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Delta \log A_{i,t}}_{\text{(I) technical efficiency}} + \underbrace{\sum_{f\in\mathcal{F}_c}\left[\sum_{i\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}\mu_{i,t-1}\Omega_{if,t-1}\Delta\log L_{fi,t} - \tilde{\Lambda}^{Y_c}_{f,t-1}\Delta\log L_{f,t}\right]}_{\text{(II) factor reallocation}} \\ &+ \underbrace{\sum_{i\in\mathcal{I}_c}\sum_{j\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}(\mu_{i,t-1}-1)\Omega_{ij,t-1}\Delta\log x_{ij,t}}_{\text{(III) reallocation in domestic intermediates}} \\ &+ \underbrace{\sum_{i\in\mathcal{I}_c}\sum_{j\notin\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}(\mu_{i,t-1}-1)\Omega_{ij,t-1}\Delta\log x_{ij,t}}_{\text{(IV) reallocation in foreign intermediates}} + \mathcal{R}_{c,t}, \end{aligned} \tag{6}$$

where $\mathcal{R}_{c,t} = \mathcal{O}(\|\Delta\|^2)$ collects second-order terms.

See Proof in Appendix C.3.

Interpretation. Term (I) measures technical efficiency. It aggregates sectoral productivity growth holding fixed the production structure inherited from period $t-1$. Term (II) measures factor reallocation. It captures the effect of changes in the allocation of primary factors across sectors relative to the change in aggregate factor supplies. Terms (III) and (IV) capture the contributions of changes in domestic and imported intermediate-input use when purchasing sectors are distorted. Appendix A.2 shows presents stylized examples to isolate and clarify the economic interpretation of each term in the decomposition.

It is useful to decompose the factor reallocation term further as

$$\text{(II)} = \underbrace{\sum_{f\in\mathcal{F}_c}\sum_{i\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}(\mu_{i,t-1}-1)\Omega_{if,t-1}\Delta\log L_{fi,t}}_{\text{(IIa) factor reallocation (distortion-driven)}} + \underbrace{\sum_{f\in\mathcal{F}_c}\left[\sum_{i\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}\Omega_{if,t-1}\Delta\log L_{fi,t} - \tilde{\Lambda}^{Y_c}_{f,t-1}\Delta\log L_{f,t}\right]}_{\text{(IIb) factor reallocation (composition)}}.$$

Term (IIa) is non-zero only when sectors are distorted, while term (IIb) can be non-zero even in competitive economies. Term (IIb) vanishes in continuous time because infinitesimal factor reallocations aggregate exactly. Over finite discrete intervals, however, the log change of aggregate factor use generally differs from the weighted average of sectoral log changes, so term (IIb) need not be zero when sectors grow at different rates.

Baqaee–Farhi Decompositions. For comparison, I also report the aggregate TFP and welfare decompositions implied by Baqaee and Farhi (2024). Their results are derived as local first-order perturbations and are stated in differential form. To align the representation with the empirical analysis and with the discrete-time decomposition proposed above, I express their counterparts using finite log changes between $t-1$ and t, with all weights evaluated at the beginning of the period. Under this convention, the resulting expressions should be interpreted as first-order approximations.

Theorem 2 (Baqaee–Farhi Decomposition of Aggregate TFP Growth). *Up to a first-order approximation, aggregate TFP growth in country c satisfies*

$$\Delta \log TFP_{c,t} \approx \underbrace{\sum_{i\in\mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1} \Delta \log A_{i,t}}_{\textit{technical efficiency}} \underbrace{\underbrace{-\sum_{i\in\mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1} \Delta \log \mu_{i,t}}_{\textit{sectoral distortions}} \underbrace{-\sum_{f\in\mathcal{F}_c} \tilde{\Lambda}^{Y_c}_{f,t-1} \Delta \log \Lambda^{Y_c}_{f,t-1}}_{\textit{income shares}}}_{\textit{domestic reallocation}}$$
$$\underbrace{+\sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}^{Y_c}_{i,t-1} - \lambda^{Y_c}_{i,t-1}\right)\left(\Delta \log q_{ci,t} - \Delta \log \lambda^{Y_c}_{i,t}\right)}_{\textit{reallocation in foreign intermediates}}. \quad (7)$$

See Proof in Appendix C.4.

Whereas aggregate TFP is a production-side object tied to domestic GDP, welfare is an expenditure-side object. The welfare decomposition therefore differs from the TFP decomposition because it incorporates all margins that affect the country's consumption possibilities, including foreign productivity, distortions embodied in traded goods, and international income flows.

Theorem 3 (Baqaee–Farhi Decomposition of Welfare Growth). *Up to a first-order approximation, welfare growth in country c satisfies*

$$\Delta \log W_{c,t} \approx \underbrace{\underbrace{\sum_{i\in\mathcal{I}} \tilde{\lambda}^{W_c}_{i,t-1} \Delta \log A_{i,t}}_{\textit{technical efficiency}} + \underbrace{\sum_{f\in\mathcal{F}} \tilde{\Lambda}^{W_c}_{f,t-1} \Delta \log L_{f,t}}_{\textit{factors}}}_{\textit{technology}} \tag{8}$$

$$\underbrace{- \underbrace{\sum_{i\in\mathcal{I}} \tilde{\lambda}^{W_c}_{i,t-1} \Delta \log \mu_{i,t}}_{\textit{distortions}} + \underbrace{\sum_{f\in\mathcal{F}^*} \left(\Lambda^{c}_{f,t-1} - \tilde{\Lambda}^{W_c}_{f,t-1}\right) \Delta \log \Lambda_{f,t}}_{\textit{income shares}} + \underbrace{\frac{\Delta T_{c,t}}{GNE_{c,t}}}_{\textit{transfers}}}_{\Delta\ \textit{allocation}}.$$

Here, $\mathcal{F}^*$ denotes the set of all factors (real and fictitious), where fictitious factors f^* capture exposures to wedges and satisfy $\tilde{\Lambda}^{W_c}_{f} = 0$.

See Proof in Appendix C.5.

Unlike the TFP decomposition in Theorem 1, which is limited to domestic production, the welfare decomposition also captures other determinants of the country's consumption possibilities. A country can therefore experience declining TFP and rising welfare simultaneously when those other determinants—including foreign technological progress, factor accumulation, and international transfers—more than offset the domestic productivity loss.

Weighting convention. The baseline decomposition evaluates all weights at the initial period. This convention has a convenient economic interpretation: each term measures the contribution of a given margin to aggregate TFP growth relative to the production structure inherited from the previous period. In this sense, the decomposition answers a sequence of counterfactual questions of the form: starting from the initial period, how much would aggregate TFP have changed if only this margin had moved as observed?

In applied growth accounting, midpoint (Törnqvist) weights are often preferred because they provide a more accurate second-order approximation to the underlying continuous-time Divisia representation. I instead use base-period weights because midpoint weights absorb interaction terms—arising from simultaneous changes in primitives and endogenous equilibrium weights—into the measured contributions of each component. For the purposes of this paper, separating those interaction effects is useful both for interpretation and for highlighting the role of sectoral reallocation. Appendix B formalizes this point and compares alternative weighting conventions. In the empirical analysis, I report robustness to these alternative weighting choices.

5 Data and Estimation

This section describes the data sources and estimation procedures used to measure aggregate productivity and welfare.

Data sources. I rely on four main data sources. The World Input-Output Database (WIOD) provides sector-level data on input-output linkages, final demand, and factor usage. I use WIOD to construct input-output matrices and expenditure shares, and to estimate sector-level distortions. These data are complemented by the KLEMS Growth and Productivty Accounts, which provide industry-level measures of output, capital, labor, and intermediate inputs. KLEMS contains more detailed information on the composition of production factors and is also used in the construction of WIOD, which ensures consistency across sources. To estimate sector-specific depreciation rates, I combine KLEMS capital-stock data with asset-specific depreciation rates from the Bureau of Economic Analysis (BEA) Fixed Asset Tables. Finally, I use World Bank data on GDP and CPI deflators to construct real GDP and real gross national expenditure (GNE). Further details on data construction and adjustments are provided in Appendices E.1–E.7.

The analysis spans 1970–2010 and uses annual data.[12] The units of observation are 25 countries or regions (24 individual countries plus a rest-of-world aggregate), each comprising 23 production sectors and two primary factors (capital and labor). The resulting world production network therefore contains 575 producers in each year.

Estimation. To compute and decompose aggregate TFP and welfare, I estimate three key inputs for each sector-country-year observation:

(i) *Distortions*. A central challenge in measuring aggregate TFP in distorted economies is to quantify distortions. In the baseline specification, I estimate distortions non-parametrically for each sector i as

$$\mu_i = 1 + \frac{\text{wedge margin}_i}{1 - \text{wedge margin}_i},$$
$$\text{Wedge margin}_i := \frac{p_i y_i - \sum_{j \in \mathcal{I}} p_j x_{ij} - w_i \ell_i - (\delta_i + r_i) k_i}{p_i y_i},$$

where $p_i y_i$ denotes gross revenue, $\sum_j p_j x_{ij}$ is expenditure on intermediate inputs, $w_i \ell_i$ is labor compensation, and $(\delta_i + r_i) k_i$ is the gross user cost of capital. The

[12] The end year is 2010 because WIOD extends only to 2014, while capital-stock series are truncated for many countries beginning in 2010.

wedge margin measures the share of revenue not absorbed by input costs. It can reflect monopolistic markups, taxes, adjustment frictions, or other frictions that drive a wedge between price and marginal cost. By construction, $\mu_i = 1$ under marginal-cost pricing, while $\mu_i > 1$ indicates positive distortions. This approach allows distortions to vary flexibly across sectors, countries, and over time. Appendix D provides additional details and discusses alternative measures used for robustness. Appendix E.8 reports distortions estimates, which point to a clear upward trend in Spain and no noticeable trend for Italy. As Appendices E.10 (Spain) and G (Italy) show, these estimates imply reasonable income shares: in Spain, the income share of wedges rose from roughly 4% to 12% over the sample period, whereas in Italy it remained constant at about 5%.

(ii) *Net-of-depreciation user cost of capital*. To compute distortions, I must estimate both net-of-depreciation user costs of capital and depreciation rates. While data on output, intermediate input expenditure, and labor compensation are directly observed, capital costs must be imputed or estimated. I estimate net-of-depreciation user costs using the method of van Vlokhoven (2022), described in Appendix D.1. This approach exploits cross-sectional variation in input choices to infer implicit user costs. A key identifying assumption is that net-of-depreciation user costs are equalized across sectors within each country.

(iii) *Depreciation rates*. Sector-specific depreciation rates are computed as weighted averages of asset-specific rates, where the weights reflect the composition of the sector's capital stock:

$$\delta_{ict} = \sum_j \frac{K_{jict}}{K_{ict}} \delta_{jt},$$

where i indexes sectors, c countries, t years, and j asset types. Here, K_{jict} denotes the stock of asset j in sector i of country c at time t, and K_{ict} is the sector's total capital stock. The term δ_{jt} is the depreciation rate associated with asset type j in year t. The procedure accommodates variation in asset compositions across sectors and over time. Appendix E.7 provides all implementation details.

Aggregate TFP and welfare. With these estimates in hand, I compute cost-based domestic factor shares as

$$\tilde{\Lambda}^{Y_c}_{f,t} = \sum_{i \in \mathcal{I}} \Omega_{Y_c,i,t} \tilde{\Psi}_{if,t}$$

and construct aggregate TFP using Definition 1. I measure welfare growth as the growth rate of real GNE per capita. Appendix E.6 details the mapping to the data.

6 Productivity and Welfare in Spain

This section applies the productivity and welfare decompositions derived above to Spain over 1970–2010. I show that the deterioration in productivity was less severe than standard measures imply, decompose its underlying sources, and relate the findings to Spain's structural transformation, trade integration, and the European convergence process. I then turn to the implications for welfare.

6.1 The Evolution of Aggregate Productivity

Figure 3 plots aggregate TFP in Spain from 1970 to 2010. The dashed line shows a TFP series based on the traditional Solow residual, which does not account for distortions. The solid line shows the distortion-adjusted TFP series, which incorporates distortions estimated using wedge margins and thus provides a more accurate measure of productivity. Relative to the conventional measure, the distortion-adjusted series implies both a later turning point and a smaller decline. The conventional measure peaks in 1988, whereas the adjusted series continues to rise until 1995. Moreover, the decline in aggregate productivity is smaller in the distortion-adjusted series than in the conventional measure (3.5 versus 6 percentage points).

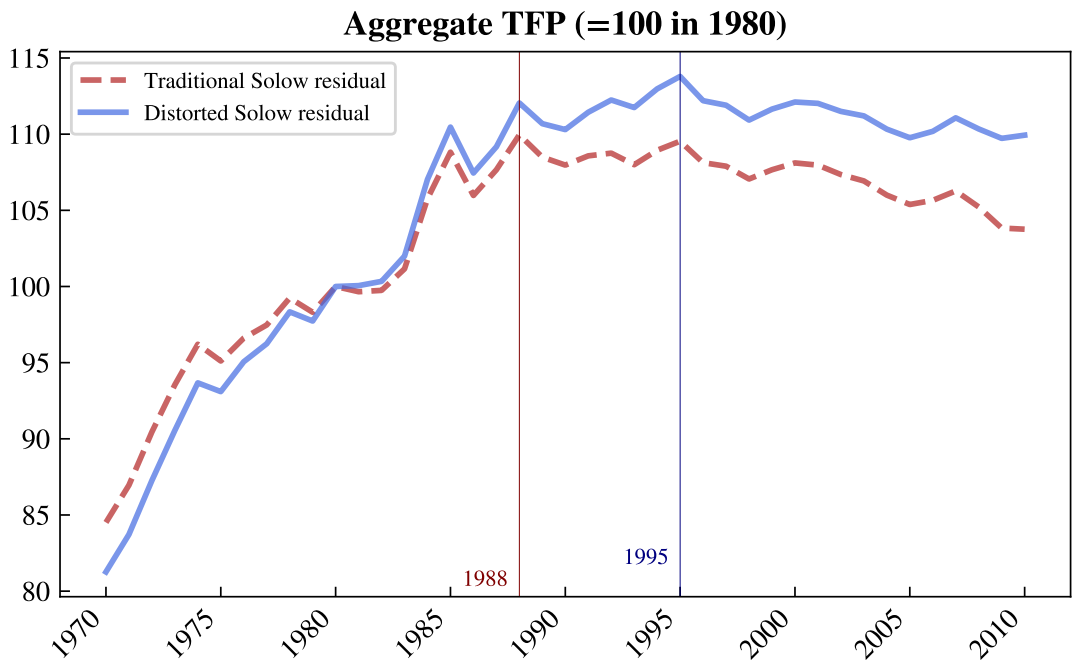


Figure 3: The Evolution of Aggregate TFP in Spain, 1970–2010.

Figure Notes. The index based on the traditional Solow residual assumes efficiency and therefore does not account for distortions. The index based on the distorted Solow residual incorporates distortions, which are estimated non-parametrically using wedge margins. Both indices are normalized to 100 in 1980. Vertical lines indicate the years in which the respective indices peak.

The differences between the two series reflect how factor growth is weighted in the two measures. The traditional Solow residual nets out factor growth using revenue-based factor shares, with labor weighted by labor compensation over value added and capital by the residual share of value added. The distortion-adjusted measure instead uses cost-based shares, with labor weighted by labor compensation over total cost and capital by capital cost over total cost. Because revenue-based and cost-based shares differ in the presence of distortions, part of the decline in the conventional Solow residual reflects secular changes in wedges rather than lower technical efficiency.

6.2 Understanding the Evolution of TFP

Figure 4 presents the cumulative decomposition of aggregate TFP over the full sample period based on Theorem 1. The left panel reports the contribution of each component separately, splitting factor reallocation into its distortion-driven and composition sub-components. The right panel groups these contributions into four broader channels: technical efficiency, factor reallocation, and domestic and foreign intermediates.

The decomposition shows that technical efficiency was the main driver of TFP growth, accounting for essentially all the cumulative gain (+103%). Factor reallocation made a modest negative contribution (−7%), driven primarily by the composition effect. Domestic intermediates contributed positively (+4%), while foreign intermediates played almost no role (−1%).

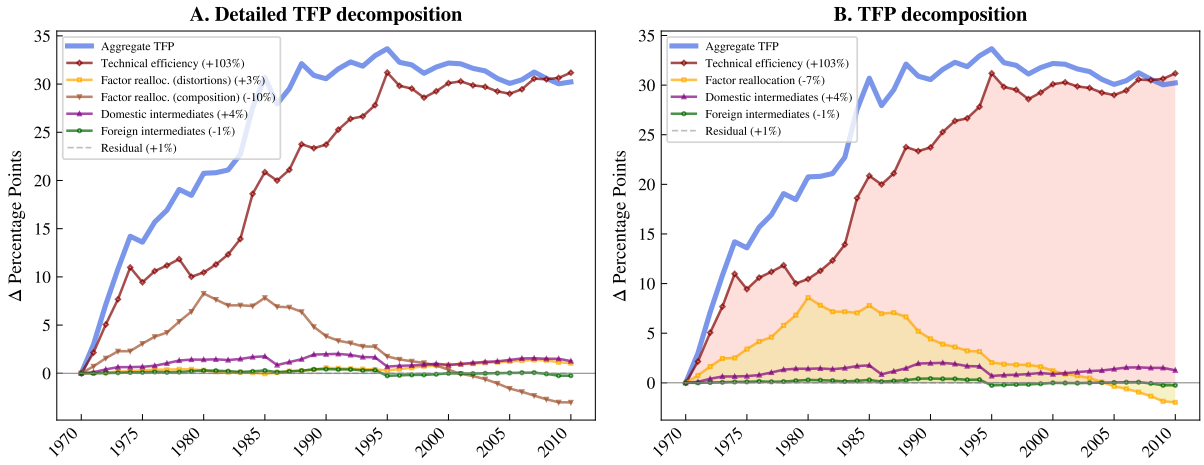


Figure 4: Cumulative Decomposition of Aggregate TFP in Spain, 1970–2010.

Figure Notes. The figure reports the cumulative decomposition based on Theorem 1. The left panel displays the individual components, with factor reallocation split into distortion-driven and composition subcomponents. The right panel shows the four main channels together with the residual. The decomposition leaves zero residual, but because reported shares are rounded to integers, the residual absorbs the rounding discrepancies so that the displayed components sum to 100.

Table 1 provides a more detailed breakdown by period. In addition to the full sample period, it reports the decomposition for the peak-to-end phase, 1995–2010. The comparison reveals sharp differences across the two time periods, which I return to after examining the underlying structural dynamics.

Table 1: Detailed Decomposition of Aggregate TFP in Spain by subperiod, 1970–2010.

Component	**1970–2010** *(Overall period)*	**1995–2010** *(Peak to end)*
Δ Aggregate TFP	+30.22 pp (100%)	−3.45 pp (100%)
Δ Technical efficiency	+31.18 pp (+103%)	−0.01 pp (+0%)
Δ Factor reallocation	−1.97 pp (−7%)	−4.02 pp (+117%)
Distortion-driven	+1.05 pp (+3%)	+0.75 pp (−22%)
Labor	+0.49 pp (+2%)	+0.66 pp (−19%)
Capital	+0.55 pp (+2%)	+0.08 pp (−2%)
Composition	−3.02 pp (−10%)	−4.77 pp (+138%)
Labor	+0.40 pp (+1%)	−3.29 pp (+96%)
Capital	−3.42 pp (−11%)	−1.47 pp (+43%)
Δ Domestic intermediates	+1.26 pp (+4%)	+0.57 pp (−17%)
Δ Foreign intermediates	−0.25 pp (−1%)	+0.01 pp (−0%)
Δ Residual	+0.00 pp (+0%)	+0.00 pp (−0%)

Table Notes. Gains and losses are expressed in percentage points ("pp"). Percentages in parentheses indicate the share of total the total change in aggregate TFP explained by each component.

Structural transformation and sectoral dynamics. To build intuition for these results, it is useful to examine the underlying sectoral dynamics. Over 1970–2010, the Spanish economy underwent a profound structural transformation, with the employment share of services rising from roughly 40 percent to 75 percent. Meanwhile, agriculture and manufacturing—the two sectors with the strongest productivity growth—contracted sharply, with their employment shares falling from 27 percent to 5 percent and from 21 percent to 13 percent, respectively (Appendix Figure 12). Within services, employment growth was concentrated in real estate and business services, wholesale and retail trade, and hotels and restaurants. As shown in Appendix Figure 13, these expanding service sectors exhibited weak productivity growth—negative in two of the three cases. Taken together, these patterns suggest that employment growth was concentrated in sectors with poor productivity performance.

These shifts unfolded alongside Spain's integration into European and global markets. Spain's accession to the European Community in 1986, the completion of the Single Market in 1993, and the convergence process leading to the adoption of the euro in 1999 were accompanied by strong capital inflows and major changes in the composition of domestic demand. Trade integration intensified especially between the mid-1980s and 2000: over this period, the foreign share of household consumption rose

from 2 percent to 11 percent, while the foreign share of intermediate inputs increased from 8 percent to 18 percent (Appendix Figure 16). To the extent that foreign goods displaced domestic production in tradable sectors while resources continued to move toward non-tradable services, trade integration may have reinforced this reallocation.

Two caveats are worth noting. First, the sectoral TFP measures presented here may capture not only changes in technical efficiency but also changes in within-sector resource allocation occurring across firms but that cannot be directly measured in the available data. Sectoral productivity declines may therefore reflect either technological stagnation or worsening within-sector misallocation.

Second, measured TFP declines in service sectors may partly reflect measurement problems with real output, rather than genuine declines in efficiency. Such problems are particularly important in activities such as real estate, hospitality, and public administration, where output is difficult to quantify and quality adjustment is less straightforward than in goods-producing sectors. If price indices do not fully account for quality change, real output growth in these sectors may be understated, and measured sectoral productivity growth will be understated as well. This caveat is therefore relevant for interpreting the weak productivity performance of expanding service sectors, and for the technical-efficiency component of the decomposition more broadly.

Zooming into the Productivity Decline. I now focus on the 1995–2010 subperiod—from the peak of aggregate TFP to the end of the sample—to isolate the sources of Spain's productivity decline. The structural transformation documented above was already underway before 1995, but its aggregate implications changed thereafter. Before 1995, strong technical efficiency gains across sectors more than offset the adverse composition effects of reallocation toward low-productivity services. After 1995, however, technical efficiency growth largely disappeared, while the reallocation continued. The construction boom was also concentrated in this later period: construction's employment share was roughly stable until 1995, then rose sharply to around 15 percent by 2007 before falling back to 6 percent by 2010. Over the same period, TFP in construction declined, so this temporary but pronounced shift of resources into the sector amplified the adverse composition effects of reallocation.

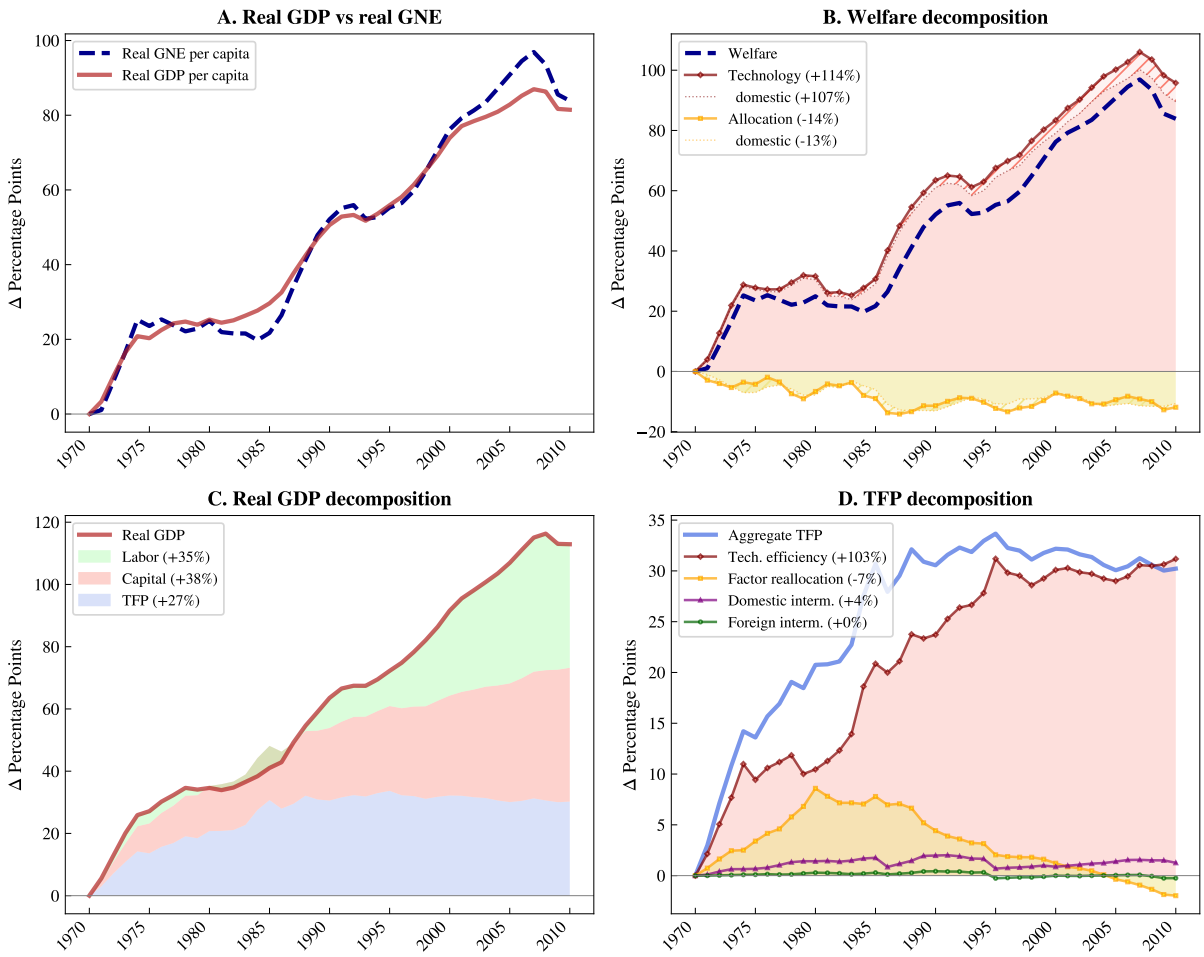
As Table 1 shows, the 3.5 percentage point decline in aggregate TFP was driven entirely by adverse factor reallocation, which contributed 4 percentage points, or 117 percent of the total decline. Most of this drag came from the composition term, which contributed 4.8 percentage points and was only partly offset by a positive distortion-driven term of 0.8 percentage points. Labor reallocation contributed 3.3 percentage points of the decline, while capital reallocation contributed 1.5 percentage points, so both factors moved in a way that reduced aggregate TFP, with labor reallocation

playing the larger role. The positive distortion-driven term implies that the observed reallocation was not concentrated in the most distorted sectors; otherwise, the decline would have been larger. Thus, Spain's aggregate productivity declined not because structural transformation accelerated after 1995, but because the technical-efficiency gains that had previously offset this adverse reallocation effects largely disappeared.

6.3 Growth Accounting and Welfare

Figure 5 presents a comprehensive view of growth, productivity, and welfare in Spain. Panel A compares real GDP per capita (deflated by the GDP deflator) with real GNE per capita (deflated by the CPI), both expressed as cumulative percentage-point changes from 1970. The two series track each other closely through the mid-1990s, after which GNE per capita slightly outpaces GDP per capita, consistent with Spain's increasing reliance on imported consumption goods documented in the Appendix. By 2007, real GNE per capita exceeds real GDP per capita by roughly 15 percentage points.

Figure 5: Growth Accounting and Welfare in Spain, 1970–2010.



Panel B reports the Baqaee–Farhi welfare decomposition, aggregated into two broad channels: technology (technical efficiency plus factor deepening) and allocation (changes in distortions, income share shifts, and transfers). Technology accounts for approximately 98 percentage points of Spain's welfare gain, while allocation reduces welfare by about 14 percentage points. Within each channel, the figure distinguishes domestic and foreign contributions. Domestic technology dominates, while foreign technology makes a modest positive contribution. The allocation drag is also predominantly domestic, reflecting the adverse effects of structural transformation documented earlier. The key message of the panel is that Spain's welfare improvement were driven overwhelmingly by technological forces, especially domestic but also foreign ones, while allocation worked in the opposite direction and partially offset those gains.

Panel C presents a standard decomposition of real GDP into TFP growth and the contributions of capital and labor. Capital deepening was the largest contributor to GDP growth, reflecting Spain's sustained investment boom during the convergence process toward euro adoption. TFP contributed roughly 30 percentage points, accounting for about 27% of cumulative GDP growth. Labor's contribution was essentially zero until 1995, but rose markedly thereafter until accounting for 35% of cumulative GDP growth by the end of the period. The decomposition shows that Spain's TFP decline was largely masked at the GDP level by continued capital accumulation: even as productivity stagnated after 1995, output per capita kept rising because Spain continued to invest heavily and to increase work hours.

Panel D reproduces the proposed TFP decomposition from Figure 4 to provide a unified view of the Spanish experience. Taken together, the four panels show that Spain's living standards continued to improve despite the productivity slowdown, creating a disconnect between measured TFP and welfare. That disconnect was sustained by capital deepening, rising labor input, and positive foreign contributions to living standards. Adverse domestic reallocation, however, reduced both productivity and the welfare gains that these forces would otherwise have delivered.

6.4 Counterfactuals

To quantify the role of structural transformation, I construct a counterfactual in which all sectoral weights—Domar weights, welfare weights, and factor shares—are held constant at their 1995 levels. Factor and intermediate input growth is assumed to be proportional across sectors, so that the production structure is preserved exactly. Sectoral technology evolves as observed in the data and is treated as exogenous.

A caveat on interpretation. This exercise is accounting-based rather than a fully-specified general-equilibrium counterfactual. In reality, structural transformation was likely intertwined with the forces shaping sectoral productivity, relative prices, and demand. The resource shift toward services may have reflected Baumol's cost disease, whereby faster productivity growth in agriculture and manufacturing raises the relative cost and expenditure share of low-productivity services. It may also have reflected changes in international comparative advantages and demand-side forces such as rising incomes and non-homothetic preferences. A counterfactual that holds sectoral weights fixed while allowing technology to evolve therefore attributes too much of the post-1995 divergence to reallocation alone. The results should thus be interpreted as an upper bound on the productivity and welfare cost of structural transformation.

Figure 6 presents the results. By construction, the actual and counterfactual series coincide through 1995. From that point onward, however, the two paths diverge sharply. Counterfactual TFP (Panel A) rises by approximately 25 percentage points between 1995 and 2010, in contrast to the actual decline of about 3.5 percentage points, generating a gap of roughly 29 percentage points by the end of the sample. Counterfactual welfare (Panel B) similarly exceeds actual welfare by about 25 percentage points in 2010. These results indicate that Spain's post-1995 productivity decline is best understood as reflecting the sectoral reallocation of resources rather than a broad-based deterioration in within-sector technical efficiency.

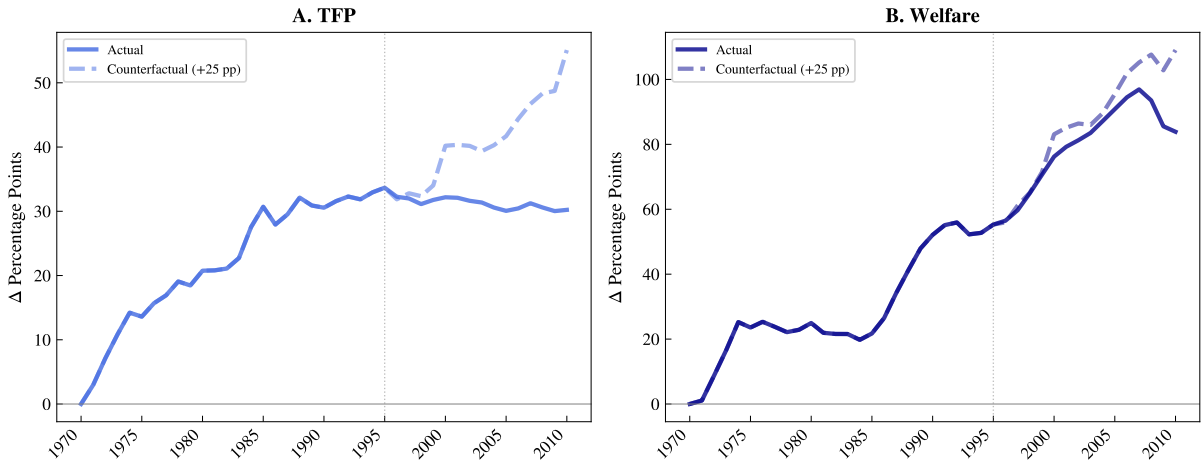


Figure 6: Counterfactual TFP and Welfare in Spain, 1970–2010.

Figure Notes. The counterfactual holds all sectoral weights fixed at 1995 levels with proportional factor and intermediate input growth, while letting sectoral technology paths evolved as observed. The vertical dashed line marks the peak year. These results should be interpreted as an upper bound—see text for discussion.

6.5 Comparison with the Baqaee–Farhi TFP Decomposition

Figure 7 contrasts the proposed decomposition (Panel A) with the Baqaee–Farhi first-order decomposition (Panel B). Both decompose the same aggregate TFP and use the same gross-output Solow residuals to measure sectoral productivity. The key difference is that the proposed decomposition leaves no residual, whereas the Baqaee–Farhi decomposition accumulates a residual of approximately −4 percentage points over 1970–2010. This gap arises from how the non-technology terms are constructed. In the proposed decomposition, factor reallocation and intermediate linkages are algebraic rearrangements of the discrete-time GDP identity, so the components sum to aggregate TFP by construction. In the Baqaee–Farhi decomposition, by contrast, the distortion, income-share, and trade terms are first-order approximations derived for infinitesimal perturbations. When applied to finite annual changes, these terms do not fully recover aggregate TFP, and the resulting gap—about 14% of cumulative TFP growth over this 40-year period—appears as a residual.

Figure 7: Comparison of TFP Decompositions for Spain, 1970–2010.

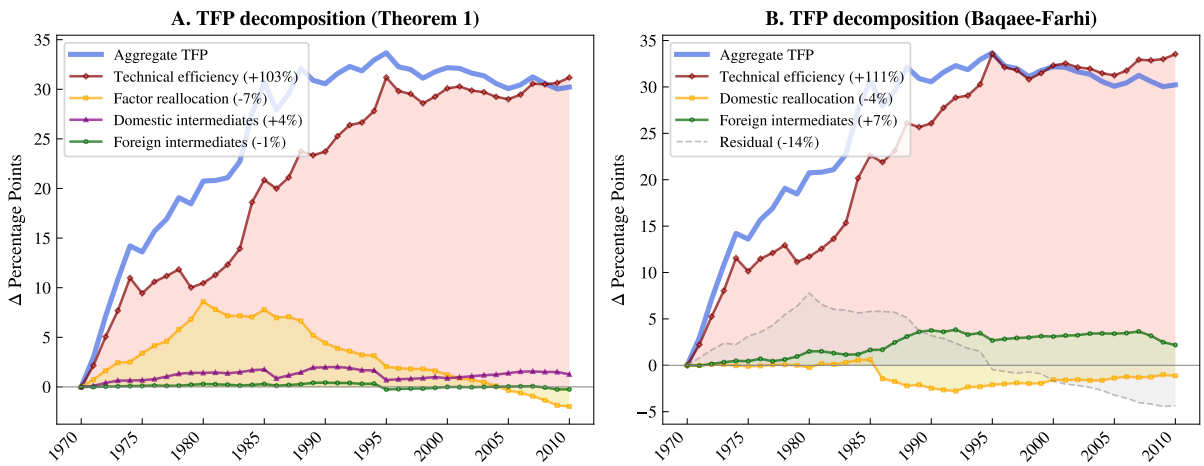


Figure Notes. Theorem 1 (Panel A) vs. Baqaee–Farhi's Theorem 2 (Panel B). Both decompose the same aggregate TFP. The proposed decomposition has zero residual, while the Baqaee–Farhi approximation accumulates a residual. Reported shares are rounded to integers in both decompositions. As a result, the residual shown in the left panel absorbs the rounding discrepancies so that the displayed components sum to 100.

The comparison is economically informative. The Baqaee–Farhi residual closely resembles the factor allocation term in the proposed decomposition, although the magnitudes are not identical because the two decompositions use different weighting schemes and define their components differently. This similarity suggests that between-sector factor reallocation is the main channel that the first-order approximation does not fully capture. By isolating factor reallocation as a distinct channel, and further decomposing it into distortion-driven and composition subcomponents, the proposed decomposition makes it possible to show that factor reallocation accounts for essentially all of Spain's post-1995 TFP decline, with the composition effect emerging as the dominant force.

Robustness. All results in this section are based on distortions estimated using wedge margins and base-period weights. In Appendix F, I show that the results are robust along both dimensions. First, using midpoint (Törnqvist) weights instead of base-period weights leaves the results essentially unchanged. Second, three alternative wedge estimation methods—gross margins, accounting profits, and cost shares—yield qualitatively similar findings: in all cases, TFP peaks around 1995, the post-peak decline is less pronounced than under the traditional Solow residual, and technical efficiency remains the dominant driver of long-run TFP growth. The factor reallocation drag is present across all specifications and remains the key explanation for Spain's TFP decline. Appendix E.10 also examines the role of production networks in shaping income shares, showing that while networks do not affect productivity measurement, they substantially alter the estimated distribution of income across labor, capital, and wedges because of double marginalization.

Italy. To streamline the exposition, I focus on Spain in the main text and relegate the analysis of Italy to Appendix G. Nevertheless, the results for Italy display both important parallels and important differences. Under both the traditional and distorted indices, aggregate TFP in Italy peaks in 2001 and declines by approximately seven percentage points through 2010. Unlike Spain, where the traditional and distorted indices diverge substantially in both timing and magnitude, the two measures yield nearly identical trajectories for Italy, suggesting that distortions play a smaller role in shaping measured productivity there—mainly through a level effect associated with a roughly constant aggregate wedge over time. At the same time, the disconnect between measured TFP and welfare is even more pronounced in Italy than in Spain, underscoring that factors other than domestic aggregate TFP also affect living standards.

The contrast between the two countries is instructive. Spain and Italy both experienced productivity declines, yet for fundamentally different reasons: reallocation in Spain and declines in technical efficiency in Italy. This suggests that the Southern European productivity slowdown cannot be attributed to a single cause, even when the aggregate symptoms appeared similar.

7 Conclusion

In this paper, I reexamine the evolution of aggregate productivity and its relationship to welfare in Spain and Italy between 1970 and 2010. Using a growth-accounting framework tailored to open and distorted economies with input-output linkages, I show that the trajectory of aggregate TFP in Spain has been less dismal than previously thought. When productivity is measured using cost-based factor shares, the decline

appears later and milder than under the traditional Solow residual. The reason is that distortions rose secularly over the period, progressively biasing the traditional Solow residual downward. By contrast, the traditional and distortion-adjusted TFP series in Italy are nearly identical, because the aggregate wedge remained roughly constant over time and therefore only affected the level of measured productivity.

To understand the sources of these productivity declines, I develop a novel decomposition of aggregate TFP growth that isolates factor reallocation as a distinct channel alongside technical efficiency and changes in domestic and foreign intermediate linkages. The comparison between Spain and Italy is instructive. In Spain, the post-1995 decline is fully accounted for by the reallocation of labor and capital toward low-productivity service sectors. In Italy, by contrast, the decline is associated primarily with falling technical efficiency. This heterogeneity cautions against single-cause explanations of the Southern European productivity slowdown, even when aggregate productivity trends appear similar across countries.

A central finding of the paper is the systematic divergence between productivity and welfare—the TFP–welfare disconnect. Despite declining domestic productivity, welfare continued to rise in both Spain and Italy. In Spain, the welfare decomposition shows that the adverse effects of domestic reallocation were more than offset by factor accumulation and by technological progress at home and abroad. Access to foreign technology, transmitted through lower prices of imported final and intermediate goods, sustained welfare growth even as the reallocation of resources depressed measured TFP. The disconnect is even more pronounced in Italy, underscoring that domestic aggregate TFP is not, by itself, a sufficient statistic for living standards in open economies.

While the analysis extends standard growth-accounting approaches to incorporate distortions, openness to trade, and production networks, it remains subject to important limitations. Most notably, sector-level data do not allow me to quantify the within-sector reallocations that others in this literature have found to be important (see, e.g., Gopinath et al., 2017), so part of what is measured here as technical efficiency may instead reflect compositional changes within sectors.

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Appendices

"The TFP–Welfare Disconnect: Evidence from Southern Europe"

Luis Pérez

Southern Methodist University

March 30, 2026

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A TFP and Welfare: Mechanisms and Examples

This appendix formalizes the economic intuition behind the TFP–welfare disconnect and provides simple theoretical examples to clarify the different components of the TFP decomposition introduced in Section 2.

A.1 The TFP–Welfare Disconnect

This section formalizes the TFP–welfare disconnect discussed in Section 2.2 using a small open-economy model. The goal is not to decompose aggregate TFP into channels, but to show—using exact growth accounting—that changes in international relative prices can generate a divergence between aggregate TFP and welfare through equilibrium reallocation, even in the absence of distortions or intermediate inputs.

Environment. Consider a small open economy with two production sectors, indexed by $i \in \{1, 2\}$, three factors of production (labor and two sector-specific factors), and a representative household.

Each sector produces a tradable good using a constant-returns-to-scale technology:

$$y_i = A_i L_{\ell,i}^{\phi} F_i^{1-\phi}, \quad \phi \in (0, 1), \tag{A.1}$$

where A_i denotes sector-specific productivity, $L_{\ell,i}$ is labor employed in sector i, F_i is a sector-specific fixed factor, and ϕ is a factor-intensity parameter that is common across sectors. Without loss of generality, fixed factors are normalized to one, yielding

$$y_i = A_i L_{\ell,i}^{\phi}. \tag{A.2}$$

Foreign producers supply perfect substitutes for each good at exogenous world prices (p_1^w, p_2^w). The economy is perfectly competitive and undistorted.

The representative household has Cobb–Douglas preferences

$$U(c_1, c_2) = c_1^{\alpha} c_2^{1-\alpha}, \qquad \alpha \in (0, 1), \tag{A.3}$$

and faces the budget constraint

$$\sum_i p_i^w c_i \leq w L_\ell + \sum_i r_i, \tag{A.4}$$

where w is the wage, L_ℓ is aggregate labor supply, which is supplied inelastically, and r_i denotes the rents accruing to fixed factors.

Equilibrium. An equilibrium is a set of prices $(\boldsymbol{p}^w, \boldsymbol{r}, w) \in \mathbb{R}^5_{++}$ and allocations $(\boldsymbol{c}, \boldsymbol{y}, \boldsymbol{L}_\ell, \mathbf{nx}) \in \mathbb{R}^8_+$ such that:

1. World prices (p_1^w, p_2^w) are exogenous.
2. Producers maximize profits taking prices as given.
3. The household maximizes utility subject to its budget constraint.
4. Labor and goods markets clear:

$$\sum_i L_{\ell,i} = L_\ell, \qquad c_i + \mathrm{nx}_i = y_i.$$

5. The balance-of-payments condition holds:

$$\sum_i p_i^w \mathrm{nx}_i = 0.$$

Characterization. Any equilibrium of the small open economy is characterized by:

$$\begin{aligned}
&(p_1^w, p_2^w) \gg \mathbf{0} \text{ given,} \\
&w = \phi p_i^w A_i L_{\ell,i}^{\phi-1}, \qquad i = 1, 2, \\
&r_i = p_i^w y_i - w L_{\ell,i}, \\
&L_{\ell,i} = \frac{(A_i \xi_i)^{\frac{1}{1-\phi}}}{\sum_{j\in\{1,2\}} (A_j \xi_j)^{\frac{1}{1-\phi}}} L_\ell, \qquad \xi_i \equiv \frac{p_i^w}{p_1^w}, \quad \xi_1 = 1, \\
&y_i = A_i L_{\ell,i}^{\phi}, \\
&c_i = \frac{\alpha_i}{\xi_i}\Big(w L_\ell + r_1 + r_2\Big), \qquad \alpha_i = \begin{cases} \alpha, & i = 1 \\ 1-\alpha, & i = 2 \end{cases}, \\
&\mathrm{nx}_i = y_i - c_i.
\end{aligned}$$

World relative prices are summarized by $\xi_2 = p_2^w / p_1^w$, which is the terms of trade.

Special Cases. When world relative prices coincide with autarky prices, the no-trade allocation is an equilibrium. Labor is allocated according to expenditure shares, and production and consumption coincide. As $\phi \to 1$, the model converges to a Ricardian environment with full specialization. Sector-specific fixed factors prevent complete specialization for $\phi < 1$, ensuring that changes in relative prices induce *partial* reallocation of labor across sectors. These special cases clarify how the terms of trade discipline factor allocation in equilibrium.

Exact Aggregate TFP Accounting. Nominal output (GDP) is

$$PY = \sum_i p_i^w y_i.$$

Real output growth is defined using a Divisia quantity index with base-period weights:

$$\Delta \log Y_t = \sum_i \lambda_{i,t-1} \Delta \log y_{i,t}, \qquad \lambda_{i,t-1} \equiv \frac{p_{i,t-1}^w y_{i,t-1}}{P_{t-1} Y_{t-1}}.$$

Since the economy is undistorted, aggregate TFP growth is given by the traditional Solow residual:

$$\Delta \log \mathrm{TFP}_t = \Delta \log Y_t - \sum_f \Lambda_{f,t-1} \Delta \log L_{f,t},$$

where $\Lambda_{f,t}$ denotes revenue-based factor shares. In this environment, aggregate labor supply and sector-specific fixed factors are inelastic and constant over time, implying $\Delta \log L_{f,t} = 0$ for all primary factors f, and therefore

$$\Delta \log \mathrm{TFP}_t = \Delta \log Y_t.$$

Using the sectoral production function,

$$\Delta \log y_{i,t} = \Delta \log A_{i,t} + \phi \Delta \log L_{\ell,i,t}.$$

Substituting this expression into real output growth, aggregate TFP growth can be written exactly as

$$\Delta \log \mathrm{TFP}_t = \sum_i \lambda_{i,t-1} \Delta \log A_{i,t} + \phi \sum_i \lambda_{i,t-1} \Delta \log L_{\ell,i,t}.$$

The first term captures sectoral productivity growth at fixed allocation weights, and the second the effect of labor reallocation across sectors. Importantly, this expression is an exact accounting identity *given the index choice* and involves no approximation.

Welfare change. With Cobb-Douglas preferences, welfare changes are given by

$$\Delta \log U_t = \alpha \Delta \log c_{1,t} + (1-\alpha) \Delta \log c_{2,t}.$$

In an open economy, welfare depends on both income and world prices, and therefore welfare need not move proportionally with domestic TFP.

Proposition 1 (Trade-induced negative TFP growth and the TFP–welfare disconnect)**.** *There exist parametrizations of the small open economy such that:*

(i) If world prices are constant, aggregate TFP growth is positive.

(ii) For a sufficiently large change in the terms of trade, aggregate TFP growth is negative.

(iii) Despite negative aggregate TFP growth, equilibrium welfare strictly increases.

Proof. See Appendix C.1. □

Interpretation. The proposition highlights a reallocation mechanism. A deterioration in the terms of trade shifts labor toward the sector with relatively lower productivity growth. As a result, the fixed-weight component of aggregate TFP growth remains positive or modest, while the reallocation term is negative and dominates, leading to a decline in aggregate TFP.

Welfare nevertheless rises because the household benefits from improved consumption possibilities at world prices. The example therefore shows that negative aggregate TFP growth does not imply declining welfare in open economies, and that sustained TFP–welfare divergence can arise from equilibrium reallocation.

A.2 Decomposing Aggregate TFP: Illustrative Examples

This appendix illustrates how the aggregate TFP decomposition introduced in Section 2.3 maps into simple economic environments. Four examples isolate, in turn, the composition effect (IIb), the distortion-driven reallocation channel (IIa), domestic intermediate reallocation (III), and the foreign intermediates reallocation (IV). In each example, aggregate TFP is computed from the Divisia definition

$$\Delta \log \mathrm{TFP} = \Delta \log Y - \sum_f \tilde{\Lambda}_f \Delta \log L_f,$$
$$\Delta \log Y = \sum_i \lambda_i \Delta \log y_i - \sum_{i \in \mathcal{I}_c} \sum_j \lambda_i \Omega_{ij} \Delta \log x_{ij},$$

where the second sum subtracts all intermediates inputs j used by domestic sectors i, both domestic ($j \in \mathcal{I}_c$) and imported ($j \notin \mathcal{I}_c$). Each table below reports all terms from both decompositions (Baqaee–Farhi vs. the decomposition proposed in this paper) side by side. A dash (—) indicates that a term has no counterpart in that decomposition. The residual absorbs whatever is not identified. All weights are evaluated at $t-1$.

The Baqaee-Farhi decomposition is

$$\Delta \log \text{TFP} = \underbrace{\sum_{i \in \mathcal{I}_c} \tilde{\lambda}_i \Delta \log A_i}_{\text{technical efficiency}} \underbrace{- \underbrace{\sum_{i \in \mathcal{I}_c} \tilde{\lambda}_i \Delta \log \mu_i}_{\text{sectoral distortions}} - \underbrace{\sum_{f \in \mathcal{F}_c} \tilde{\Lambda}_f \Delta \log \Lambda_f}_{\text{income shares}}}_{\text{domestic reallocation}}$$
$$+ \underbrace{\sum_{i \in \mathcal{I} - \mathcal{I}_c} \left(\tilde{\lambda}_i - \lambda_i\right)\left(\Delta \log q_{ci} - \Delta \log \lambda_i\right)}_{\text{reallocation in foreign intermediates}} + \underbrace{\mathcal{R}.}_{\text{residual}}$$

In the examples below, the cost-based Domar weights are $\tilde{\lambda}_i = \mu_i \lambda_i$ because production networks only have one layer, cost-based aggregate factor shares are $\tilde{\Lambda}_f = \sum_i \tilde{\lambda}_i \Omega_{if}$, and revenue-based factor shares are $\Lambda_f = \sum_i \lambda_i \Omega_{if}$.

Example I: Open economy with no distortions and no intermediate goods. Consider the small open economy of Appendix A.1. Since there are no distortions ($\boldsymbol{\mu} = \mathbf{1}$) or intermediate goods and factors are in fixed supply, the TFP decomposition reduces to:

$$\Delta \log \text{TFP}_t = \underbrace{\sum_{i \in \{1,2\}} \lambda_{i,t-1} \Delta \log A_{i,t}}_{\text{(I) technical efficiency}} + \underbrace{\sum_{i \in \{1,2\}} \lambda_{i,t-1} \Omega_{i\ell,t-1} \Delta \log L_{\ell i,t}}_{\text{(IIb) factor reallocation (composition)}} + \underbrace{\mathcal{R}_t.}_{\text{residual}}$$

With a Cobb–Douglas production function, $\Omega_{i\ell} = \phi$ is uniform across sectors and constant over time, and the aggregate labor share satisfies $\tilde{\Lambda}_\ell = \Lambda_\ell = \sum_i \lambda_i \Omega_{i\ell} = \phi$, which is constant over time. Under the following parametrizations, we have the following equilibrium allocations:

Parameter	**Description**	**Value**
α	weight on good 1 in Cobb-Douglas utility	0.5
ϕ	labor share in production (all sectors)	0.7
L_ℓ	total labor endowment	1
(L_{F_1}, L_{F_2})	endowments of fixed factors	$(1,1)$
(μ_1, μ_2)	sectoral distortions	$(1,1)$
$(A_{1,t-1}, A_{2,t-1})$	base-period productivities	$(1,1)$
$(A_{1,t}, A_{2,t})$	last-period productivities	$(0.99, 1.02)$
$(p_{1,t-1}, p_{2,t-1})$	base-period world prices	$(1,1)$
$(p_{1,t}, p_{2,t})$	last-period world prices	$(1, 0.8)$

	Equilibria under selected parametrizations					
	$L_{\ell 1}$	$L_{\ell 2}$	w	y_1	y_2	GDP
$t-1$	0.5	0.5	0.8618	0.6156	0.6156	1.2311
t	0.6557	0.3443	0.7866	0.7368	0.4835	1.1236

The following table reports the two TFP growth decompositions.

Term	**This paper**	**Baqaee–Farhi 2024**
Aggregate TFP growth	−0.0308	−0.0308
Technical efficiency	+0.0049	+0.0049
Factor reallocation	−0.0357	—
Distortion driven	0	—
Composition effect	−0.0357	—
Domestic reallocation	—	0
Sectoral distortions	—	0
Income shares	—	0
Domestic intermediates reallocation	0	—
Foreign intermediates reallocation	0	0
Residual	0	−0.0357

In this example, aggregate TFP falls by roughly 3 percentage points. The 3.6 percentage-point compositional loss—arising from labor reallocating toward the sector with lower productivity growth—is fully identified in the decomposition proposed in this paper. In the Baqaee–Farhi decomposition, by contrast, this effect is not separately identified as a reallocation channel and is absorbed by the residual.

Example II: Closed economy with distortions and no intermediate goods. Consider an economy like the one in the previous example, except that it is now closed and distorted through markup wedges in output markets, so $\boldsymbol{\mu} \neq \mathbf{1}$. In this case, the TFP decomposition reduces to:

$$\Delta \log \mathrm{TFP}_t = \underbrace{\sum_{i\in\{1,2\}} \lambda_{i,t-1} \Delta \log A_{i,t}}_{\text{(I) technical efficiency}} + \underbrace{\sum_i \lambda_{i,t-1}(\mu_{i,t-1}-1)\Omega_{i\ell,t-1}\Delta \log L_{\ell i,t}}_{\text{(IIa) factor reallocation (distortion-driven)}}$$
$$+ \underbrace{\sum_i \lambda_{i,t-1}\Omega_{i\ell,t-1}\Delta \log L_{\ell i,t}}_{\text{(IIb) factor reallocation (composition)}} + \underbrace{\mathcal{R}_t}_{\text{residual}}.$$

Under the following parametrizations, we have the following equilibrium allocations:

Parameter	Description	Value
α	weight on good 1 in Cobb-Douglas utility	0.5
ϕ	labor share in production (all sectors)	0.7
L_ℓ	total labor endowment	1
(L_{F_1}, L_{F_2})	endowments of fixed factors	$(1, 1)$
$(\mu_{1,t-1}, \mu_{2,t-1})$	base-period sectoral distortions	$(1, 1.5)$
$(\mu_{1,t}, \mu_{2,t})$	last-period sectoral distortions	$(1, 2.5)$
$(A_{1,t-1}, A_{2,t-1})$	base-period productivities	$(1, 1)$
$(A_{1,t}, A_{2,t})$	last-period productivities	$(1, 1.02)$

	Equilibria under selected parametrizations							
	$L_{\ell 1}$	$L_{\ell 2}$	p_1	p_2	w	y_1	y_2	GDP
$t-1$	0.6	0.4	1	1.3283	0.8159	0.6994	0.5268	1.3990
t	0.7143	0.2857	1	1.8623	0.7743	0.7902	0.4244	1.5804

The following table reports the two TFP growth decompositions.

Term	This paper	Baqaee–Farhi 2024
Aggregate TFP growth	−0.0468	−0.0468
Technical efficiency	+0.0099	+0.0149
Factor reallocation	−0.0568	—
Distortion driven	−0.0393	—
Composition effect	−0.0175	—
Domestic reallocation	—	−0.2611
Sectoral distortions	—	−0.3831
Income shares	—	+0.1220
Domestic intermediates reallocation	0	—
Foreign intermediates reallocation	0	0
Residual	0	+0.1994

In this example, aggregate TFP falls by roughly 5 percentage points. The decomposition proposed in this paper cleanly separates the two reallocation channels, with a near-zero residual. The Baqaee–Farhi decomposition leaves a large residual (+0.1994). The technical-efficiency terms also differ across decompositions because Baqaee-Farhi weights sectoral productivity growth using cost-based Domar weights.

Example III: Closed economy with distortions and intermediate goods. Consider an economy like the one in the previous example, except that instead of sector-specific fixed factors in production it uses a domestic intermediate good. Specifically, sector 1 produces a final good using labor and a domestic intermediate according to $y_1 = A_1 L_1^{\phi} x_{12}^{1-\phi}$, and sector 2 produces using only labor, $y_2 = A_2 L_2$. The output of sector 2 is

purchased entirely by sector 1, which uses it as an intermediate good, so $x_{12} = y_2$. The wage is the numeraire ($w = 1$), and aggregate labor is fixed ($\Delta \log L = 0$).

In this case, the TFP decomposition reduces to:

$$\Delta \log \mathrm{TFP}_t = \underbrace{\sum_{i \in \{1,2\}} \lambda_{i,t-1} \Delta \log A_{i,t}}_{\text{(I) technical efficiency}} + \underbrace{\lambda_{1,t-1}(\mu_{1,t-1} - 1)\Omega_{12,t-1} \Delta \log x_{12,t}}_{\text{(III) reallocation in domestic intermediates}} + \underbrace{\mathcal{R}_t}_{\text{residual}}.$$

Under the following parametrizations, we have the following equilibrium allocations:

Parameter	**Description**	**Value**
α	weight on good 1 in Cobb-Douglas utility	0.5
ϕ	labor share in production of sector 1	0.5
L	total labor endowment	1
(μ_1, μ_2)	base-period sectoral distortions	$(1.5, 1)$
$(A_{1,t-1}, A_{2,t-1})$	base-period productivities	$(1, 1)$
$(A_{1,t}, A_{2,t})$	last-period productivities	$(1, 1.1)$

	Equilibria under selected parametrizations							
	L_1	L_2	p_1	p_2	w	y_1	$y_2 = x_{12}$	GDP
$t-1$	0.5	0.5	3	1	1	0.500	0.5	1.5
t	0.5	0.5	2.860	0.909	1	0.524	0.550	1.5

The following table reports the two TFP growth decompositions.

Term	**This paper**	**Baqaee–Farhi 2024**
Aggregate TFP growth	+0.0447	+0.0447
Technical efficiency	+0.0318	+0.0318
Factor reallocation	0	—
Distortion driven	0	—
Composition effect	0	—
Domestic reallocation	—	0
Sectoral distortions	—	0
Income shares	—	0
Domestic intermediates reallocation	+0.0159	—
Foreign intermediates reallocation	0	0
Residual	0	+0.0159

In this example, aggregate TFP increases by roughly 4.5 percentage points. The decomposition proposed in this paper attributes 1.6 percentage points to changes in domestic intermediate use and leaves a near-zero residual. The Baqaee–Farhi decomposition has no counterpart for domestic intermediates and assigns this component to the residual.

Example IV: Open economy with foreign intermediate goods. Consider now a small open economy with a single sector producing a tradable final good according to $y = AL^{\phi}m^{1-\phi}$, where m denotes foreign intermediates. Imported intermediates are purchased at exogenous world price p_m^w, and the tradable good sells at exogenous world price p^w, since there is a perfect substitute available abroad. Since the firm is a price-taker—both p^w and p_m^w are exogenous—$\mu > 1$ cannot represent market power in the output market. Instead, μ is a cost wedge: a domestic distortion, such as an output tax or an input tax, that drives a wedge between the world price and marginal cost, leading the firm to under-employ inputs relative to the competitive benchmark. The revenue from the wedge is rebated lump-sum to the household. The first-order condition from the firm's problem gives

$$m = \left(\frac{(1-\phi)Ap^w}{\mu p_m^w}\right)^{1/\phi}, \qquad w = \frac{\phi p^w y}{\mu L}.$$

The representative household supplies labor inelastically ($L = 1$), owns the firm, and has utility $U = c$. Household income equals $wL + \pi + T$. In equilibrium, profits are zero ($\pi = 0$) and the lump-sum rebate is $T = p^w(\mu - 1)y/\mu$. The trade balance requires the country to export part of y to finance imports:

$$p^w(y - c) = p_m^w m \qquad \Longrightarrow \qquad c = y - \frac{p_m^w}{p^w}m,$$

so GDP $= p^w c = p^w y - p_m^w m$.

For Cobb-Douglas, $\Omega_m = p^w m/(p^w y) = (1-\phi)/\mu$ and $\lambda_Y = p^w y/\text{GDP}$, which are constant across periods. In this case, the TFP decomposition reduces to

$$\Delta \log \text{TFP}_t = \underbrace{\lambda_{t-1}^Y \Delta \log A_t}_{\text{(I) technical efficiency}} + \underbrace{\lambda_{m,t-1}(\mu_{t-1} - 1)\Omega_{m,t-1}\Delta \log m_t}_{\text{(IV) reallocation in foreign intermediates}} + \underbrace{\mathcal{R}_t}_{\text{residual}}.$$

Foreign-intermediate reallocation term in the Baqaee–Farhi decomposition. Baqaee and Farhi define q_{cm} as the domestic net output of good m—production of m in country c, which is zero since m is foreign, minus domestic intermediate use of m:

$$q_{cm} = 0 - x_{ym} = -m.$$

So $q_{cm} = -m < 0$, consistently with GDP $= p^w y + p_m^w(-m) = p^w y - p_m^w m$. The revenue-based Domar weight of the foreign sector is $\lambda_m = -p_m^w q_{cm}/\text{GDP} = p_m^w m/\text{GDP} = \lambda^Y \Omega_m = \lambda^Y(1-\phi)/\mu$, so $\Delta \log \lambda_m = 0$. The cost-based Domar weight is $\tilde{\lambda}_m = \lambda^Y \tilde{\Omega}_m = \lambda^Y \mu \Omega_m = \lambda^Y(1-\phi)$, computed using the domestic wedge μ, not the foreign sector's. Notably, $\tilde{\lambda}_m$ equals the gross output share of m in production and is independent of μ. The gap

between cost-based and revenue-based weights is therefore

$$\tilde{\lambda}_m - \lambda_m = \lambda^Y(1-\phi) - \frac{\lambda^Y(1-\phi)}{\mu} = \lambda^Y(1-\phi)\frac{\mu-1}{\mu} = \lambda^Y(\mu-1)\Omega_m.$$

Since $\Delta \log q_{cm} = \Delta \log(-m) = \Delta \log m$ and $\Delta \log \lambda_m = 0$, we have

$$\left(\tilde{\lambda}_m^{Y_c} - \lambda_m^{Y_c}\right)\left(\Delta \log q_{c,m} - \Delta \log \lambda_m^{Y_c}\right) = \lambda^Y(\mu-1)\Omega_m \Delta \log m,$$

so the Baqaee–Farhi foreign-intermediate reallocation term is identical to the term in the decomposition proposed in this paper. Under the following parametrizations, we have the following equilibrium allocations:

Parameter	**Description**	**Value**
ϕ	labor share gross output	0.5
L	total labor endowment	1
A	productivity	1
μ	cost wedge	1.5
p^w	world output price	1
$(p^w_{m,t-1}, p^w_{m,t})$	world input price	$(0.333, 0.300)$

	Equilibria under selected parametrizations				
	m	y	c	w	GDP
$t-1$	1	1	0.667	0.333	0.667
t	1.235	1.111	0.741	0.370	0.741

The following table reports the two TFP growth decompositions.

Term	**This paper**	**Baqaee–Farhi 2024**
Aggregate TFP growth	+0.0527	+0.0527
Technical efficiency	0	0
Factor reallocation	0	—
Distortion driven	0	—
Composition effect	0	—
Domestic reallocation	—	0
Sectoral distortions	—	0
Income shares	—	0
Domestic intermediates reallocation	0	—
Foreign intermediates reallocation	+0.0527	+0.0527
Residual	0	0

In this example, aggregate TFP increases by roughly 5 percentage points, and both decompositions identify the full TFP gain with zero residual.

B Local Decompositions and Discrete-Time Accounting

This appendix clarifies the relationship between the discrete-time decomposition proposed in Theorem 1 and the local first-order decompositions of Baqaee and Farhi (2024). The decomposition in the main text is derived directly in discrete time using finite log changes and base-period weights. By contrast, the Baqaee–Farhi formulas are *local* differential identities derived for *infinitesimal* perturbations around an equilibrium. Implementing them over a finite interval between two observed equilibria therefore requires a discretization, and hence a weighting convention. This appendix shows how alternative conventions arise, what interaction terms they absorb, and why the baseline empirical analysis uses base-period weights.

When primitives evolve smoothly as functions of time, the distinction between local and discrete-time accounting is essentially innocuous: the local accounting identity integrates exactly to a Divisia representation, and the choice of how equilibrium weights are evaluated is immaterial at first order. In many empirical settings, however, underlying primitives change in a non-smooth manner—for example, following abrupt changes in the terms of trade, technology adoption, trade liberalization, or lumpy investment and capacity adjustment. In such environments, equilibrium allocations and weights can shift substantially over short horizons, and large reallocations interact with underlying heterogeneity in ways that are not well captured by a single first-order approximation.

Because equilibria are observed only at discrete dates, the continuous-time accounting identity cannot be implemented directly and must be approximated using endpoint data. Since equilibrium weights move endogenously between observations, different weighting conventions correspond to different approximations of the same underlying identity and therefore imply different residual terms and economic interpretations.

I first integrate the local TFP decomposition between two observed equilibria to obtain an *exact but unobservable* representation in terms of line integrals. I then discuss three canonical discretizations—base-period, end-period, and midpoint (Tornqvist) weights—and show how interaction terms arise mechanically from discretization when changes in primitives and equilibrium weights are finite. Finally, I explain the empirical choice made in the paper: base-period weights. Relative to midpoint weights, this convention is less accurate as a pure numerical approximation, but it preserves a cleaner counterfactual interpretation of each component and leaves higher-order interaction and reallocation/composition effects in the residual rather than absorbing them into the measured contributions.

B.1 Local GE Accounting Identities and Finite Changes

Fix a country c. Let $\boldsymbol{\theta}$ denote the vector of primitives (technologies, wedges, etc.) that pin down equilibrium allocations and prices. Equilibrium objects—such as Domar weights and factor shares—are functions of $\boldsymbol{\theta}$ through the equilibrium mapping.

The first-order TFP decomposition can be written as the differential identity

$$\begin{aligned} \mathrm{d}\log \mathrm{TFP}_c = \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}(\boldsymbol{\theta})\mathrm{d}\log A_i - \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}(\boldsymbol{\theta})\mathrm{d}\log \mu_i - \sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}(\boldsymbol{\theta})\mathrm{d}\log \Lambda_f^{Y_c} \\ + \sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}_i^{Y_c}(\boldsymbol{\theta}) - \lambda_i^{Y_c}(\boldsymbol{\theta})\right)\left(\mathrm{d}\log q_{ci} - \mathrm{d}\log \lambda_i^{Y_c}\right). \end{aligned} \tag{B.1}$$

For notational convenience, define

$$\omega_i(\boldsymbol{\theta}) := \tilde{\lambda}_i^{Y_c}(\boldsymbol{\theta}) - \lambda_i^{Y_c}(\boldsymbol{\theta}), \qquad g_{ci}(\boldsymbol{\theta}) := \log q_{ci}(\boldsymbol{\theta}) - \log \lambda_i^{Y_c}(\boldsymbol{\theta}),$$

so that the trade term can be written compactly as

$$\sum_{i\in\mathcal{I}-\mathcal{I}_c} \omega_i(\boldsymbol{\theta})\mathrm{d}g_{ci}.$$

Equation (B.1) is a *local* first-order GE accounting identity. It describes how aggregate TFP responds to *infinitesimal perturbations* in primitives, but it does not by itself specify how those infinitesimal changes should be aggregated over a finite interval between two observed equilibria.

B.2 Exact Finite-Change Representation

Let the economy be observed in equilibrium at dates $t-1$ and t, associated with primitives $\boldsymbol{\theta}_{t-1}$ and $\boldsymbol{\theta}_t$. To integrate the differential identity (B.1) over a finite change, introduce auxiliary parameter $\tau \in [0,1]$ and choose any $\mathcal{C}^1$ path $\boldsymbol{\theta}(\tau)$ such that $\boldsymbol{\theta}(0) = \boldsymbol{\theta}_{t-1}$ and $\boldsymbol{\theta}(1) = \boldsymbol{\theta}_t$.[13]

Along this path define

$$\tilde{\lambda}_i^{Y_c}(\tau) := \tilde{\lambda}_i^{Y_c}(\boldsymbol{\theta}(\tau)), \qquad a_i(\tau) := \log A_i(\boldsymbol{\theta}(\tau)), \qquad m_i(\tau) := \log \mu_i(\boldsymbol{\theta}(\tau)),$$

$$\tilde{\Lambda}_f^{Y_c}(\tau) := \tilde{\Lambda}_f^{Y_c}(\boldsymbol{\theta}(\tau)), \qquad l_f(\tau) := \log \Lambda_f^{Y_c}(\boldsymbol{\theta}(\tau)), \qquad \omega_i(\tau) := \omega_i(\boldsymbol{\theta}(\tau)), \qquad g_i(\tau) := g_{ci}(\boldsymbol{\theta}(\tau)).$$

[13] The path is a mathematical device that connects the two observed equilibria. It need not represent within-period dynamics. A canonical choice is linear interpolation in primitives: $\boldsymbol{\theta}(\tau) = (1-\tau)\boldsymbol{\theta}_{t-1} + \tau\boldsymbol{\theta}_t$.

Integrating (B.1) along the path yields the exact finite-change representation

$$\Delta \log \mathrm{TFP}_{c,t} = \int_0^1 \left(\sum_{i \in \mathcal{I}_c} \tilde{\lambda}_i^{Y_c}(\tau)\, \mathrm{d}a_i(\tau) - \sum_{i \in \mathcal{I}_c} \tilde{\lambda}_i^{Y_c}(\tau)\, \mathrm{d}m_i(\tau) - \sum_{f \in \mathcal{F}_c} \tilde{\Lambda}_f(\tau)\, \mathrm{d}\ell_f(\tau) + \sum_{i \in \mathcal{I} - \mathcal{I}_c} \omega_i(\tau)\, \mathrm{d}g_i(\tau) \right), \tag{B.2}$$

where $\Delta \log \mathrm{TFP}_{c,t} := \log \mathrm{TFP}_c(\boldsymbol{\theta}_t) - \log \mathrm{TFP}_c(\boldsymbol{\theta}_{t-1})$.

B.3 Weighting Conventions in Discrete Time

Each term in (B.2) has the generic structure

$$\int_0^1 w(\tau)\, \mathrm{d}x(\tau), \tag{B.3}$$

where x is a (log) variable and w the associated equilibrium weight.

With discrete data, the path $\tau \mapsto w(\tau)$ is unobserved, and the integral must be approximated using endpoint information. There are three canonical discretizations:

1. **Base-period weights**:

$$\int_0^1 w(\tau)\, \mathrm{d}x(\tau) \approx w(0)\, \Delta x.$$

2. **End-period weights**:

$$\int_0^1 w(\tau)\, \mathrm{d}x(\tau) \approx w(1)\, \Delta x.$$

3. **Midpoint (Tornqvist) weights**:

$$\int_0^1 w(\tau)\, \mathrm{d}x(\tau) \approx \overline{w}\, \Delta x, \qquad \overline{w} := \frac{1}{2}\Big(w(0) + w(1)\Big).$$

All three conventions coincide in the continuous-time limit. Differences arise only because empirical implementations use finite changes between discrete dates. In much of index-number and growth-accounting practice, midpoint weights are the default because they provide a second-order approximation to the underlying Divisia integral. That property is attractive when the main objective is numerical approximation. In the present setting, however, the objective is also economic attribution: I want the decomposition to preserve a clear base-period counterfactual interpretation and to

leave interaction effects associated with changing equilibrium weights in the residual rather than absorbing them into each measured component.

B.4 Interaction Terms and Second-Order Accuracy

The role of weighting choices becomes transparent through a second-order expansion.

Lemma 1 (Second-order discretization)**.** *If w and x are $\mathcal{C}^3$ on $[0,1]$, then*

$$\int_0^1 w(\tau)\,\mathrm{d}x(\tau) = w(0)\,\Delta x + \frac{1}{2}\Delta w \Delta x + \mathcal{O}(\|\Delta\|^3),$$

where $\|\Delta\| := |\Delta w| + |\Delta x|$. Equivalently,

$$\int_0^1 w(\tau)\,\mathrm{d}x(\tau) = \overline{w}\,\Delta x + \mathcal{O}(\|\Delta\|^3).$$

This Lemma shows that midpoint (Tornqvist) weights achieve second-order accuracy by absorbing the interaction term, $\frac{1}{2}\Delta w\,\Delta x$, into each measured contribution $\overline{w}\,\Delta x$. Under the base-period convention, by contrast, this term is left in the residual rather than redistributed across components.

Proof. Since $x \in \mathcal{C}^3$, we can write $\mathrm{d}x(\tau) = x'(\tau)\,\mathrm{d}\tau$, so

$$\int_0^1 w(\tau)\,\mathrm{d}x(\tau) = \int_0^1 w(\tau)\,x'(\tau)\,\mathrm{d}\tau. \tag{B.3}$$

Add and subtract $w(0)$ to obtain the exact decomposition:

$$\int_0^1 w(\tau)\,\mathrm{d}x(\tau) = w(0)\,\Delta x + \int_0^1 \Big(w(\tau) - w(0)\Big)x'(\tau)\,\mathrm{d}\tau. \tag{B.4}$$

We now approximate the remainder integral. By Taylor's theorem, for each $\tau \in [0,1]$ there exist $\xi_w(\tau), \xi_x(\tau) \in (0,\tau)$ such that

$$w(\tau) = w(0) + w'(0)\tau + \frac{1}{2}w''(0)\tau^2 + \frac{1}{6}w'''(\xi_w(\tau))\tau^3, \tag{B.5}$$

$$x'(\tau) = x'(0) + x''(0)\tau + \frac{1}{2}x'''(\xi_x(\tau))\tau^2. \tag{B.6}$$

Multiplying these expansions and integrating term-by-term yields

$$\int_0^1 \Big(w(\tau) - w(0)\Big)x'(\tau)\,\mathrm{d}\tau = \frac{1}{2}\,w'(0)x'(0) + \mathcal{O}(\|\Delta\|^3), \tag{B.7}$$

where $\mathcal{O}(\|\Delta\|^3)$ collects all third-order and higher terms.

Finally, relate $w'(0)$ and $x'(0)$ to endpoint changes. By the fundamental theorem of calculus,

$$\Delta w = w(1) - w(0) = \int_0^1 w'(\tau)\,\mathrm{d}\tau = w'(0) + O(\|\Delta\|^2), \tag{B.8}$$

$$\Delta x = x(1) - x(0) = \int_0^1 x'(\tau)\,\mathrm{d}\tau = x'(0) + O(\|\Delta\|^2). \tag{B.9}$$

Multiplying (B.8) and (B.9) yields

$$w'(0)x'(0) = \Delta w\,\Delta x + O(\|\Delta\|^3). \tag{B.10}$$

Combining (B.4), (B.7), and (B.10) yields

$$\int_0^1 w(\tau)\,\mathrm{d}x(\tau) = w(0)\,\Delta x + \frac{1}{2}\,\Delta w\,\Delta x + \mathcal{O}(\|\Delta\|^3), \tag{B.11}$$

which proves the first claim.

For the equivalent midpoint form, note the identity

$$\overline{w}\,\Delta x = w(0)\,\Delta x + \frac{1}{2}\Delta w\,\Delta x, \qquad \overline{w} := \frac{1}{2}\Big(w(0) + w(1)\Big), \tag{B.12}$$

and substitute into the previous equation to obtain

$$\int_0^1 w(\tau)\,\mathrm{d}x(\tau) = \overline{w}\,\Delta x + \mathcal{O}(\|\Delta\|^3). \tag{B.13}$$

This completes the proof. □

B.5 Discrete-time Approximation of the Baqaee–Farhi decomposition

Applying Lemma 1 term-by-term to (B.2) yields

$$\Delta \log \mathrm{TFP}_{c,t} = \underbrace{\sum_{i\in\mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1}\Delta\log A_{i,t} - \sum_{i\in\mathcal{I}_c} \tilde{\lambda}^{Y_c}_{i,t-1}\Delta\log \mu_{i,t} - \sum_{f\in\mathcal{F}_c} \tilde{\Lambda}^{Y_c}_{f,t-1}\Delta\log \Lambda^{Y_c}_{f,t} + \sum_{i\in\mathcal{I}-\mathcal{I}_c} \omega_{i,t-1}\Delta g_{i,t}}_{\Delta\log \mathrm{TFP}^{\mathrm{FO}}_{c,t}} + \mathcal{R}^{\mathrm{BF}}_{c,t},$$

where

$$\mathcal{R}^{\mathrm{BF}}_{c,t} = \frac{1}{2}\mathcal{C}_{c,t} + \mathcal{O}(\|\Delta\|^3),$$

and

$$\mathcal{C}_{c,t} = \sum_{i \in \mathcal{I}_c} \Delta \tilde{\lambda}_{i,t}^{Y_c} \Delta \log A_{i,t} - \sum_{i \in \mathcal{I}_c} \Delta \tilde{\lambda}_{i,t}^{Y_c} \Delta \log \mu_{i,t} - \sum_{f \in \mathcal{F}_c} \Delta \tilde{\Lambda}_{f,t}^{Y_c} \Delta \log \Lambda_{f,t}^{Y_c} + \sum_{i \in \mathcal{I} - \mathcal{I}_c} \Delta \omega_{i,t} \Delta g_{i,t}.$$

This expression makes precise the distinction between base-period and midpoint implementations of the Baqaee–Farhi decomposition. The base-period approximation isolates the first-order contribution of each margin relative to the initial equilibrium, while the residual $\mathcal{R}_{c,t}^{\mathrm{BF}}$ collects the interaction terms generated by finite changes in both primitives and equilibrium weights. Midpoint weights improve numerical accuracy, but they do so by absorbing the leading interaction term into the reported contribution of each component. The weighting-convention issue is therefore not confined to the residual alone: it affects the economic interpretation of the empirical decomposition.

B.6 Relation to the Baseline TFP Decomposition

The mechanical effect of alternative weighting conventions is similar in both decompositions: they change the allocation of finite changes across measured components and the residual. Midpoint (Tornqvist) weights absorb second-order interaction terms into the components, while base-period weights leave those terms in the residual. The conceptual difference is that, for the Baqaee–Farhi formulas, weighting conventions arise when a local differential decomposition is implemented over a finite interval in the data, whereas in Theorem 1 the use of base-period weights is built directly into the discrete-time decomposition.

I use base-period weights in the baseline because they preserve a clear counterfactual interpretation: each term answers how aggregate TFP would have changed, conditional on the production structure inherited from the previous period, if only that margin had changed as observed. Alternative weighting schemes would instead define alternative discrete-time decompositions, with different allocations of higher-order terms across components and the residual.

C Omitted Proofs

C.1 Proposition 1 (The TFP–welfare disconnect)

Proof. Let the original equilibrium (at $t-1$) be parametrized by

$$(p_1^w, p_2^w, \alpha, \phi, L_\ell, A_1, A_2) = (1, 1, 0.5, 0.7, 1, 1, 1).$$

Then $L_{\ell,1,t-1} = L_{\ell,2,t-1} = \lambda_{1,t-1} = \lambda_{2,t-1} = 0.5$. Consider two alternative transitions:

(i) No terms-of-trade change. Maintain all parameters the same, except technological parameters. Let $(A_{1,t}, A_{2,t}) = (0.99, 1.02)$. Using the equilibrium characterization, $\Delta \log \mathrm{TFP}_t > 0$ and $\Delta \log U_t > 0$.

(ii) Terms-of-trade change. Maintain all parameters the same, except technological parameters and the terms of trade. Let $(A_{1,t}, A_{2,t}) = (0.99, 1.02)$ and $p_2^w = 0.8$. Using the equilibrium characterization, labor reallocates toward the sector with negative productivity growth, $\Delta \log \mathrm{TFP}_t < 0$ and $\Delta \log U_t > 0$.

This verifies (i)–(iii). □

C.2 Proposition 2 (Bias in the Traditional Solow Residual)

Proposition 2 (Bias in Traditional Solow Residual)**.** *Assume that there is a representative producer, producing output Y using technology $Y = AF(L_1, \ldots, L_N)$, where $A > 0$ is a productivity parameter and F is a constant returns-to-scale production function which satisfies standard regularity conditions (i.e., continuity, differentiability, and Inada conditions.) Moreover assume that the representative producer is monopolistic and prices its output charging a markup $\mu \geq 1$ over marginal cost MC; that is, $P = \mu \times MC$. Then, the traditional Solow residual, defined as output growth minus revenue-based input growth, can be written as*

$$\mathrm{d}\log Y - \sum_{f \in \mathcal{F}} \Lambda_n \mathrm{d}\log L_n = \mathrm{d}\log A + \left(\frac{\mu - 1}{\mu}\right)\Big\{\mathrm{d}\log Y - \mathrm{d}\log A\Big\}, \tag{2}$$

where $\Lambda_n \equiv w_n L_n/(PY)$ is the revenue-based share of input n.

Proof. Taking logs and differentiating both sides of the production function with respect to time yields

$$\mathrm{d}\log Y = \mathrm{d}\log A + \sum_{n \in N} \left(\frac{\partial F}{\partial L_n} \frac{L_n}{F}\right) \mathrm{d}\log L_n, \tag{C.1}$$

where $\mathrm{d}\log X \equiv \frac{\mathrm{d}\log X}{\mathrm{d}t}$.

The cost-minimization problem of the representative producer may be written as

$$\begin{aligned} \min_{\boldsymbol{L}\geq \boldsymbol{0}} \quad & C(\boldsymbol{w}) \equiv \sum_{n\in N} w_n L_n \\ \text{s.t.} \quad & Y = AF(L_1,\ldots,L_N) \geq \overline{Y}, \end{aligned}$$

where w_n is the price of input n, L_n is the quantity demanded of input n, and $\overline{Y}$ is a minimal-output constraint.

Letting λ denote the Lagrange multiplier associated with the constraint, a generic first-order condition for an interior demand reads as:

$$w_n = \lambda A \frac{\partial F}{\partial L_n} \qquad \Longleftrightarrow \qquad w_n L_n = \lambda A \frac{\partial F}{\partial L_n} L_n.$$

By the Envelope Theorem, it follows that $\lambda = MC$. Hence, we have

$$\begin{aligned} w_n L_n &= MC \times A \frac{\partial F}{\partial L_n} L_n \\ &= \frac{\mu}{P} \times A \frac{\partial F}{\partial L_n} L_n \qquad \text{(using the price-setting condition)} \end{aligned}$$

Rearranging and doing simple algebraic manipulations, it follows that

$$\frac{\partial F}{\partial L_n} \frac{L_n}{F} = \mu \times \frac{w_n L_n}{PY} \equiv \tilde{\Lambda}_n,$$

where $\tilde{\Lambda}_n \equiv \mu \times w_n L_n/(PY)$ and $\Lambda_n \equiv w_n L_n/(PY)$ are the cost- and revenue-based shares of input n, respectively.

The fact that $\tilde{\Lambda}_n$ is the cost-based share of input n, follows from noting that with a CRS production function, the profit rate s_π is given by

$$s_\pi = 1 - \frac{1}{\mu},$$

which implies that the markup can be written as

$$\mu = (1 - s_\pi)^{-1} = \left(1 - \frac{\sum_{k\in N} w_k L_k}{PY}\right)^{-1} = \frac{PY}{\sum_{k\in N} w_k L_k}.$$

Hence,

$$\tilde{\Lambda}_n = \mu \times \frac{w_n L_n}{PY} = \frac{w_n L_n}{\sum_{k\in N} w_k L_k},$$

which is the cost share of input n.

Plugging cost-based shares into expression (C.1) and re-arranging, we obtain that the distorted Solow residual correctly measures TFP growth:

$$\mathrm{d}\log Y - \sum_{n\in N} \tilde{\Lambda}_n \mathrm{d}\log L_n = \mathrm{d}\log A. \tag{C.2}$$

To obtain the bias associated with the traditional Solow residual, defined as output growth minus revenue-weighted input growth, we can algebraically manipulate expression (C.2). That is, we can write

$$\begin{aligned}\mathrm{d}\log Y - \sum_{n\in N} \Lambda_n \mathrm{d}\log L_n &= \mathrm{d}\log A + \sum_{n\in N} \left(\tilde{\Lambda}_n - \Lambda_n\right)\mathrm{d}\log L_n \\ &= \mathrm{d}\log A + (\mu - 1)\sum_{n\in N} \Lambda_n \mathrm{d}\log L_n.\end{aligned}$$

The equation in the main text is the discrete-time analog of this last expression. □

C.3 Theorem 1 (Decomposition of Aggregate TFP Growth)

Proof. Nominal GDP of country c equals the value added by domestic producers:

$$P_{Y_c} Y_c = \sum_{i\in\mathcal{I}_c} p_i y_i - \sum_{i\in\mathcal{I}_c}\sum_{j\in\mathcal{I}} p_j x_{ij}.$$

The Divisia quantity index Y_c is defined by holding prices at their period-$(t-1)$ values and computing the corresponding change in real output. Taking a first-order approximation around $t-1$,

$$P_{Y_c,t-1}\left(Y_{c,t} - Y_{c,t-1}\right) \approx \sum_{i\in\mathcal{I}_c} p_{i,t-1}\left(y_{i,t} - y_{i,t-1}\right) - \sum_{i\in\mathcal{I}_c}\sum_{j\in\mathcal{I}} p_{j,t-1}\left(x_{ij,t} - x_{ij,t-1}\right) + O(\|\Delta\|^2).$$

Dividing by $P_{Y_c,t-1} Y_{c,t-1}$, using $\Delta x_t / x_{t-1} \approx \Delta \log x_t$, and defining

$$\lambda^{Y_c}_{i,t-1} \equiv \frac{p_{i,t-1} y_{i,t-1}}{P_{Y_c,t-1} Y_{c,t-1}}, \qquad \Omega_{ij,t-1} \equiv \frac{p_{j,t-1} x_{ij,t-1}}{p_{i,t-1} y_{i,t-1}},$$

yields

$$\Delta\log Y_{c,t} = \sum_{i\in\mathcal{I}_c} \lambda^{Y_c}_{i,t-1}\,\Delta\log y_{i,t} - \sum_{i\in\mathcal{I}_c}\sum_{j\in\mathcal{I}} \lambda^{Y_c}_{i,t-1}\,\Omega_{ij,t-1}\,\Delta\log x_{ij,t} + O(\|\Delta\|^2). \tag{C.3}$$

Each sector $i \in \mathcal{I}_c$ produces gross output according to a constant-returns-to-scale production function

$$y_{i,t} = A_{i,t} F_i(\{L_{fi,t}\}_{f \in \mathcal{F}_c}, \{x_{ij,t}\}_{i \in \mathcal{I}}).$$

A first-order log-linear approximation around $t-1$ gives

$$\Delta \log y_{i,t} = \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \varepsilon_{if,t-1} \Delta \log L_{fi,t} + \sum_{j \in \mathcal{I}} \varepsilon_{ij,t-1} \Delta \log x_{ij,t} + r_{i,t} \tag{C.4}$$

where

$$\varepsilon_{if} \equiv \frac{\partial \log y_i}{\partial \log L_{fi}}, \qquad \varepsilon_{ij} \equiv \frac{\partial \log y_i}{\partial \log x_{ij}},$$

and $r_{i,t} = \mathcal{O}(\|\Delta\|^2)$ collects second-order terms. More explicitly,

$$r_{i,t} = \frac{1}{2} \sum_{f \in \mathcal{F}_c} \Delta \varepsilon_{if,t} \Delta \log L_{fi,t} + \frac{1}{2} \sum_{j \in \mathcal{I}} \Delta \varepsilon_{ij,t} \Delta \log x_{ij,t} + O(\|\Delta\|^3),$$

where $\Delta \varepsilon_{if,t}$ and $\Delta \varepsilon_{ij,t}$ denote the changes in output elasticities between $t-1$ and t.

Sector i's price is $p_i = \mu_i \cdot \mathrm{mc}_i$. Cost minimization implies

$$p_i \frac{\partial y_i}{\partial L_{fi}} = \mu_i w_f, \qquad p_i \frac{\partial y_i}{\partial x_{ij}} = \mu_i p_j.$$

Multiplying by L_{fi}/y_i and x_{ij}/y_i, respectively, yields

$$\varepsilon_{if} = \frac{L_{fi}}{y_i} \frac{\partial y_i}{\partial L_{fi}} = \frac{\mu_i w_f L_{fi}}{p_i y_i} = \mu_i \Omega_{if} \equiv \tilde{\Omega}_{if},$$

and

$$\varepsilon_{ij} = \frac{x_{ij}}{y_i} \frac{\partial y_i}{\partial x_{ij}} = \frac{\mu_i p_j x_{ij}}{p_i y_i} = \mu_i \Omega_{ij} \equiv \tilde{\Omega}_{ij}.$$

Substituting $\varepsilon_{if} = \tilde{\Omega}_{if}$ and $\varepsilon_{ij} = \tilde{\Omega}_{ij}$ into (C.4), we obtain

$$\Delta \log y_{i,t} = \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \tilde{\Omega}_{if,t-1} \Delta \log L_{fi,t} + \sum_{j \in \mathcal{I}} \tilde{\Omega}_{ij,t-1} \Delta \log x_{ij,t} + r_{i,t}. \tag{C.5}$$

The remainder is now

$$r_{i,t} = \frac{1}{2} \sum_{f} \Delta \tilde{\Omega}_{if,t} \Delta \log L_{fi,t} + \frac{1}{2} \sum_{j} \Delta \tilde{\Omega}_{ij,t} \Delta \log x_{ij,t} + O(\|\Delta\|^3),$$

which equals zero for Cobb-Douglas production functions.

Substituting (C.5) into (C.3) and collecting terms gives

$$\Delta \log Y_{c,t} = \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \tilde{\Omega}_{if,t-1} \Delta \log L_{fi,t} + \sum_{j \in \mathcal{I}} \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} (\tilde{\Omega}_{ij,t-1} - \Omega_{ij,t-1}) \Delta \log x_{ij,t} + \mathcal{R}^Y_{c,t},$$

where

$$\mathcal{R}^Y_{c,t} = \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} r_{i,t} + \mathcal{O}(\|\Delta\|^2).$$

Using $\tilde{\Omega}_{if,t-1} = \mu_{i,t-1} \Omega_{if,t-1}$ and $\tilde{\Omega}_{ij,t-1} = \mu_{i,t-1} \Omega_{ij,t-1}$, we have

$$\Delta \log Y_{c,t} = \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Delta \log A_{i,t} + \sum_{f \in \mathcal{F}_c} \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \mu_{i,t-1} \Omega_{if,t-1} \Delta \log L_{fi,t} \tag{C.6}$$

$$+ \sum_{j \in \mathcal{I}} \sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} (\mu_{i,t-1} - 1) \Omega_{ij,t-1} \Delta \log x_{ij,t} + \mathcal{R}^Y_{c,t}. \tag{C.7}$$

By definition 1,

$$\Delta \log \mathrm{TFP}_{c,t} = \Delta \log Y_{c,t} - \sum_{f \in \mathcal{F}_c} \tilde{\Lambda}^{Y_c}_{f,t-1} \Delta \log L_{f,t}.$$

Subtracting $\sum_f \tilde{\Lambda}^{Y_c}_{f,t-1} \Delta \log L_{f,t}$ from both sides of (C.6), splitting the intermediate-input term into domestic inputs ($j \in \mathcal{I}_c$) and imported inputs ($j \notin \mathcal{I}_c$), and collecting the remainder into $\mathcal{R}_{c,t}$, yields

$$\Delta \log \mathrm{TFP}_{c,t} = \underbrace{\sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \Delta \log A_{i,t}}_{\text{(I) technical efficiency}} + \underbrace{\sum_{f \in \mathcal{F}_c} \left[\sum_{i \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} \mu_{i,t-1} \Omega_{if,t-1} \Delta \log L_{fi,t} - \tilde{\Lambda}^{Y_c}_{f,t-1} \Delta \log L_{f,t} \right]}_{\text{(II) factor reallocation}}$$
$$+ \underbrace{\sum_{i \in \mathcal{I}_c} \sum_{j \in \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} (\mu_{i,t-1} - 1) \Omega_{ij,t-1} \Delta \log x_{ij,t}}_{\text{(III) reallocation in domestic intermediates}}$$
$$+ \underbrace{\sum_{i \in \mathcal{I}_c} \sum_{j \notin \mathcal{I}_c} \lambda^{Y_c}_{i,t-1} (\mu_{i,t-1} - 1) \Omega_{ij,t-1} \Delta \log x_{ij,t}}_{\text{(IV) reallocation in foreign intermediates}} + \mathcal{R}_{c,t}. \tag{7}$$

To decompose term (II), write

$$\lambda_{i,t-1}^{Y_c}\mu_{i,t-1} = \lambda_{i,t-1}^{Y_c} + \lambda_{i,t-1}^{Y_c}(\mu_{i,t-1} - 1).$$

Then

$$\text{(II)} = \underbrace{\sum_{f\in\mathcal{F}}\sum_{i\in\mathcal{I}_c}\lambda_{i,t-1}^{Y_c}(\mu_{i,t-1}-1)\Omega_{if,t-1}\Delta\log L_{fi,t}}_{\text{(IIa)}} + \underbrace{\sum_{f\in\mathcal{F}}\left[\sum_{i\in\mathcal{I}_c}\lambda_{i,t-1}^{Y_c}\Omega_{if,t-1}\Delta\log L_{fi,t} - \tilde{\Lambda}_{f,t-1}^{Y_c}\Delta\log L_{f,t}\right]}_{\text{(IIb)}}.$$

To see why term (IIb) vanishes in continuous time, note that $\lambda_i\Omega_{if} = w_f L_{fi}/(P_{Y_c}Y_c)$. Hence,

$$\sum_{i\in\mathcal{I}_c}\lambda_i^{Y_c}\Omega_{if}\,\mathrm{d}\log L_{fi} = \frac{w_f}{P_{Y_c}Y_c}\sum_{i\in\mathcal{I}_c}\mathrm{d}L_{fi} = \frac{w_f\,\mathrm{d}L_f}{P_{Y_c}Y_c} = \tilde{\Lambda}_f^{Y_c}\mathrm{d}\log L_f.$$

In discrete time, the analogous identity does not generally hold for log changes over a finite interval because the log change of aggregate factor use is not equal to the share-weighted average of sectoral log changes unless all sectors grow proportionally. As a result, term (IIb) can be non-zero over finite horizons.

Finally, the residual is

$$\mathcal{R}_{ct} = \frac{1}{2}\sum_{i\in\mathcal{I}_c}\lambda_{i,t-1}^{Y_c}\left(\sum_{f\in\mathcal{F}_c}\Delta\tilde{\Omega}_{if,t}\Delta\log L_{fi,t} + \sum_{j\in\mathcal{I}}\Delta\tilde{\Omega}_{ij,t}\Delta\log x_{ij,t}\right) + O(\|\Delta\|^3),$$

which is zero when all sectors have Cobb–Douglas production functions. □

C.4 Theorem 2 (Baqaee–Farhi Decomposition of Aggregate TFP Growth)

The first part of the proof follows Baqaee and Farhi (2024)'s proof of Theorem 1 taking more detailed steps, and then uses the definition of aggregate TFP growth to provide a first-order decomposition of this object.

Proof. The nominal output of country c can be obtained as the revenue generated by domestic producers via wedges and the income that accrues to primary factors. That is,

$$P_{Y_c}Y_c = \sum_{i\in\mathcal{I}_c}\left(1 - \frac{1}{\mu_i}\right)p_i y_i + \sum_{f\in\mathcal{F}_c} w_f L_f.$$

Totally differentiating this expression gives

$$\mathrm{d}P_{Y_c}Y_c + P_{Y_c}\mathrm{d}Y_c = \sum_{i\in\mathcal{I}_c}\left\{\mathrm{d}\left(1-\frac{1}{\mu_i}\right)p_iy_i + \left(1-\frac{1}{\mu_i}\right)\mathrm{d}p_iy_i + \left(1-\frac{1}{\mu_i}\right)p_i\mathrm{d}y_i\right\} + \sum_{f\in\mathcal{F}_c}\left(\mathrm{d}w_fL_f + w_f\mathrm{d}L_f\right).$$

Using simple algebraic manipulations, we can write

$$P_{Y_c}Y_c\left(\frac{\mathrm{d}P_{Y_c}}{P_{Y_c}} + \frac{\mathrm{d}Y_c}{Y_c}\right) = \sum_{i\in\mathcal{I}_c}\left(1-\frac{1}{\mu_i}\right)p_iy_i\left\{\frac{\mathrm{d}(1-1/\mu_i)}{1-1/\mu_i} + \frac{\mathrm{d}p_i}{p_i} + \frac{\mathrm{d}y_i}{y_i}\right\} + \sum_{f\in\mathcal{F}_c}w_fL_f\left(\frac{\mathrm{d}w_f}{w_f} + \frac{\mathrm{d}L_f}{L_f}\right),$$

Using $\mathrm{d}\log x \equiv \mathrm{d}x/x$, the linearity of $\mathrm{d}\log$, and rearranging, we have

$$\mathrm{d}\log Y_c = \sum_{i\in\mathcal{I}_c}\left(1-\frac{1}{\mu_i}\right)\lambda_i^{Y_c}\mathrm{d}\log\left(\left[1-\frac{1}{\mu_i}\right]p_iy_i\right) + \sum_{f\in\mathcal{F}_c}\Lambda_f^{Y_c}\left(\mathrm{d}\log w_f + \mathrm{d}\log L_f\right) - \mathrm{d}\log P_{Y_c}.$$

By Sheppard's lemma, the change in the price of a domestic good i is given by

$$\mathrm{d}\log p_i = \mathrm{d}\log\mu_i - \mathrm{d}\log A_i + \sum_{j\in\mathcal{I}_c}\tilde{\Omega}_{ij}\mathrm{d}\log p_j + \sum_{j\in\mathcal{I}-\mathcal{I}_c}\tilde{\Omega}_{ij}\mathrm{d}\log p_j + \sum_{f\in\mathcal{F}_c}\tilde{\Omega}_{if}\mathrm{d}\log w_f.$$

Using recursion on p_i, we can write

$$\begin{aligned}
\mathrm{d}\log p_i &= \left(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\right) + \sum_{j\in\mathcal{I}_c}\tilde{\Omega}_{ij}\Bigg\{\left(\mathrm{d}\log\mu_j - \mathrm{d}\log A_j\right) + \sum_{k\in\mathcal{I}_c}\tilde{\Omega}_{jk}\mathrm{d}\log p_k + \sum_{k\in\mathcal{I}-\mathcal{I}_c}\tilde{\Omega}_{jk}\mathrm{d}\log p_k \\
&\qquad + \sum_{f\in\mathcal{F}_c}\tilde{\Omega}_{jf}\mathrm{d}\log w_f\Bigg\} + \sum_{j\in\mathcal{I}-\mathcal{I}_c}\tilde{\Omega}_{ij}\mathrm{d}\log p_j + \sum_{f\in\mathcal{F}_c}\tilde{\Omega}_{if}\mathrm{d}\log w_f \\
&= \left(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\right) + \sum_{j\in\mathcal{I}_c}\tilde{\Omega}_{ij}\left\{\left(\mathrm{d}\log\mu_j - \mathrm{d}\log A_j\right) + \sum_{k\in\mathcal{I}_c}\tilde{\Omega}_{jk}\left(\mathrm{d}\log\mu_k - \mathrm{d}\log A_k\right) + \cdots\right\} \\
&\qquad + \sum_{j\in\mathcal{I}-\mathcal{I}_c}\tilde{\Omega}_{ij}\mathrm{d}\log p_j + \sum_{f\in\mathcal{F}_c}\tilde{\Omega}_{if}\mathrm{d}\log w_f.
\end{aligned}$$

Since the world's GDP was taken as the *numeraire*, we have that $\lambda_i = (p_iy_i)/\mathrm{GDP} = p_iy_i$ for any $i \in \mathcal{I}$, and $\Lambda_f = (w_fL_f)/\mathrm{GDP} = w_fL_f$ for any $f \in \mathcal{F}$. Totally differentiating the second expression and rearranging gives $\mathrm{d}\log w_f = \mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f$. Hence,

$$\begin{aligned}
\mathrm{d}\log p_i &= \left(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\right) + \sum_{j\in\mathcal{I}_c}\tilde{\Omega}_{ij}\left\{\left(\mathrm{d}\log\mu_j - \mathrm{d}\log A_j\right) + \sum_{k\in\mathcal{I}_c}\tilde{\Omega}_{jk}\left(\mathrm{d}\log\mu_k - \mathrm{d}\log A_k\right) + \cdots\right\} \\
&\qquad + \sum_{j\in\mathcal{I}-\mathcal{I}_c}\tilde{\Omega}_{ij}\mathrm{d}\log p_j + \sum_{f\in\mathcal{F}_c}\tilde{\Omega}_{if}\left(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\right).
\end{aligned}$$

By definition,

$$
\begin{aligned}
\mathrm{d}\log P_{Y_c} &= \sum_{i\in\mathcal{I}} \Omega_{Y_c,i}\mathrm{d}\log p_i \\
&= \sum_{i\in\mathcal{I}_c} \Omega_{Y_c,i}\mathrm{d}\log p_i + \sum_{i\in\mathcal{I}-\mathcal{I}_c} \Omega_{Y_c,i}\mathrm{d}\log p_i \\
&= \sum_{i\in\mathcal{I}_c} \Omega_{Y_c,i}\mathrm{d}\log p_i - \sum_{i\in\mathcal{I}-\mathcal{I}_c} \lambda_i^{Y_c}\mathrm{d}\log p_i \qquad \text{(since } \Omega_{Y_c,i} = -\lambda_i^{Y_c} \text{ for imported intermediates)} \\
&= \sum_{i\in\mathcal{I}_c} \Omega_{Y_c,i}\Bigg[\Big(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\Big) + \sum_{j\in\mathcal{I}_c}\tilde{\Omega}_{ij}\Bigg\{\Big(\mathrm{d}\log\mu_j - \mathrm{d}\log A_j\Big) + \sum_{k\in\mathcal{I}_c}\tilde{\Omega}_{jk}\Big(\mathrm{d}\log\mu_k - \mathrm{d}\log A_k\Big) + \cdots\Bigg\} \\
&\qquad + \sum_{j\in\mathcal{I}-\mathcal{I}_c}\tilde{\Omega}_{ij}\mathrm{d}\log p_j + \sum_{f\in\mathcal{F}_c}\tilde{\Omega}_{if}\Big(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\Big)\Bigg] - \sum_{i\in\mathcal{I}-\mathcal{I}_c}\lambda_i^{Y_c}\mathrm{d}\log p_i \\
&= \sum_{i\in\mathcal{I}_c} \Omega_{Y_c,i}\Big(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\Big) + \sum_{i\in\mathcal{I}_c}\Omega_{Y_c,i}\sum_{j\in\mathcal{I}_c}\tilde{\Omega}_{ij}\Big(\mathrm{d}\log\mu_j - \mathrm{d}\log A_j\Big) + \cdots \\
&\qquad + \sum_{i\in\mathcal{I}_c}\Omega_{Y_c,i}\Bigg\{\sum_{i\in\mathcal{I}-\mathcal{I}_c}\tilde{\Omega}_{ii}\mathrm{d}\log p_i + \sum_{f\in\mathcal{F}_c}\tilde{\Omega}_{if}\mathrm{d}\Big(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\Big)\Bigg\} - \sum_{i\in\mathcal{I}-\mathcal{I}_c}\lambda_i^{Y_c}\mathrm{d}\log p_i \\
&= \sum_{i\in\mathcal{I}_c}\sum_{i\in\mathcal{I}_c}\Omega_{Y_c,i}\Big[(I-\tilde{\Omega})^{-1}\Big]_{ii}\Big(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\Big) + \sum_{i\in\mathcal{I}_c}\sum_{i\in\mathcal{I}-\mathcal{I}_c}\Omega_{Y_c,i}\Big[(I-\tilde{\Omega})^{-1}\Big]_{ii}\mathrm{d}\log p_i \\
&\qquad + \sum_{i\in\mathcal{I}_c}\sum_{f\in\mathcal{F}_c}\Omega_{Y_c,i}\Big[(I-\tilde{\Omega})^{-1}\Big]_{if}\Big(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\Big) - \sum_{i\in\mathcal{I}-\mathcal{I}_c}\lambda_i^{Y_c}\mathrm{d}\log p_i \\
&= \sum_{i\in\mathcal{I}_c}\sum_{i\in\mathcal{I}_c}\Omega_{Y_c,i}\tilde{\Psi}_{ii}\Big(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\Big) + \sum_{i\in\mathcal{I}-\mathcal{I}_c}\sum_{i\in\mathcal{I}_c}\Omega_{Y_c,i}\tilde{\Psi}_{ii}\mathrm{d}\log p_i \\
&\qquad + \sum_{f\in\mathcal{F}_c}\sum_{i\in\mathcal{I}_c}\Omega_{Y_c,i}\tilde{\Psi}_{if}\Big(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\Big) - \sum_{i\in\mathcal{I}-\mathcal{I}_c}\lambda_i^{Y_c}\mathrm{d}\log p_i \\
&= \sum_{i\in\mathcal{I}_c}\tilde{\lambda}_i^{Y_c}\Big(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\Big) + \sum_{f\in\mathcal{F}_c}\tilde{\Lambda}_f^{Y_c}\Big(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\Big) + \sum_{i\in\mathcal{I}-\mathcal{I}_c}\Big(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\Big)\mathrm{d}\log p_i.
\end{aligned}
$$

Substituting this expression of $\mathrm{d}\log P_{Y_c}$ into that of $\mathrm{d}\log Y_c$ gives

$$
\begin{aligned}
\mathrm{d}\log Y_c = &\sum_{i\in\mathcal{I}_c}\left(1-\frac{1}{\mu_i}\right)\lambda_i^{Y_c}\mathrm{d}\log\left(\left[1-\frac{1}{\mu_i}\right]p_i y_i\right) + \sum_{f\in\mathcal{F}_c}\Lambda_f^{Y_c}\Big(\mathrm{d}\log w_f + \mathrm{d}\log L_f\Big) \\
&- \Bigg\{\sum_{i\in\mathcal{I}_c}\tilde{\lambda}_i^{Y_c}\Big(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\Big) + \sum_{f\in\mathcal{F}_c}\tilde{\Lambda}_f^{Y_c}\Big(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\Big) + \sum_{i\in\mathcal{I}-\mathcal{I}_c}\Big(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\Big)\mathrm{d}\log p_i\Bigg\}.
\end{aligned}
$$

Next note that since $\Lambda_f = (w_f L_f)/\mathrm{GDP} = w_f L_f$, we can write

$$
\sum_{f\in\mathcal{F}_c}\Lambda_f^{Y_c}\Big(\mathrm{d}\log w_f + \mathrm{d}\log L_f\Big) \qquad \Longleftrightarrow \qquad \sum_{f\in\mathcal{F}_c}\Lambda_f^{Y_c}\mathrm{d}\log\Lambda_f.
$$

Moreover, since all the income generated in the production of Y_c accrues to domestic factors or wedges, and $-\mathrm{d}\log GDP = 0$, we can write

$$
\sum_{i\in\mathcal{I}_c}\left(1-\frac{1}{\mu_i}\right)\lambda_i^{Y_c}\mathrm{d}\log\left(\left[1-\frac{1}{\mu_i}\right]p_i y_i\right) \equiv \Lambda_{f^*}^{Y_c}\,\mathrm{d}\log\Lambda_{f^*},
$$

where f^* denotes fictitious factor f^* (i.e., wedges.) By using this notation, one can write the above expressions more compactly. Letting $\mathcal{F}_c^*$ denote the set of all domestic production factors (fictitious and real), we can write

$$\sum_{i\in\mathcal{I}_c}\left(1-\frac{1}{\mu_i}\right)\lambda_i^{Y_c}\mathrm{d}\log\left(\left[1-\frac{1}{\mu_i}\right]p_iy_i\right)+\sum_{f\in\mathcal{F}_c}\Lambda_f^{Y_c}\mathrm{d}\log\Lambda_f\equiv\sum_{f\in\mathcal{F}_c^*}\Lambda_f^{Y_c}\mathrm{d}\log\Lambda_f.$$

Hence, we have

$$\begin{aligned}\mathrm{d}\log Y_c=&\sum_{f\in\mathcal{F}_c^*}\Lambda_f^{Y_c}\mathrm{d}\log\Lambda_f-\sum_{f\in\mathcal{F}_c}\tilde{\Lambda}_f^{Y_c}\mathrm{d}\log\Lambda_f+\sum_{f\in\mathcal{F}_c}\tilde{\Lambda}_f^{Y_c}\mathrm{d}\log L_f\\&+\sum_{i\in\mathcal{I}_c}\tilde{\lambda}_i^{Y_c}\mathrm{d}\log A_i-\sum_{i\in\mathcal{I}_c}\tilde{\lambda}_i^{Y_c}\mathrm{d}\log\mu_i+\sum_{i\in\mathcal{I}-\mathcal{I}_c}\left(\tilde{\lambda}_i^{Y_c}-\lambda_i^{Y_c}\right)\mathrm{d}\log p_i.\end{aligned}$$

Using the identity $\lambda_i^{Y_c}=(p_iq_{ci})/\mathrm{GDP}_c$ for $i\in\mathcal{I}-\mathcal{I}_c$, we can obtain

$$\mathrm{d}\log p_i=\mathrm{d}\log\lambda_i^{Y_c}-\mathrm{d}\log q_{ci}+\mathrm{d}\log\mathrm{GDP}_c,$$

which allows to write

$$\begin{aligned}\mathrm{d}\log Y_c=&\sum_{f\in\mathcal{F}_c^*}\Lambda_f^{Y_c}\mathrm{d}\log\Lambda_f-\sum_{f\in\mathcal{F}_c}\tilde{\Lambda}_f^{Y_c}\mathrm{d}\log\Lambda_f+\sum_{f\in\mathcal{F}_c}\tilde{\Lambda}_f^{Y_c}\mathrm{d}\log L_f+\sum_{i\in\mathcal{I}_c}\tilde{\lambda}_i^{Y_c}\mathrm{d}\log A_i\\&-\sum_{i\in\mathcal{I}_c}\tilde{\lambda}_i^{Y_c}\mathrm{d}\log\mu_i+\sum_{i\in\mathcal{I}-\mathcal{I}_c}\left(\tilde{\lambda}_i^{Y_c}-\lambda_i^{Y_c}\right)\left(\mathrm{d}\log\lambda_i^{Y_c}-\mathrm{d}\log q_{ci}+\mathrm{d}\log\mathrm{GDP}_c\right).\end{aligned}$$

With simple algebraic manipulations we can obtain

$$
\begin{aligned}
\mathrm{d}\log Y_c = &\sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}\mathrm{d}\log L_f + \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log A_i - \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log \mu_i - \sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}\left(\mathrm{d}\log \Lambda_f^{Y_c} + \mathrm{d}\log \mathrm{GDP}_c\right) \\
&+ \sum_{f\in\mathcal{F}_c^*} \Lambda_f^{Y_c}\left(\mathrm{d}\log \Lambda_f^{Y_c} + \mathrm{d}\log \mathrm{GDP}_c\right) + \sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\right)\left(\mathrm{d}\log q_{ci} - \mathrm{d}\log \lambda_i^{Y_c}\right) \\
&- \sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\right)\mathrm{d}\log \mathrm{GDP}_c \\
= &\sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}\mathrm{d}\log L_f + \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log A_i - \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log \mu_i - \sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}\mathrm{d}\log \Lambda_f^{Y_c} \\
&\underbrace{\sum_{f\in\mathcal{F}_c^*} \Lambda_f^{Y_c}\mathrm{d}\log \Lambda_f^{Y_c}}_{\substack{= 0 \text{ since the revenue share gained by} \\ \text{one factor is lost by other(s)}}} + \sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\right)\left(\mathrm{d}\log q_{ci} - \mathrm{d}\log \lambda_i^{Y_c}\right) \\
&+ \left[\underbrace{\sum_{f\in\mathcal{F}_c^*} \Lambda_f^{Y_c}}_{\substack{= 1 \text{ since all income accrues to} \\ \text{domestic factors and wedges}}} - \sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c} - \sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\right)\right]\mathrm{d}\log \mathrm{GDP}_c.
\end{aligned}
$$

Noting that

$$
\sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c} + \sum_{i\in\mathcal{I}-\mathcal{I}_c} \tilde{\lambda}_i^{Y_c} = 1 + \sum_{i\in\mathcal{I}-\mathcal{I}_c} \lambda_i^{Y_c}
$$

gives

$$
\begin{aligned}
\mathrm{d}\log Y_c = &\sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}\mathrm{d}\log L_f + \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log A_i - \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log \mu_i - \sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}\mathrm{d}\log \Lambda_f^{Y_c} \\
&+ \sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\right)\left(\mathrm{d}\log q_{ci} - \mathrm{d}\log \lambda_i^{Y_c}\right).
\end{aligned}
$$

Using definition 1, we obtain

$$
\begin{aligned}
\mathrm{d}\log \mathrm{TFP}_c \approx &\sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log A_i - \sum_{i\in\mathcal{I}_c} \tilde{\lambda}_i^{Y_c}\mathrm{d}\log \mu_i - \sum_{f\in\mathcal{F}_c} \tilde{\Lambda}_f^{Y_c}\mathrm{d}\log \Lambda_f^{Y_c} \\
&+ \sum_{i\in\mathcal{I}-\mathcal{I}_c} \left(\tilde{\lambda}_i^{Y_c} - \lambda_i^{Y_c}\right)\left(\mathrm{d}\log q_{ci} - \mathrm{d}\log \lambda_i^{Y_c}\right).
\end{aligned}
$$

To align this expression with the discrete-time decomposition used in the empirical analysis, I express the local differential result in finite log changes between $t-1$ and t, with all weights evaluated at the initial period. □

C.5 Theorem 3 (Baqaee–Farhi Decomposition of Welfare Growth)

This is an alternative proof to Baqaee and Farhi (2024)'s proof of Theorem 2.

Proof. Nominal gross national expenditure of country c is

$$P_{W_c} W_c = \sum_{f \in \mathcal{F}} \Phi_{cf} w_f L_f + \sum_{i \in \mathcal{I}} \Phi_{ci} \left(1 - \frac{1}{\mu_i}\right) p_i y_i + T_c.$$

Totally differentiating this expression assuming constant ownership shares gives

$$\mathrm{d}\log W_c = \sum_{f \in \mathcal{F}} \Lambda_f^c \left(\mathrm{d}\log w_f + \mathrm{d}\log L_f\right) + \sum_{i \in \mathcal{I}} \lambda_i^c \mathrm{d}\log\left(\Phi_{ci}\left[1 - \frac{1}{\mu_i}\right] p_i y_i\right) + \frac{\mathrm{d}T_c}{\mathrm{GNE}_c} - \mathrm{d}\log P_{W_c}.$$

An implication of Sheppard's lemma is that

$$\begin{aligned}
\mathrm{d}\log p_i &= \mathrm{d}\log \mu_i - \mathrm{d}\log A_i + \sum_{j \in \mathcal{I}} \tilde{\Omega}_{ij} \mathrm{d}\log p_j + \sum_{f \in \mathcal{F}} \tilde{\Omega}_{if} \mathrm{d}\log w_f \\
&= \left(\mathrm{d}\log \mu_i - \mathrm{d}\log A_i\right) + \sum_{j \in \mathcal{I}} \tilde{\Omega}_{ij} \left\{\left(\mathrm{d}\log \mu_j - \mathrm{d}\log A_j\right) + \sum_{k \in \mathcal{I}} \tilde{\Omega}_{jk}\left(\mathrm{d}\log \mu_k - \mathrm{d}\log A_k\right) + \cdots\right\} \\
&\quad + \sum_{f \in \mathcal{F}} \tilde{\Omega}_{if} \mathrm{d}\log w_f + \sum_{j \in \mathcal{I}} \tilde{\Omega}_{ij} \left\{\sum_{f \in \mathcal{F}} \tilde{\Omega}_{jf} \mathrm{d}\log w_f + \cdots\right\}.
\end{aligned}$$

Using the definition of $\mathrm{d}\log P_{W_c}$, we can write

$$\begin{aligned}
\mathrm{d}\log P_{W_c} &= \sum_{i \in \mathcal{I}} \Omega_{W_c,i} \Bigg\{\left(\mathrm{d}\log \mu_i - \mathrm{d}\log A_i\right) + \sum_{j \in \mathcal{I}} \tilde{\Omega}_{ij} \Bigg[\left(\mathrm{d}\log \mu_j - \mathrm{d}\log A_j\right) \\
&\qquad + \sum_{k \in \mathcal{I}} \tilde{\Omega}_{jk}\left(\mathrm{d}\log \mu_k - \mathrm{d}\log A_k\right) + \cdots + \sum_{f \in \mathcal{F}} \tilde{\Omega}_{jf} \mathrm{d}\log w_f + \cdots\Bigg]\Bigg\} + \sum_{f \in \mathcal{F}} \tilde{\Omega}_{if} \mathrm{d}\log w_f \\
&= \sum_{i \in \mathcal{I}} \sum_{i \in \mathcal{I}} \Omega_{W_c,i} \left[(I - \tilde{\Omega})^{-1}\right]_{ii} \left(\mathrm{d}\log \mu_i - \mathrm{d}\log A_i\right) + \sum_{i \in \mathcal{I}} \sum_{f \in \mathcal{F}} \Omega_{W_c,i} \left[(I - \tilde{\Omega})^{-1}\right]_{if} \mathrm{d}\log w_f \\
&= \sum_{i \in \mathcal{I}} \sum_{i \in \mathcal{I}} \Omega_{W_c,i} \tilde{\Psi}_{ii} \left(\mathrm{d}\log \mu_i - \mathrm{d}\log A_i\right) + \sum_{f \in \mathcal{F}} \sum_{i \in \mathcal{I}} \Omega_{W_c,i} \tilde{\Psi}_{if} \mathrm{d}\log w_f \\
&= \sum_{i \in \mathcal{I}} \tilde{\lambda}_i^{W_c} \left(\mathrm{d}\log \mu_i - \mathrm{d}\log A_i\right) + \sum_{f \in \mathcal{F}} \tilde{\Lambda}_f^{W_c} \mathrm{d}\log w_f.
\end{aligned}$$

Substituting this last expression of $\mathrm{d}\log P_{W_c}$ into $\mathrm{d}\log W_c$ gives

$$
\begin{aligned}
\mathrm{d}\log W_c &= \sum_{f\in\mathcal{F}} \Lambda_f^c \left(\mathrm{d}\log w_f + \mathrm{d}\log L_f\right) + \sum_{i\in\mathcal{I}} \lambda_i^c \mathrm{d}\log\left(\Phi_{ci}\left[1-\frac{1}{\mu_i}\right]p_i y_i\right) + \frac{\mathrm{d}T_c}{\mathrm{GNE}_c} \\
&\qquad - \left\{\sum_{i\in\mathcal{I}} \tilde{\lambda}_i^{W_c}\left(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\right) + \sum_{f\in\mathcal{F}} \tilde{\Lambda}_f^{W_c}\mathrm{d}\log w_f\right\} \\
&= \sum_{f\in\mathcal{F}} \Lambda_f^c \mathrm{d}\log\Lambda_f + \sum_{i\in\mathcal{I}} \lambda_i^c \mathrm{d}\log\left(\Phi_{ci}\left[1-\frac{1}{\mu_i}\right]p_i y_i\right) + \frac{\mathrm{d}T_c}{\mathrm{GNE}_c} \\
&\qquad - \left\{\sum_{i\in\mathcal{I}} \tilde{\lambda}_i^{W_c}\left(\mathrm{d}\log\mu_i - \mathrm{d}\log A_i\right) + \sum_{f\in\mathcal{F}} \tilde{\Lambda}_f^{W_c}\left(\mathrm{d}\log\Lambda_f - \mathrm{d}\log L_f\right)\right\} \\
&= \sum_{i\in\mathcal{I}} \tilde{\lambda}_i^{W_c}\mathrm{d}\log A_i + \sum_{f\in\mathcal{F}} \tilde{\Lambda}_f^{W_c}\mathrm{d}\log L_f - \sum_{i\in\mathcal{I}} \tilde{\lambda}_i^{W_c}\mathrm{d}\log\mu_i + \frac{\mathrm{d}T_c}{\mathrm{GNE}_c} \\
&\qquad + \sum_{f\in\mathcal{F}} \left(\Lambda_f^c - \tilde{\Lambda}_f^{W_c}\right)\mathrm{d}\log\Lambda_f + \sum_{i\in\mathcal{I}} \lambda_i^c \mathrm{d}\log\left(\Phi_{ci}\left[1-\frac{1}{\mu_i}\right]p_i y_i\right)
\end{aligned}
$$

Noting that $\lambda_i = \frac{p_i y_i}{\mathrm{GNE}} = p_i y_i$ and $\mathrm{d}\log\mathrm{GNE} = 0$, we can write

$$
\sum_{i\in\mathcal{I}} \lambda_i^c \mathrm{d}\log\left(\Phi_{ci}\left[1-\frac{1}{\mu_i}\right]p_i y_i\right) \equiv \Lambda_{f^*}^c \mathrm{d}\log\Lambda_{f^*},
$$

where f^* stands for wedges, a fictitious factor, $\tilde{\Lambda}_{f^*}^c$ is the share of income/expenditure of country c that comes from domestic and foreign wedges, and Λ_{f^*} is the share of world's GDP/GNE owned by country c. Letting $\mathcal{F}^*$ denote the set of all (fictitious and real) factors, and noting that $\tilde{\Lambda}_{f^*}^{W_c} = 0$ for $f^* \in \mathcal{F}^*$, we can write

$$
\mathrm{d}\log W_c \approx \sum_{i\in\mathcal{I}} \tilde{\lambda}_i^{W_c}\mathrm{d}\log A_i + \sum_{f\in\mathcal{F}} \tilde{\Lambda}_f^{W_c}\mathrm{d}\log L_f - \sum_{i\in\mathcal{I}} \tilde{\lambda}_i^{W_c}\mathrm{d}\log\mu_i + \sum_{f\in\mathcal{F}^*} \left(\Lambda_f^c - \tilde{\Lambda}_f^{W_c}\right)\mathrm{d}\log\Lambda_f + \frac{\mathrm{d}T_c}{\mathrm{GNE}_c}.
$$

As with the TFP decomposition, I express the local differential result in finite log changes between $t-1$ and t, with all weights evaluated at the initial period. □

D Estimation of Distortions

This appendix describes the methods used to estimate the distortions underlying the empirical results. Two main approaches are considered: (i) the margins approach and (ii) the production-function approach.

D.1 The Margins Approach

Following Domowitz et al. (1986), I estimate distortions using gross margins, defined as the ratio of gross operating surplus to total sales. The gross margin of producer i is

$$\text{Gross Margin}_i := \frac{\text{Gross Operating Surplus}_i}{\text{Sales}_i} = \frac{p_i y_i - \sum_{j \in \mathcal{I}} p_j x_{ij} - \sum_{f \in \mathcal{F}^V} w_f l_{if}}{p_i y_i}, \quad \text{(D.1)}$$

where $p_i y_i$ denotes sales, $\sum_j p_j x_{ij}$ is intermediate input expenditures, and $\sum_{f \in \mathcal{F}^V} w_f l_{if}$ is the cost of variable factors of production.

The gross margins approach avoids assumptions about production technologies or market structure, and it requires minimal data: revenues, intermediate input costs, and compensation for variable factors such as labor. These data are readily available in sources such as KLEMS and WIOD. Empirically, Anderson et al. (2020) show that gross margins co-move closely with markups computed from measures of marginal costs, suggesting that gross margins provide a reliable proxy for markups in terms of their distribution, even if not necessarily in levels.

Margins and Distortions. In my preferred specification, distortions μ_i represent wedges between marginal cost and price, capturing market imperfections such as market power or distorting taxes. To quantify these, I compute wedge margins, which adjust gross margins by accounting for expenditures on fixed or quasi-fixed factors such as capital:

$$\text{Wedge margin}_i := \frac{p_i y_i - \sum_{j \in \mathcal{I}} p_j x_{ij} - \sum_{f \in \mathcal{F}^V} w_f l_{if} - \sum_{f \in \mathcal{F}^F} \delta_f \ell_{if} - \sum_{f \in \mathcal{F}^F} r_f \ell_{if}}{p_i y_i}, \quad \text{(D.2)}$$

where δ_f is the depreciation rate of factor f, and r_f is its net-of-depreciation rental rate.

Wedge margins approximate pure-profit margins more accurately than gross margins, because they include the user costs of fixed factors. Under the model's assumption of constant returns to scale, wedge margins are theoretically consistent measures of distortions. More generally, the following identity holds:

$$\text{Wedge margin} = 1 - \frac{\text{RS}}{\mu},$$

where RS are the returns to scale.[14] With constant returns to scale, this implies:

$$\mu_i = \frac{1}{1 - \text{wedge margin}_i}. \tag{D.3}$$

Estimating wedge margins is challenging. It requires producer-level capital stocks disaggregated by asset type, as well as asset-specific depreciation rates and user costs. I use capital data from KLEMS and WIOD, and depreciation rates from the Bureau of Economic Analysis (BEA).[15]

I estimate user costs of capital using the method of van Vlokhoven (2022), which exploits cross-sectional variation in input choices. Specifically, I estimate the following OLS regression:

$$\frac{p_i y_i}{\text{COGS}_i} = \overline{\psi} + \overline{\psi w^{\text{gross}}} \frac{p_i^K K_i}{\text{COGS}_i} + \varepsilon_i,$$

where COGS_i denotes the costs of goods sold—intermediate input expenditure plus labor compensation—and $p_i^K K_i$ is the nominal capital stock. The intercept $\overline{\psi}$ and slope $\overline{\psi w^{\text{gross}}}$ imply an estimate of the gross user cost of capital,

$$\overline{w^{\text{gross}}} = \overline{\psi w^{\text{gross}}} / \overline{\psi}.$$

OLS provides unbiased estimates under the assumption of constant marginal cost and equalized user costs across producers.

D.1.1 The Accounting-Profits Approach

The accounting-profits approach can be viewed as a special case of the wedge-margins approach in which the net-of-depreciation rental rates of fixed factors are set equal to zero. Equivalently, all operating surplus net of intermediate and variable-input expenditure is treated as distortion-related surplus.

D.2 The Production Function Approach

The production function approach to markup estimation dates back to Hall (1986, 1988), who derived an expression for the markup using the first-order condition for a variable input of a cost-minimizing producer operating under price-taking behavior in that

[14] This relation follows under cost minimization and standard regularity conditions on production functions, including differentiability and quasi-concavity (see Hasenzagl and Perez, 2023).

[15] Depreciation rates from the BEA are more reliable than those provided by KLEMS or implied by accounting data. The BEA accounts for tax incentives and accounting practices that may lead firms to overstate depreciation and, unlike KLEMS, it does not back out depreciation rates assuming zero profits.

input market. The core insight is that a producer's markup can be expressed as the output elasticity of a variable input multiplied by the inverse of that input's revenue share. To see this, suppose producer i uses technology

$$y_i = F_i(A_i, \mathbf{x}_i, \boldsymbol{\ell}_i),$$

where y denotes output, A is Hicks-neutral productivity, $\mathbf{x}$ is a vector of variable inputs, and $\boldsymbol{\ell}$ is a vector of pre-determined or quasi-fixed inputs.

The producer minimizes total costs subject to an output constraint:

$$\begin{aligned} \min_{\mathbf{x}_i, \boldsymbol{\ell}_i} \quad & \mathbf{p}_i' \mathbf{x}_i + \mathbf{w}_i' \boldsymbol{\ell}_i + \text{other costs}_i \\ \text{s.t.} \quad & y_i \geq \overline{y}_i. \end{aligned}$$

Let λ_i be the Lagrange multiplier on the constraint. The first-order condition with respect to variable input x_{ij} (with $x_{ij} > 0$) yields

$$p_j = \lambda_i \frac{\partial y_i}{\partial x_{ij}},$$

where λ_i denotes marginal cost. Thus, defining the markup as the ratio of output price to marginal costs,

$$\mu_i := \frac{p_i}{\lambda_i} = \frac{\partial y_i}{\partial x_{ij}} \frac{p_i}{p_j}.$$

Multiplying and dividing by x_{ij}/y_i, we obtain

$$\mu_i = \frac{\partial y_i}{\partial x_{ij}} \frac{x_{ij}}{y_i} \frac{p_i y_i}{p_j x_{ij}} \equiv \epsilon(y_i, x_{ij}) \times \frac{p_i y_i}{p_j x_{ij}}, \tag{D.4}$$

where $\epsilon(y_i, x_{ij})$ is the output elasticity of x_{ij}.

To implement this expression, one must estimate $\epsilon(y_i, x_{ij})$. Two approaches are widely used: the control-function approach and the cost-shares approach.

D.2.1 The Control-Function Approach

The control-function approach requires specifying a functional form for the production function, typically Cobb-Douglas or translog. Consider the Cobb-Douglas case with one variable input x and one pre-determined input ℓ. The associated log-regression is:

$$\log y_{it} = \log A_{it} + \beta_x \log x_{it} + \beta_\ell \log \ell_{it} + \varepsilon_{it}, \tag{D.5}$$

where $\log A_{it}$ is the (unobserved) productivity term, β_x is the elasticity to be estimated (i.e., $\epsilon(y, x)$), and ε_{it} captures measurement error or unanticipated shocks to output or productivity that are observed neither by the econometrician nor the producer.

This approach suffers two well-known econometric challenges. First, $\log A_{it}$ is observed by the producer but not by the econometrician, leading to simultaneity bias. Following Ackerberg et al. (2015), the solution involves a two-step control-function procedure. The first stage non-parametrically controls for unobserved productivity, and the second stage estimates production-function parameters using moment conditions. This yields a consistent estimate of $\beta_x = \varepsilon(y_i, x_{ij})$. Second, most datasets report revenues and expenditures rather than physical quantities, introducing price-related bias. See De Loecker et al. (2016) for a discussion on how to address these biases.

D.2.2 The Cost-Shares Approach

The cost-shares approach developed by Foster et al. (2008) offers a fully nonparametric alternative. It avoids estimating a production function by assuming constant returns to scale and that all inputs are variable. Under these assumptions, the elasticity of a variable input is given by its share in total costs:

$$\epsilon(y_i, x_{ij}) = \frac{p_j x_{ij}}{\sum_{j \in \mathcal{I}} p_j x_{ij} + \sum_{f \in \mathcal{F}} w_f \ell_{if}}. \tag{D.6}$$

Although I classify this method under the production-function approach, it is closely related to the margins approach. In particular, under constant returns to scale, cost minimization, full flexibility of all inputs, input-market price-taking, and the absence of non-operating costs, the cost-shares approach delivers distortion estimates equivalent to those obtained from wedge margins, provided that the same rental rates are used for fixed factors. For that reason, I treat the cost-shares approach as a production-function estimator and note its equivalence to wedge margins only as a special case.

E Data Appendix

This appendix describes the datasets used in the empirical analysis, along with the sample selection, data cleaning, and estimation procedures. The analysis relies primarily on two sources: the KLEMS Growth and Productivity Accounts database and the World Input-Output Database (WIOD). KLEMS provides industry-level data on output and factors of production, while WIOD contains annual data on global input-output linkages, capturing trade flows between producers and both intermediate and final users. Intermediate users are producers that purchase goods to incorporate them as inputs in their own production processes, whereas final users—households, non-profit organizations, governments, and firms—purchase goods for consumption or investment purposes.

E.1 KLEMS

KLEMS provides industry-level data on output, inputs, and productivity starting in 1970 for a range of countries, primarily EU member states but also non-EU economies such as the United States, Japan, and Canada. Coverage varies by release.[16,17] Despite release-specific variation, all KLEMS data are constructed following a harmonized methodology grounded in the neoclassical growth-accounting framework.[18]

The KLEMS data draw on national statistical sources and adhere to the System of National Accounts (SNA) and the European System of Accounts (ESA). A harmonization protocol ensures cross-country comparability through consistent definitions of outputs, inputs, and prices. This harmonization is one of KLEMS's main contributions. Another important advantage of KLEMS relative to national sources is its disaggregation of labor and capital.[19] A higher level of granularity is achieved by merging labor surveys, earnings data, establishment surveys, and social security records.

Nonetheless, KLEMS is subject to the usual measurement issues. Reliability may decline at finer levels of disaggregation or over long time periods.[20] Measuring output

[16]Early releases (2007, 2008, 2009) use the ISIC Rev. 3/NACE Rev. 1 industry classification whereas later releases (2012, 2016, 2017, 2021) use the ISIC Rev. 4/NACE Rev. 2 classification and provide wider country coverage. However, later releases often begin in 1995. For questions requiring long time series, earlier releases are preferable.

[17]In some cases, KLEMS provides more historical disaggregation than national sources. For example, Spanish industry data in KLEMS begin in 1970, whereas coverage from the Instituto Nacional de Estadística (INE) begins in 2000.

[18]See O'Mahony and Timmer (2009) for a methodological overview and Timmer et al. (2007) for country-level documentation.

[19]Labor is split into 18 categories by education (high, medium, low), gender, and age (15–29, 30–49, 50+). Capital is categorized as ICT and non-ICT, with further breakdowns by asset types. Capital classification evolved with ESA 2010. Starting with the 2016 release, the data begin in 1995.

[20]This is due to reliance on supplementary sources or additional assumptions.

is particularly difficult in services and non-market sectors, where hedonic pricing methods are hard to leverage and many outputs (e.g., public services or imputed rents in real estate) are not priced in markets. As a result, many studies restrict attention to the market economy, excluding real estate, public administration, education, and health. In addition, some forms of capital—especially intangible or organizational capital—are poorly captured in national accounts, which may bias productivity measures.[21]

In this paper, I use the 2007/08 and 2021 KLEMS releases. These are combined with WIOD's Socioeconomic Accounts to construct growth-accounting variables and estimate factor compensation and depreciation rates. Depreciation rates are computed using capital input files from the 2008 and 2021 KLEMS releases, which detail industry-specific capital composition. Labor compensation is inferred from labor compensation to value added ratios in the 2007/08 KLEMS release. For national-level aggregates, I use the 2007/08 and 2021 KLEMS data at the total industry level.

E.2 WIOD

The World Input-Output Database (WIOD) provides annual world input-output tables (WIOTs), which integrate national input-output tables with bilateral trade flows. A WIOT links domestic transactions among households, firms, and governments with international trade, providing a global picture of intermediate and final goods flows.

WIOD includes several releases (e.f., 2013, 2016) and initiatives (e.g., Long-Run WIOD), each with different time spans, country coverage, and industry classifications. I use the WIOD 2013 and Long-Run WIOD datasets, both of which use ISIC Rev. 3 and together span 1965–2011.[22]

The Long-Run WIOD covers 26 countries and a rest-of-world region for 1965–2000. The WIOD 2013 release includes 40 countries and a rest-of-world region for 1995–2011. Since the two datasets cover different groups of countries and industries, I harmonize the data as follows.[23] First, I expand the rest-of-world region by including countries that appear in only one of the datasets (see Table 3 for the final list of countries.) Second, I use the industry groupings for which data are available in both datasets (see Table 4 for a list.) In total, I analyze data for 24 countries and a rest-of-region, each covering 23 industries. Together, these countries account for over 85% of the world's GDP, with the rest-of-world region accounting for the remainder. Each year contains approximately 0.5 millions of observations, yielding roughly 23 million observations over 40 years.

[21] See Oulton (2017).
[22] The 2016 release adds only three years (2012–2014) and uses a different industry classification.
[23] WIOD 2013 release is more granular in that it covers 40 countries (vs. 26) and 35 industries (vs. 23.)

To illustrate the structure of WIOTs, Table 2 presents a two-country, two-industry example. Rows of the industry table represent flows from industries to other users (intermediate or final), while columns represent demands. Entry z_{ij}^{AA} denotes the value of goods from industry i in country A used by industry j in the same country. Diagonal elements represent domestic transactions; off-diagonal ones represent international trade (e.g., z_{ij}^{BA} is the value of goods from industry i in country B used by industry j in A) Final-use columns cover final demands from households (HHs), non-profit organizations (NPOs), governments (GVT), and firms demanding goods for investment (GFCF) or inventory purposes (INVEN). The non-industry rows of Table 2 capture total demand by intermediate and final users (II_FOB), taxes and subsidies on products (TXSP), CIF/FOB adjustments on exports (EXP_ADJ), direct purchases abroad by residents (PURR), purchases on domestic territory by non-residents (PURNR), value added at basic prices (VA), and international transport margins (INTTTM).

Summing all elements within an industry row or column gives gross output at basic prices, which can be computed using either:

$$x_i^r = \underbrace{\sum_{s\in\{A,B\}} \sum_{j=1}^{I_s} z_{ji}^{sr}}_{\substack{\text{intermediate}\\ \text{consumption}}} + \underbrace{\mathrm{VA}_i^r}_{\substack{\text{value}\\ \text{added}}} + \underbrace{\mathrm{tax}_i^r + \mathrm{adj}_i^r + \mathrm{trans}_i^r}_{\text{tax and trade adjustments}}, \qquad \text{(Production approach)}$$

$$x_i^r = \underbrace{\sum_{s\in\{A,B\}} \sum_{j=1}^{I_s} z_{ij}^{rs}}_{\substack{\text{sales to}\\ \text{intermediate consumers}}} + \underbrace{\sum_{s\in\{A,B\}} \sum_{v\in\{C,N,G,K,I\}} f_{i,v}^{rs}}_{\substack{\text{sales to}\\ \text{final consumers}}}. \qquad \text{(Expenditure approach)}$$

These two approaches mirror national accounting conventions.

WIOD vs. similar datasets. Compared with alternative multi-country input-output datasets such as Eora, OECD–ICIO, IDE–JETRO, and GTAP, WIOD offers the best balance for the purposes of this paper between time coverage, factor detail, and consistency with KLEMS. Its main advantages are the length of the annual series, the integration with growth-accounting data, and the relatively transparent construction of factor measures. Its main drawbacks are narrower country coverage and coarser industry disaggregation than some alternatives. For a long-run analysis of productivity and welfare, however, WIOD is the most suitable source.

Table 2: Two-country input-output table, following the conventions of WIOD's 2013 release.

			Intermediate use						Final use										Total use
			Country A			Country B			Country A					Country B					
			Industries			Industries			Final demand					Final demand					GO
			1	$\cdots$	I_A	1	$\cdots$	I_B	HHs	NPOs	GVT	GFCF	INVEN	HHs	NPOs	GVT	GFCF	INVEN	
Supply Country-Industry	Country A	Industries 1		z_{ij}^{AA}			z_{ij}^{AB}		$f_{i,\mathrm{H}}^{AA}$	$f_{i,\mathrm{N}}^{AA}$	$f_{i,\mathrm{G}}^{AA}$	$f_{i,\mathrm{K}}^{AA}$	$f_{i,\mathrm{I}}^{AA}$	$f_{i,\mathrm{H}}^{AB}$	$f_{i,\mathrm{N}}^{AB}$	$f_{i,\mathrm{G}}^{AB}$	$f_{i,\mathrm{K}}^{AB}$	$f_{i,\mathrm{I}}^{AB}$	x_1^A
		$\vdots$																	$\vdots$
		Industries I_A																	$x_{I_A}^A$
	Country B	Industries 1		z_{ij}^{BA}			z_{ij}^{BB}		$f_{i,\mathrm{H}}^{BA}$	$f_{i,\mathrm{N}}^{BA}$	$f_{i,\mathrm{G}}^{BA}$	$f_{i,\mathrm{K}}^{BA}$	$f_{i,\mathrm{I}}^{BA}$	$f_{i,\mathrm{H}}^{BB}$	$f_{i,\mathrm{N}}^{BB}$	$f_{i,\mathrm{G}}^{BB}$	$f_{i,\mathrm{K}}^{BB}$	$f_{i,\mathrm{I}}^{BB}$	x_1^B
		$\vdots$																	$\vdots$
		Industries I_B																	$x_{I_B}^B$
II_FOB				z_j^A			z_j^B		f_{H}^A	f_{N}^A	f_{G}^A	f_{K}^A	f_{I}^A	f_{H}^B	f_{N}^B	f_{G}^B	f_{K}^B	f_{I}^B	
TXSP				tax_j^A			tax_j^B		$\mathrm{tax}_{\mathrm{H}}^A$	$\mathrm{tax}_{\mathrm{N}}^A$	$\mathrm{tax}_{\mathrm{G}}^A$	$\mathrm{tax}_{\mathrm{K}}^A$	$\mathrm{tax}_{\mathrm{I}}^A$	$\mathrm{tax}_{\mathrm{H}}^B$		$\cdots$		$\mathrm{tax}_{\mathrm{I}}^B$	
EXP_ADJ				adj_j^A			adj_j^B		$\mathrm{adj}_{\mathrm{H}}^A$	$\mathrm{adj}_{\mathrm{N}}^A$	$\mathrm{adj}_{\mathrm{G}}^A$	$\mathrm{adj}_{\mathrm{K}}^A$	$\mathrm{adj}_{\mathrm{I}}^A$	$\mathrm{adj}_{\mathrm{H}}^B$		$\cdots$		$\mathrm{adj}_{\mathrm{I}}^B$	
PURR									$\mathrm{par}_{\mathrm{H}}^A$	$\mathrm{par}_{\mathrm{N}}^A$	$\mathrm{par}_{\mathrm{G}}^A$	$\mathrm{par}_{\mathrm{K}}^A$	$\mathrm{par}_{\mathrm{I}}^A$	$\mathrm{par}_{\mathrm{H}}^B$		$\cdots$		$\mathrm{par}_{\mathrm{I}}^B$	
PURNR									$\mathrm{pdn}_{\mathrm{H}}^A$	$\mathrm{pdn}_{\mathrm{N}}^A$	$\mathrm{pdn}_{\mathrm{G}}^A$	$\mathrm{pdn}_{\mathrm{K}}^A$	$\mathrm{pdn}_{\mathrm{I}}^A$	$\mathrm{pdn}_{\mathrm{H}}^B$				$\mathrm{pdn}_{\mathrm{I}}^B$	
VA				VA_j^A			VA_j^B												
INTTTM				trans_j^A			trans_j^B		$\mathrm{trans}_{\mathrm{H}}^A$	$\mathrm{trans}_{\mathrm{N}}^A$	$\mathrm{trans}_{\mathrm{G}}^A$	$\mathrm{trans}_{\mathrm{K}}^A$	$\mathrm{trans}_{\mathrm{I}}^A$	$\mathrm{trans}_{\mathrm{H}}^B$	$\mathrm{trans}_{\mathrm{N}}^B$	$\mathrm{trans}_{\mathrm{G}}^B$	$\mathrm{trans}_{\mathrm{K}}^B$	$\mathrm{trans}_{\mathrm{I}}^B$	
GO				x_j^A		x_1^B	$\cdots$	$x_{I_B}^B$											

Table Notes.

- *Abbreviations*. HHs: Households, NPOs: Non-profit organizations, GVT: Government, GFCG: Gross fixed capital formation, INVEN: Inventories, GO: Gross output, II_FOB: Total intermediate consumption, TXSP: Taxes less subsidies on products, EXP_ADJ: CIF/FOB adjustments on exports, PURR: Direct purchases abroad by residents, PURNR: Purchases on the domestic territory by non-residents, VA: Value added at basic prices, INTTTM: International transport margins, I_c: number of industries in country c.
- *Units*. In the WIOTs, entries represent USD millions.
- *Interpretations*. Red shaded cells denote empty cells (or cells populated by zeroes in the WIOTs). z_{ij}^{AA} is a generic entry of the national input-output table of country A, and it captures the demand of goods from industry j by industry i. A similar interpretation applies for z_{ij}^{BB}. z_{ij}^{AB} is the demand of goods i from country A by industry j in country B. Alternatively, one could say that z_{ij}^{AB} is the supply of goods by industry i in country A to industry j in country B. The remaining cells are left blank because their interpretation is straightforward.

E.3 Country Sample

Table 3: Country and Region Sample

Australia (AUS)	Denmark (DNK)	Ireland (IRL)	Sweden (SWE)
Austria (AUT)	Spain (ESP)	Italy (ITA)	Taiwan (TWN)
Belgium (BEL)	Finland (FIN)	Japan (JPN)	United States (USA)
Brazil (BRA)	France (FRA)	South Korea (KOR)	Rest of world (ROW)
Canada (CAN)	United Kingdom (GBR)	Mexico (MEX)	
China (CHN)	Greece (GRC)	Netherlands (NLD)	
Germany (DEU)	India (IND)	Portugal (PRT)	

Table Notes. ISO-3 country codes are listed in parenthesis. The countries listed in this table appear in both the Long-Run and the 2013 WIOD releases. Some countries (e.g., Hong-Kong) may appear in one of the aforementioned WIOD releases, but not in the other. In such cases, countries are included in the "Rest of world" region, which is appropriately harmonized across releases.

E.4 Industry Sample

Table 4: Industry Sample

Code	Name
AtB	Agriculture, Hunting, Forestry and Fishing
C	Mining and Quarrying
D15t16	Food, Beverages and Tobacco
D17t19	Textiles, Textile, Leather, and Footwear
D21t22	Pulp, Paper, Paper, Printing and Publishing
D23	Coke, Refined Petroleum and Nuclear Fuel
D24	Chemicals and Chemical Products
D25	Rubber and Plastics
D26	Other Non-Metallic Mineral
D27t28	Basic Metals and Fabricated Metal
D29	Machinery, Nec
D30t33	Electrical and Optical Equipment
D34t35	Transport Equipment
Dnec	Manufacturing, Nec; Recycling
E	Electricity, Gas and Water Supply
F	Construction
G	Wholesale and Retail Trade
H	Hotels and Restaurants
I60t63	Transport and Storage
I64	Post and Telecommunications
J	Financial Intermediation
K	Real Estate, Renting and Business Activities
LtQ	Community Social and Personal Services

Table Notes. Industry codes and names follow the ISIC Rev. 3.1 Industry Classification.

E.5 Data Harmonization and Panel Construction

Becuase the empirical analysis combines multiple releases of KLEMS and WIOD, each with different time spans, country sets, and industry classifications, I apply a uniform harmonization procedure across sources. The basic principle is a retropolation procedure in which the most recent release provides the reference levels, while earlier releases provide growth rates for the pre-overlap period. This is analogous to the splicing procedures used by statistical agencies after comprehensive revisions: newer releases are taken to improve the level measurement, while earlier releases remain informative about year-to-year changes.

KLEMS retropolation. I use the KLEMS 2021 release as the authoritative source for 1995 onward and the KLEMS 2007 release for growth rates prior to 1995. For any variable X,

$$X_t^{\text{harm}} = X_t^{\text{KL21}}, \quad t \geq 1995,$$

and

$$X_t^{\text{harm}} = X_{t+1}^{\text{harm}} \times \frac{X_t^{\text{KL07}}}{X_{t+1}^{\text{KL07}}}, \quad t < 1995.$$

This procedure anchors the series to the levels in the 2021 release while preserving the growth rates from the earlier release before 1995. At the aggregate level, the labor share is computed as

$$\Lambda_{Lt} = \frac{\text{LAB}_t}{\text{VA}_t}.$$

At the sectoral level, labor-compensation-to-value-added ratios are rescaled so that their value-added-weighted average matches the aggregate labor share exactly. This ensures consistency between sectoral and aggregate factor shares.

WIOD splicing. The harmonized input-output panel combines the WIOD 2013 release (1995–2011) with Long-Run WIOD (1965–2000). I do so following a procedure similar to the one Lewis et al. (2022) For each seller-sector i, I compute a splice factor over the overlap years 1995–2000,

$$r_i = \frac{1}{6} \sum_{t=1995}^{2000} \frac{\text{GO}_{i,t}^{\text{W13}}}{\text{GO}_{i,t}^{\text{LR}}},$$

and rescale the Long-Run WIOD observations for years prior to 1995 according to

$$\mathrm{IO}^{\mathrm{harm}}_{ij,t} = r_i \times \mathrm{IO}^{\mathrm{LR}}_{ij,t}, \qquad t < 1995.$$

For 1995 onward, I use WIOD 2013 directly. This row-level rescaling preserves the input-output shares and within-release growth rates of the Long-Run WIOD while aligning nominal levels with the WIOD 2013 scale. After splicing, I recompute the aggregate rows so that intermediate use equals the column sum of industry purchases and value added is defined residually as VA = GO − II.

Volume indices. Sectoral volume indices for gross output, intermediate inputs, and value added are taken from the WIOD Socio-Economic Accounts. I apply the same retropolation logic: WIOD 2013 provides the reference levels from 1995 onward, and the Long-Run WIOD growth rates are used to extend the series back to 1970. For sectoral labor quantity indices, I use KLEMS 2021 as the reference release and the KLEMS 2007 growth rates for the earlier period. When sectoral labor indices are unavailable in the early years, I backfill them using growth in total hours worked, preserving the level in the first year with direct coverage.

World Bank data. I use World Bank data on gross national expenditure (GNE), GDP, population, and price deflators. Both deflators and GNE/GDP series come from the World Bank's World Development Indicators.[24] Real welfare is measured using real gross national expenditure per capita, deflated by the consumer price index (CPI), while real GDP per capita is deflated using the GDP deflator. I rely on World Bank aggregates rather than WIOD-derived aggregates because the latter are best suited for annual shares and input-output networks, not for aggregate expenditure levels.

E.6 Growth-Accounting Variables

Table 5 presents the correspondence between model variables and their empirical counterparts used in the growth-accounting exercise. Because the analysis spans a long time horizon and compares alternative growth-accounting procedures, I combine data from KLEMS and WIOD. The two sources are internally consistent, since WIOD was constructed using KLEMS data and related national-accounting conventions.

The variables in Table 5 fall into three categories. *Common variables* are used in all growth-accounting exercises. *Traditional Solow Residual* variables are those used to

[24] Indicators used: NE.DAB.TOTL.CN (GNE in current LCU), NY.GDP.MKTP.CN (GDP in current LCU), FP.CPI.TOTL (CPI index), NY.GDP.DEFL.ZS (GDP deflator index), SP.POP.TOTL (population).

estimate TFP under the standard frictionless benchmark, in which the capital share is computed residually as one minus the labor share. *Distorted Solow Residual* variables instead allow for wedges. Their construction follows Section 3 and requires additional information on depreciation rates, net-of-depreciation user costs, and production networks.

Table 5: Growth-Accounting Variables.

Model Variable	Data Counterpart	Source
COMMON VARIABLES		
$\Delta \log Y_t$	$\Delta \log \text{VA_QI}_t$	KLEMS 2007 (pre-1995), KLEMS 2021 (post-1995)
$\Delta \log L_{Kt}$	$\Delta \log \text{CAP_QI}_t$	
$\Delta \log L_{Lt}$	$\Delta \log \text{LAB_QI}_t$	
TRADITIONAL SOLOW RESIDUAL		
Λ_L	LAB/VA	KLEMS 2007 (pre-1995), KLEMS 2021 (post-1995)
Λ_K	$1 - \Lambda_L$	
DISTORTED SOLOW RESIDUAL		
Λ_L	Revenue-based labor share	WIOD, KLEMS, BEA
Λ_K	Revenue-based capital share	
$\tilde{\Lambda}_L$	Cost-based labor share	
$\tilde{\Lambda}_K$	Cost-based capital share	

Table Notes. This table presents the mapping between model variables and data counterparts. Here, $\Delta \log X_t = \log X_t - \log X_{t-1}$. Common variables are those used irrespective of the TFP measure to be constructed. Variables under "Traditional Solow Residual" are the revenue-based factor shares resulting from ignoring distortions and input-output networks. Variables under "Distorted Solow Residual" are revenue- and cost-based factor shares that incorporate distortions and IO linkages.

E.7 Consumption of Fixed Capital

I estimate the consumption of fixed capital using capital stock data from KLEMS and WIOD, combined with asset-specific depreciation rates from the Fixed-Asset Tables published by the U.S. Bureau of Economic Analysis (BEA).

From KLEMS, I use two capital input files: the 2008 release for the period 1970–2005, and the 2021 release for 2005–2010. Both files report eight distinct types of real capital stocks: computing equipment ("K_IT"), communications equipment ("K_CT"), software ("K_Soft"), transport equipment ("K_TraEq"), other machinery and equipment ("K_OMach"), non-residential investment ("K_OCon"), residential structures ("K_RStruc"), and other assets ("K_Other").[25]

Using these data, I compute the share of each capital type j in the total capital stock of sector i in country c and year t as

$$\text{Share in capital stock}_{jict} = \frac{K_{jict}}{K_{ict}}.$$

Given these shares, I construct the sector-specific depreciation rate δ_{ict} as a weighted average of asset-specific depreciation rates:

$$\delta_{ict} = \sum_j \delta_{jt} \times \text{Share in capital stock}_{jict}.$$

Because the BEA reports depreciation rates at a finer level of disaggregation than the capital types provided in KLEMS, I map the BEA depreciation rates to KLEMS categories. Table 6 documents this mapping and the resulting depreciation rates for each asset type.

When data on asset-specific capital stocks are missing for a given country-sector-year observation, I impute capital shares using the unweighted mean from all other observations. Table 7 illustrates the resulting depreciation rates for a sample of sectors.

[25] KLEMS also divides capital ("K_GFCF") into ICT ("K_ICT") and non-ICT ("K_NonICT") categories.

Table 6: Asset-Specific Depreciation Rates.

Asset	Description	Depreciation Rate Before 1978	Depreciation Rate After 1978
K_IT	**Computing**	**0.2729**	**0.3119**
Office, computing, and accounting machinery		0.2729	0.3119
K_CT	**Communications**	**0.1438**	**0.1438**
Business services		0.1500	0.1500
Other industries		0.1100	0.1100
Instruments		0.1350	0.1350
Other instruments		0.1800	0.1800
K_Soft	**Software**	**0.1947**	**0.1947**
Video and audio products, computers, etc.		0.1833	0.1833
Electronic countermeasures		0.2357	0.2357
Other		0.1650	0.1650
K_TraEq	**Transport**	**0.1440**	**0.1440**
Local and interurban passenger transit		0.1232	0.1232
Trucking and warehousing		0.1725	0.1725
Transportation by air, depositiory institutions, etc.		0.0825	0.0825
Other industries		0.1100	0.1100
Other motor vehicles		0.2316	0.2316
K_OMach	**Machinery**	**0.1046**	**0.1046**
Other fabricated metal products		0.0917	0.0917
Steam engines and turbines		0.0516	0.0516
Internal combustion engines		0.2063	0.2063
Metalworking machinery		0.1225	0.1225
Special industrial machinery, n.e.c.		0.1031	0.1031
General industrial equipment		0.1072	0.1072
Electrical transmission, distribution, and industrial apparatus		0.0500	0.0500
K_OCon	**Non-residential**	**0.0262**	**0.0262**
Industrial buildings		0.0314	0.0314
Mobile offices		0.0556	0.0556
Office buildings		0.0247	0.0247
Commercial warehouses		0.0222	0.0222
Other commercial buildings		0.0262	0.0262
Educational buildings		0.0188	0.0188
Hospital and institutional buildings		0.0188	0.0188
Hotels and motels		0.0281	0.0281
All other nonfarm buildings		0.0249	0.0249
Railroad replacement track		0.0275	0.0275
Other railroad structures		0.0166	0.0166
Telecommunications		0.0237	0.0237
K_RStruc	**Residential**	**0.0418**	**0.0418**
1-to-4-unit structures-new		0.0114	0.0114
1-to-4-unit structures-additions and alterations		0.0227	0.0227
1-to-4-unit structures-major replacements		0.0364	0.0364
5-or-more-unit structures-new		0.0140	0.0140
5-or-more-unit structures-additions and alterations		0.0284	0.0284
5-or-more-unit structures-major replacements		0.0455	0.0455
Mobile homes		0.0455	0.0455
Other structures		0.0227	0.0227
Equipment		0.1500	0.1500
K_Other	**Other**	**0.1465**	**0.1465**
Medical		0.1834	0.1834
Construction		0.1550	0.1550
Industrial		0.0917	0.0917
General		0.1650	0.1650
Other		0.1375	0.1375

Table Notes. Rows in bold and with description correspond to the capital types provided by KLEMS. The depreciation rates for these stocks are computed as the unweighted mean of the depreciation rates for the assets listed in the rows with indentation below, which display the depreciation rates for the asset types provided by the BEA and which are mapped to KLEMS assets as shown.

Table 7: Illustration of sector-specific depreciation rates for Spain.

Sector	Code	1970	1980	1990	2000	2010
Hotels and Restaurants	H	0.047	0.047	0.049	0.060	0.065
Post and Telecommunications	I64	0.058	0.063	0.068	0.084	0.094

Table Notes. Sector-specific depreciation rates are computed as a weighted average of asset-specific depreciation rates, where the weights are the shares of each type of capital in the total capital of a sector for a particular year.

E.8 Distortion Estimates

This appendix outlines the data-processing steps and modeling assumptions used to estimate sectoral wedges.

Baseline wedge measure. I compute wedge margins using equation (D.2) after estimating each sector's capital consumption and the corresponding net-of-depreciation user cost. Whenever the implied margin is negative, I set it equal to zero. I then back out the corresponding wedge using equation (D.3) and winsorize the resulting distribution at the 1st and 99th percentiles.

Capital consumption is estimated using capital-composition data and BEA asset-specific depreciation rates, as described in Section E.7. For Spain, the sales-weighted average depreciation rate rises from approximately 6% in 1970 to 8% in 2010. I compute the net-of-depreciation rental rate following van Vlokhoven (2022), using a three-year rolling window. For year t in country c, I pool sector-level observations from years $t-1$, t, and $t+1$ and impose a common user cost of capital across sectors within the country-year. As an example, Figure 8 shows the unconditional distribution of wedges (left panel) and the evolution of the harmonic sales-weighted wedge for Spain (right panel), which indicates rising distortions over time.

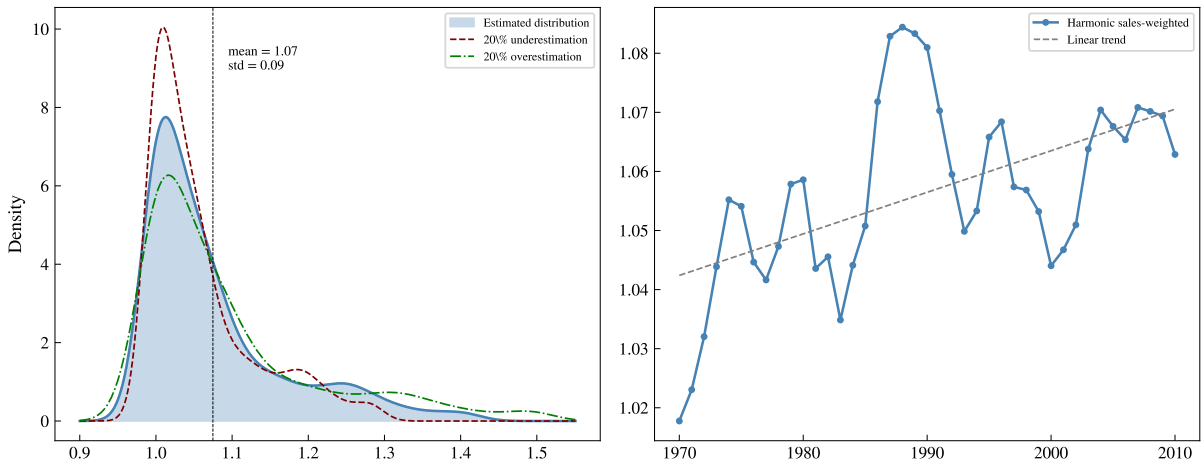


Figure 8: Wedges in Spain, 1970–2010.

Figure Notes. The left panel reports the (unconditional) distribution of wedges in Spain for the period of analysis. The dash- and dash-doted lines show how the distribution would look like if the wedge margin were underestimated or overestimated by 20%, respectively. The right panels displays the evolution of the harmonic sales-weighted wedge, and the dash line provides a linear fit.

Alternative wedge measure based on gross margins. I calculate gross margins using equation (D.1). Whenever gross margins are negative, I set them to zero. I then apply equation (D.3) and winsorize the resulting wedges at the 1st and 99th percentiles.

Figure 9 plots the unconditional distribution of wedges (left) and their evolution over time (right). Since gross margins do not deduct capital costs, estimated wedges are higher than those based on wedge margins.

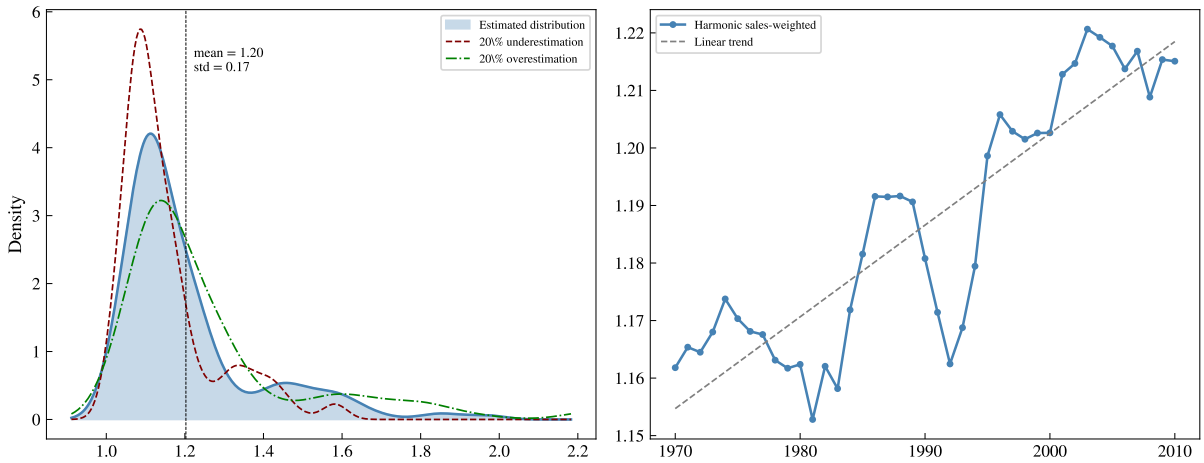


Figure 9: Wedges in Spain (gross margins), 1970–2010.

Figure Notes. The left panel report (unconditional) distributions of wedges for the period of analysis. The dash- and dash-doted lines show how the distribution would look like if gross margins were underestimated or overestimated by 20%, respectively. The right panel displays the evolution of harmonic sales-weighted wedge over time, and the dash line provides a linear fit.

Alternative wedge measure using the accounting–profits approach. This approach mirrors the wedge-margin method but assumes the net-of-depreciation user cost of capital is zero. Figure 10 shows that the resulting wedge also displays a secular increase.

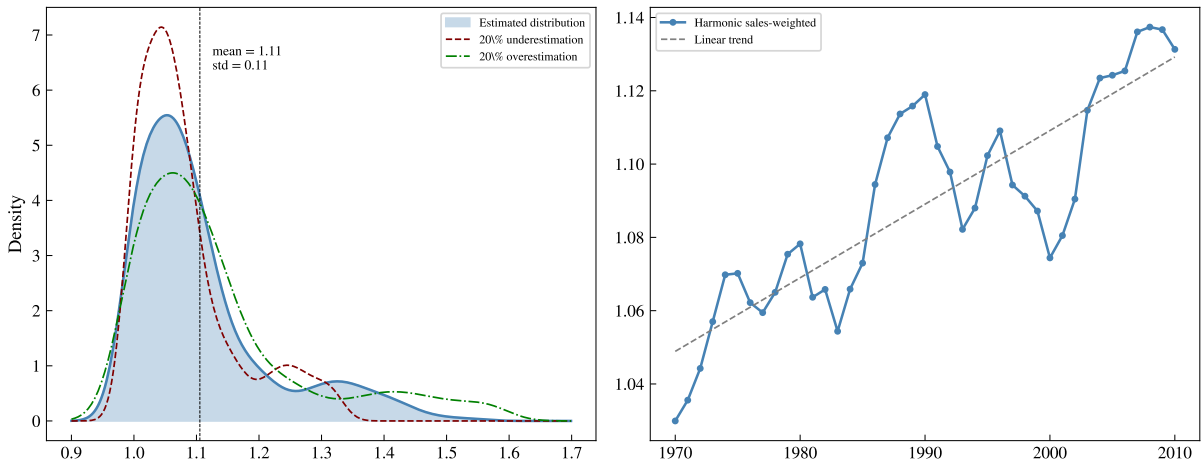


Figure 10: Wedges in Spain (accounting profits), 1970–2010.

Figure Notes. The left panel displays the (unconditional) distribution of wedges in Spain for the period of analysis. The dash- and dash-doted lines show how the distribution would look like if the wedge margin were underestimated or overestimated by 20%. The right panel shows the evolution of the sales-weighted wedge over time, and the dash line provides a linear fit.

Alternative wedge measure using the control–function approach. I estimate the elasticity of labor using the methodology of Olley and Pakes (1996) and the correction of

Ackerberg et al. (2015). Estimation is conducted under both Cobb–Douglas and translog technologies, allowing for time-varying elasticities. I impose common elasticities across countries within each industry and estimate them using 3-year rolling windows. See Table 8 for details.

Table 8: Estimation Details in the Application of Control–Function Approach.

	(1)	**(2)**	**(3)**	**(4)**
Technology	CD	CD	TL	TL
Elasticities	Constant	Time-varying	Constant	Time-varying
Productivity process	AR(1)			
Degree of polynomial	2nd			
Volume units	✓			
Outcome	Gross output			
State	Capital			
Free	Labor			
Proxy	Investment			

Because the sample size is small ($N = 25$ sectors per country), elasticities are imprecisely estimated—frequently exceeding one—I discard this approach for computing wedges. This method is better suited to firm-level data with large cross-sectional variation.

Alternative wedge measure using the cost–shares approach. As shown in previous sections, this method yields equivalent wedges to the wedge-margin approach when using the same user costs of capital. To distinguish it, I compute wedges assuming a constant gross rental rate of capital of 10%. Figure 11 shows the resulting estimates.

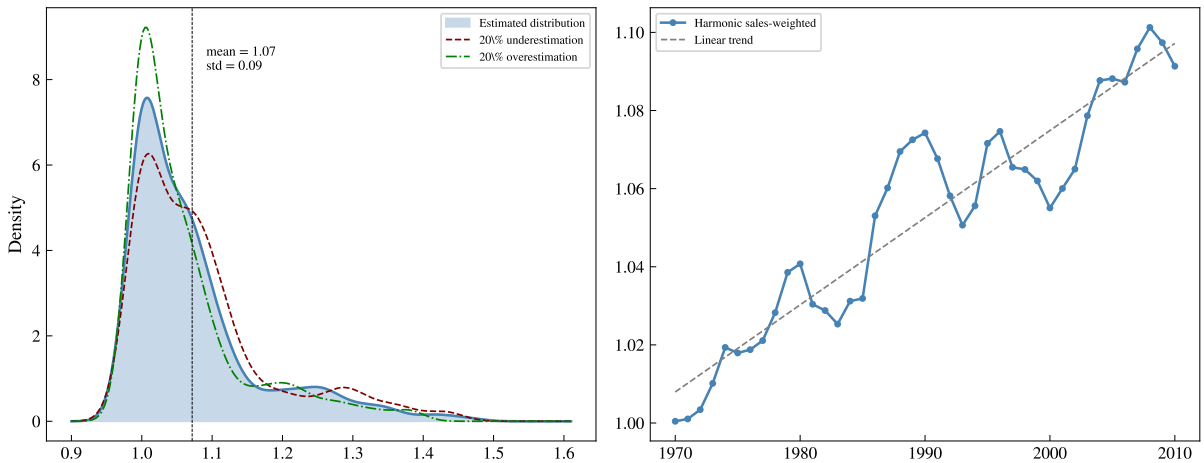


Figure 11: Wedges in Spain (costs-shares approach), 1970–2010.

Figure Notes. The left panel displays the (unconditional) distribution of wedges in Spain for the period of analysis. The dash- and dash-doted lines show how the distribution would look like if the gross-of-depreciation rental rate of capital is set to 8% and 12%, respectively. The right panel shows the evolution of the sales-weighted wedge over time, and the dash line provides a linear fit.

E.9 Sectoral Employment Shares and Productivity

Figure 12 shows the evolution of sectoral employment shares in Spain from 1970 to 2014. Labor has steadily shifted away from non-service sectors—agriculture, manufacturing, and construction—toward services, particularly professional services, health care, and accommodation and food.

Figure 12: Labor Reallocation in Spain, 1970–2014.

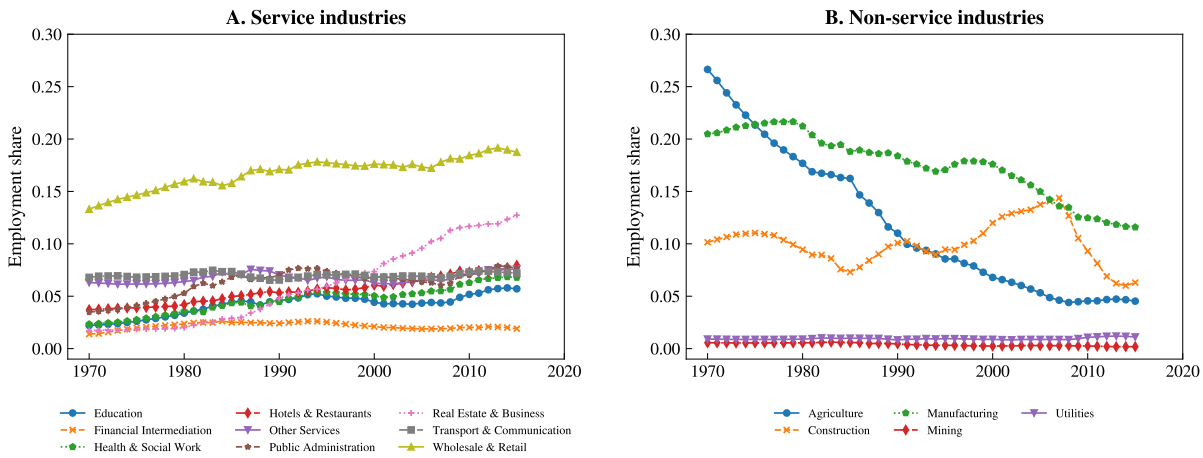


Figure Notes. Employment shares are calculated as the total hours worked by persons engaged in a particular sector ("H_EMP") divided by the economy's total. Data comes from KLEMS.

Figure 13 shows sectoral TFP indices, constructed using the traditional Solow residual in terms of value added from KLEMS. Sectors that shed labor—agriculture and manufacturing—exhibited positive TFP growth, while expanding sectors—real estate and business, and hotels and restaurants—experienced persistent TFP declines.

Figure 13: Sectoral TFP Growth in Spain, 1980–2018.

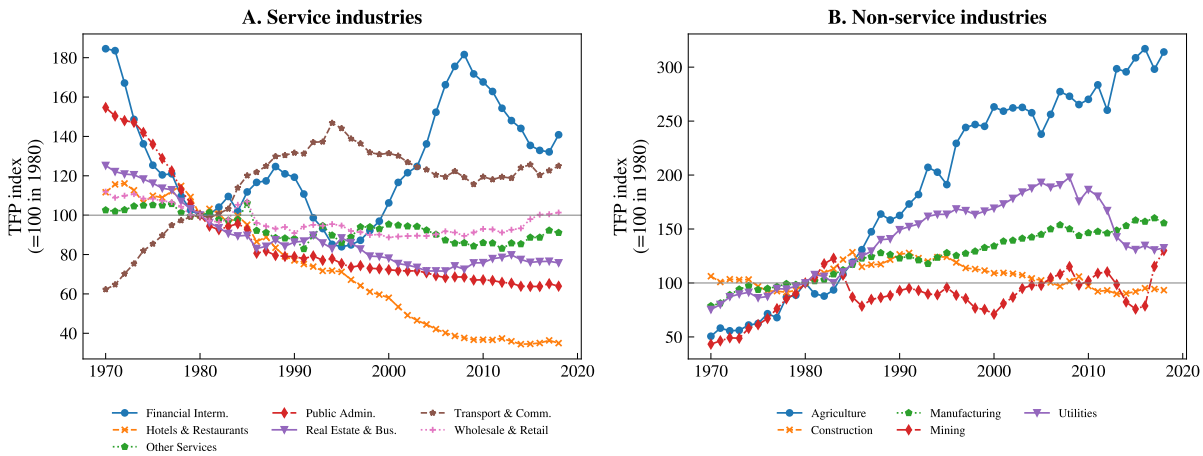


Figure Notes. TFP is normalized to 100 in 1980.

E.10 Production Networks, TFP Measurement, and Income Shares

While input-output linkages do not affect the measurement of aggregate productivity, they are crucial for understanding related outcomes, especially the sources of productivity growth and the distribution of income. Figure 14 illustrates this point: ignoring production networks leaves the TFP index unchanged (left panel) but leads to markedly different estimates of income shares (right panel) due to double marginalization—that is, the amplification of distortions as they propagate through production linkages.

Figure 14: Biases in TFP and the Income Share of Wedges in Spain, 1970–2010.

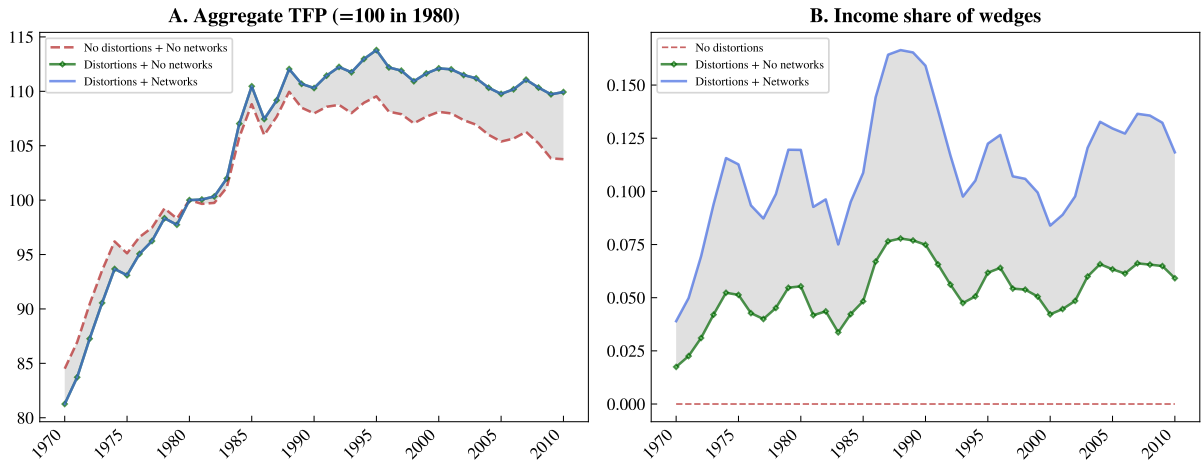


Figure Notes. The left panel plots three different TFP indices, all normalized to 100 in 1980. The right panel shows the share of income that accrues to wedges under three different assumptions: (i) there are no distortions, (ii) there are distortions but no production linkages; and (iii) there are both distortions and global production linkages.

The finding that ignoring production networks leads to substantially underestimating the share of income accruing to wedges (or profits) carries important implications for the large empirical literature on market power. In particular, profit shares computed using sales-weighted profit rates systematically understate the true profit share (see Hasenzagl and Perez, 2023, for a related discussion.)[26]

Figure 15 plots the evolution of income shares in Spain from 1970 to 2010. The labor share declined from 69% to 63.5%, consistent with the global trend documented by Karabarbounis and Neiman (2014). The capital share also fell—from 27% to 23.5%—with most of the decline occurring before the mid-1980s. In contrast, the income share accruing to wedges rose steadily, reaching 11% by 2010. While this trend is noteworthy, it should be interpreted with caution. The analysis abstracts from returns to scale, and it is plausible that rising distortions coincided with increasing returns to scale, as documented by Hasenzagl and Perez (2023) for the United States during a similar

[26] Hasenzagl and Perez (2023) generalize the production side of Baqaee and Farhi (2024) to allow for arbitrary returns to scale and market power in input markets. They provide a characterization of the aggregate profit share in terms of the input-output multiplier and economy-wide indicators of market power, including the aggregate markup, aggregate monopsony power, and aggregate returns to scale.

period. If so, the capital share shown in Figure 15 likely understates its true value, while the wedge share is overstated.

Figure 15: Evolution of Income Shares in Spain, 1970–2010.

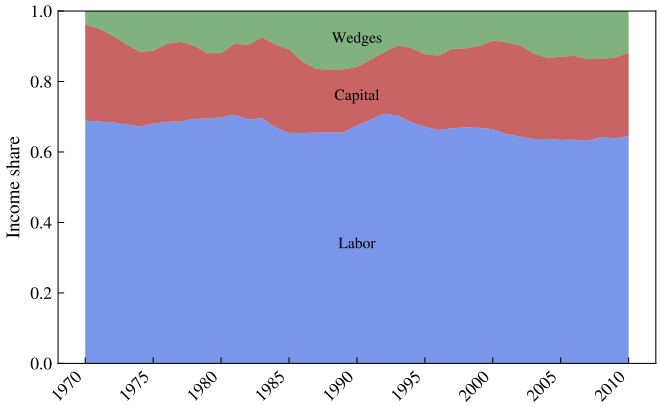


Figure Notes. Income shares are computed after subtracting taxes from total income. That is, I compute the share of labor-, capital-, and wedge income by dividing the income generated by that factor/source over net-of-taxes aggregate value added.

E.11 Trade Integration

Figure 16 shows the foreign share of household consumption (Panel A) and intermediate inputs (Panel B) in Spain and Italy form 1970 to 2010. In Spain, imported consumption goods rose from approximately 2% of total household expenditures in 1970 to 13% by 2007. Italy experienced a similar, though more gradual, increase—from around 4% to 11% over the same period. The foreign share of intermediate inputs purchased by domestic producers more than doubled in Spain from 7% to 18%; In Italy, it rose from 9% to 18%.

Figure 16: Trade Statistics in Spain and Italy, 1970–2014

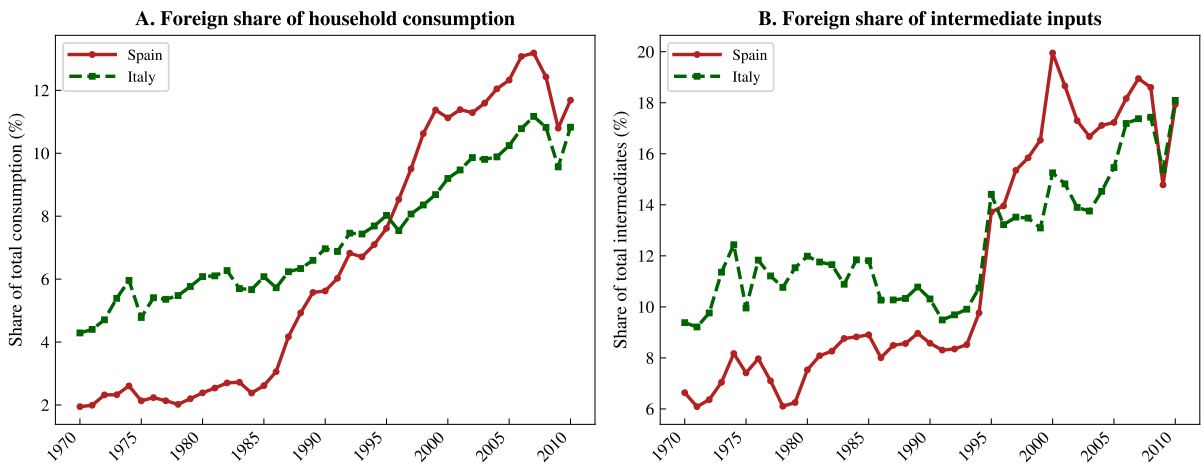


Figure Notes. Data are from WIOD.

F Robustness

This appendix examines the robustness of the results along two dimensions: the choice of aggregation weights and the method used to estimate distortions. I present most of the results for Spain only, but the main conclusions are unchanged for Italy as well.

Alternative weighting schemes. The baseline analysis uses base-period weights. As a robustness check, I recompute the relevant aggregates and decompositions using midpoint (Törnqvist) weights. The results are essentially unchanged. In particular, the timing of Spain's aggregate TFP peak, the magnitude of the TFP decline, and the relative importance of technical efficiency and factor reallocation remain very similar under the two weighting schemes. This indicates that the main conclusions are not driven by the specific choice of weights.

Figure 17 displays the aggregate TFP indices for Spain and Italy, computed under the two weighting schemes. The twos series are nearly undistinguishable. By 2010, the aggregate TFP index for Spain is 103.69 under base-period weights and 103.56 under midpoint weights; for Italy, these numbers are 107.19 and 107.11, respectively.

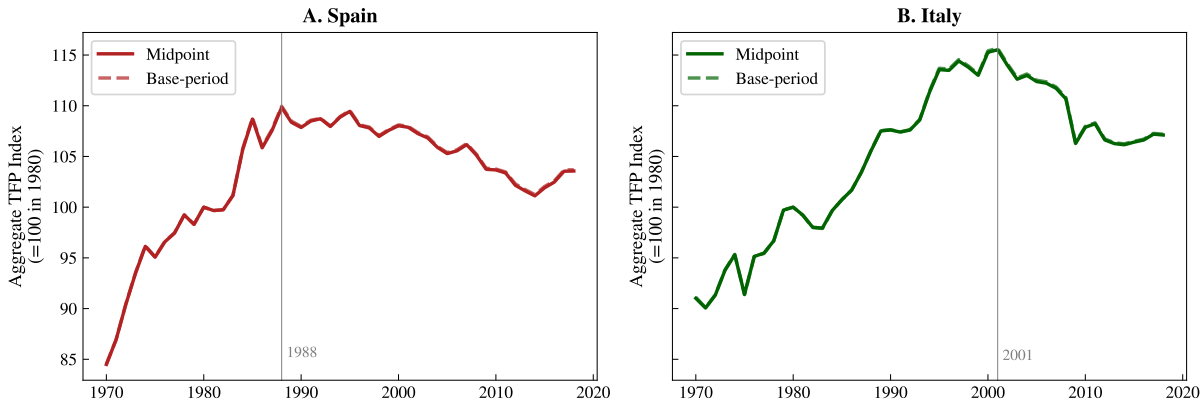
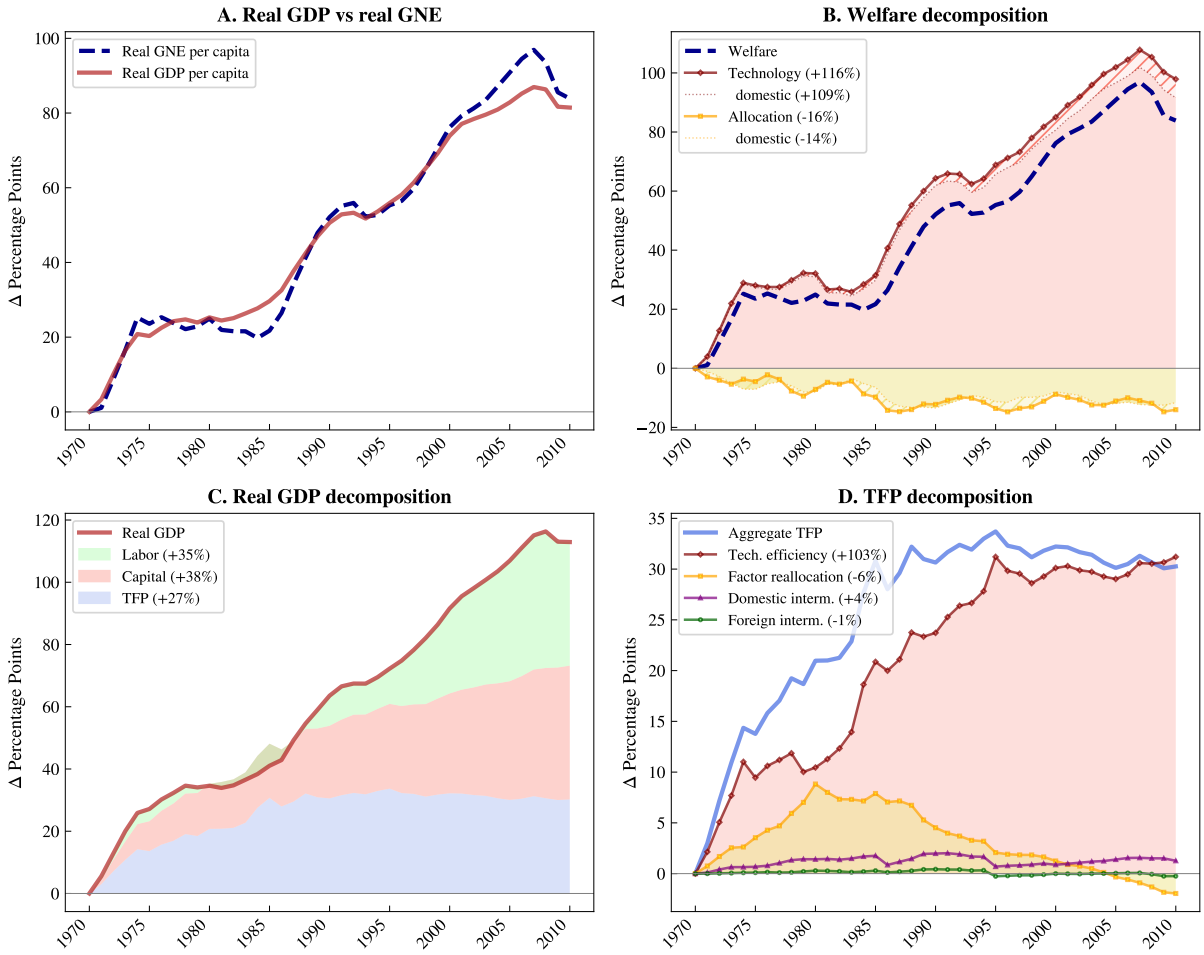


Figure 17: Aggregate TFP Indices: Base-period vs. Midpoint Weights.

Figure 18 reproduces the growth-accounting exercise and the aggregate TFP and welfare decompositions using midpoint weights. Compared with Figure 5, which reports the same objects under base-period weights, the differences are negligible. The two figures are visually indistinguishable, and the quantitative contributions change only minimally: in the welfare decomposition, the contribution of technology increases from 114% to 116%, while the contribution of allocation falls from −14% to −16%. The TFP decomposition is likewise essentially unchanged.

Figure 18: Growth Accounting and Welfare in Spain: Midpoint Weights.



Alternative distortion measures. Figure 19 plots the evolution of aggregate TFP in Spain from 1970 to 2010 under alternative distortion measures—Appendices D and E.8 provide the relevant methodological details, as well as the estimated distortions. The main conclusions are robust to these alternative measures. Under all specifications except gross-margins—which effectively imposes zero capital costs—aggregate TFP peaks in 1995, and its subsequent decline is considerably smaller than that implied by the traditional Solow residual.

Figure 20 reports corresponding robustness checks for the TFP decomposition. Panel A reproduces the baseline decomposition based on wedge margins. Panels B–D report decompositions based on alternative measures of distortions: gross margins, accounting profits, and cost shares. The qualitative message is unchanged across specifications: technical efficiency accounts for most aggregate TFP growth (from 89% to 105% depending on the method), whereas factor reallocation accounts for the entire post-1995 TFP decline.

Figure 19: Aggregate TFP in Spain with alternative measures of distortions.

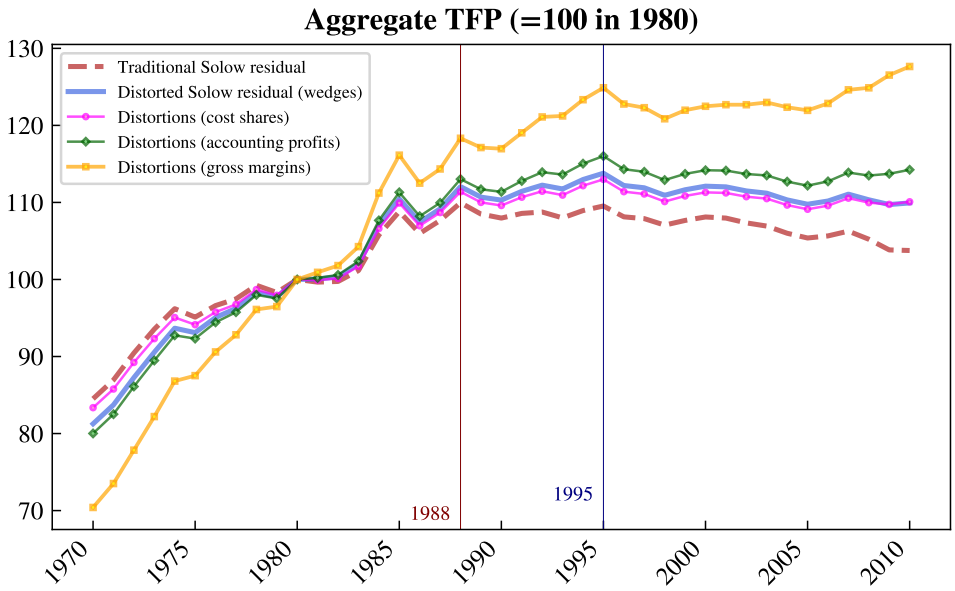
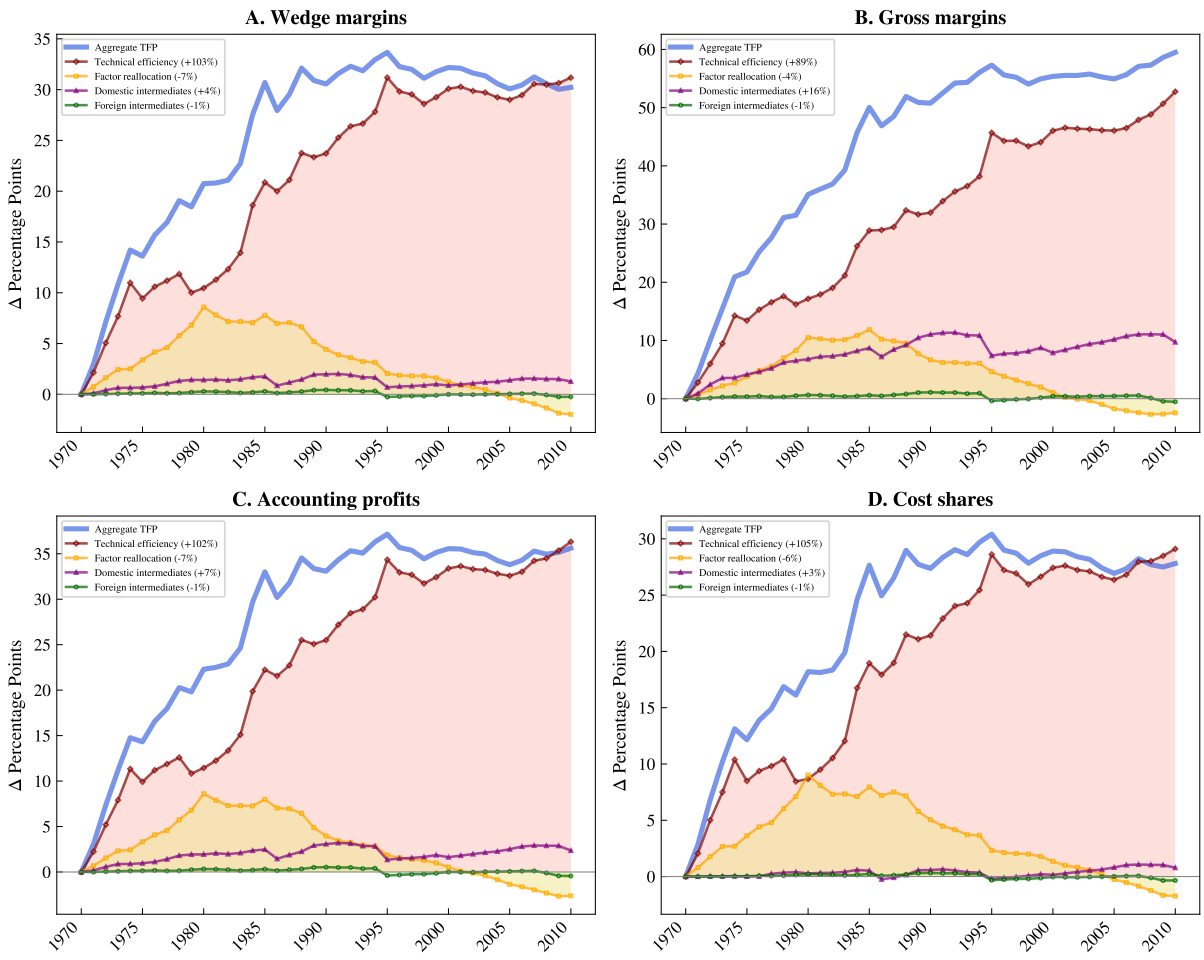


Figure 20: Decompositions of Aggregate TFP in Spain with Several Distortion Measures.



G The Case of Italy

Figure 21 plots aggregate TFP in Italy from 1970 to 2010 under several alternative distortion measures, together with the traditional Solow residual. In contrast to Spain, the traditional and distorted TFP indices are remarkably similar in Italy. Under both approaches, TFP peaks in 2001 and declines by roughly seven percentage points through 2010. This close comovement indicates that distortions play a more limited role in shaping measured productivity in Italy than in Spain, operating mainly through a level effect associated with a relatively stable aggregate wedge over time.

Figure 21: The Evolution of Aggregate TFP in Italy, 1970–2010.

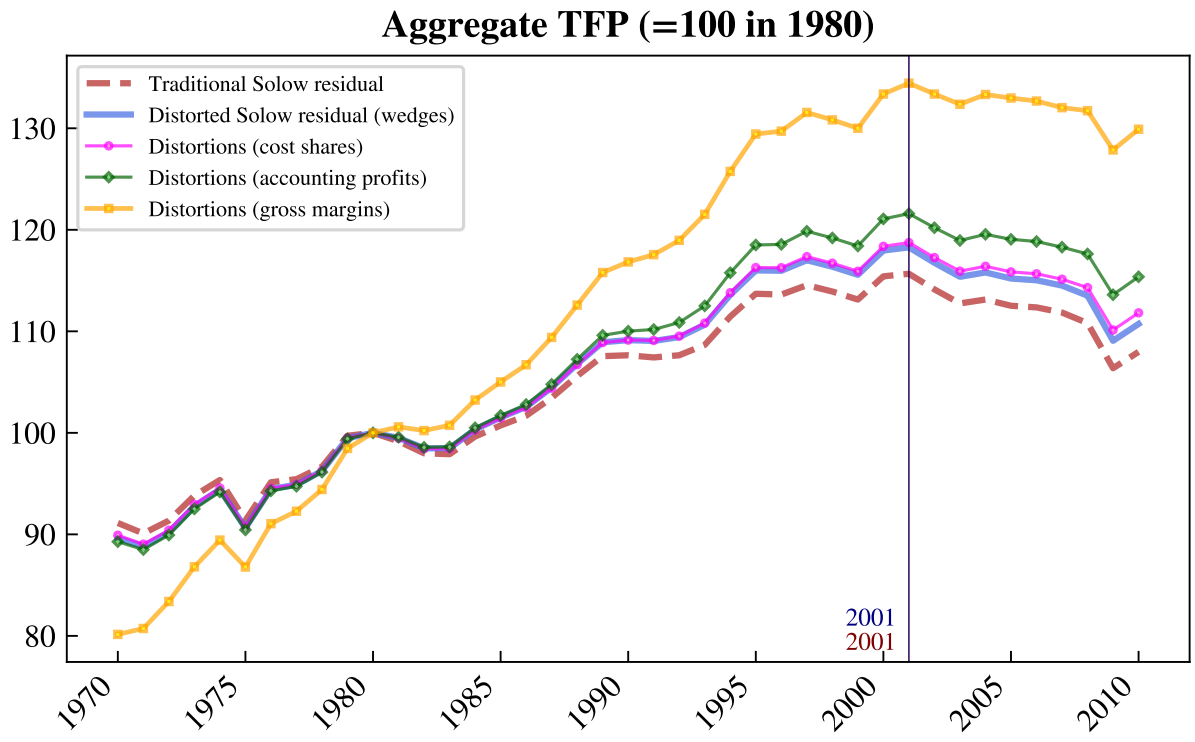


Figure 22 reports decompositions of aggregate TFP using alternative measures of distortions. The qualitative message is consistent across specifications. Declines in technical efficiency account for most of Italy's aggregate TFP slowdown, while factor reallocation plays almost no role. This stands in sharp contrast to Spain, where reallocation is the dominant force behind the TFP decline. Results are broadly similar across alternative distortion measures, with the main exception being gross margins, which effectively impose a zero user cost of capital.

Figure 22: Decompositions of Aggregate TFP in Italy with Several Distortion Measures.

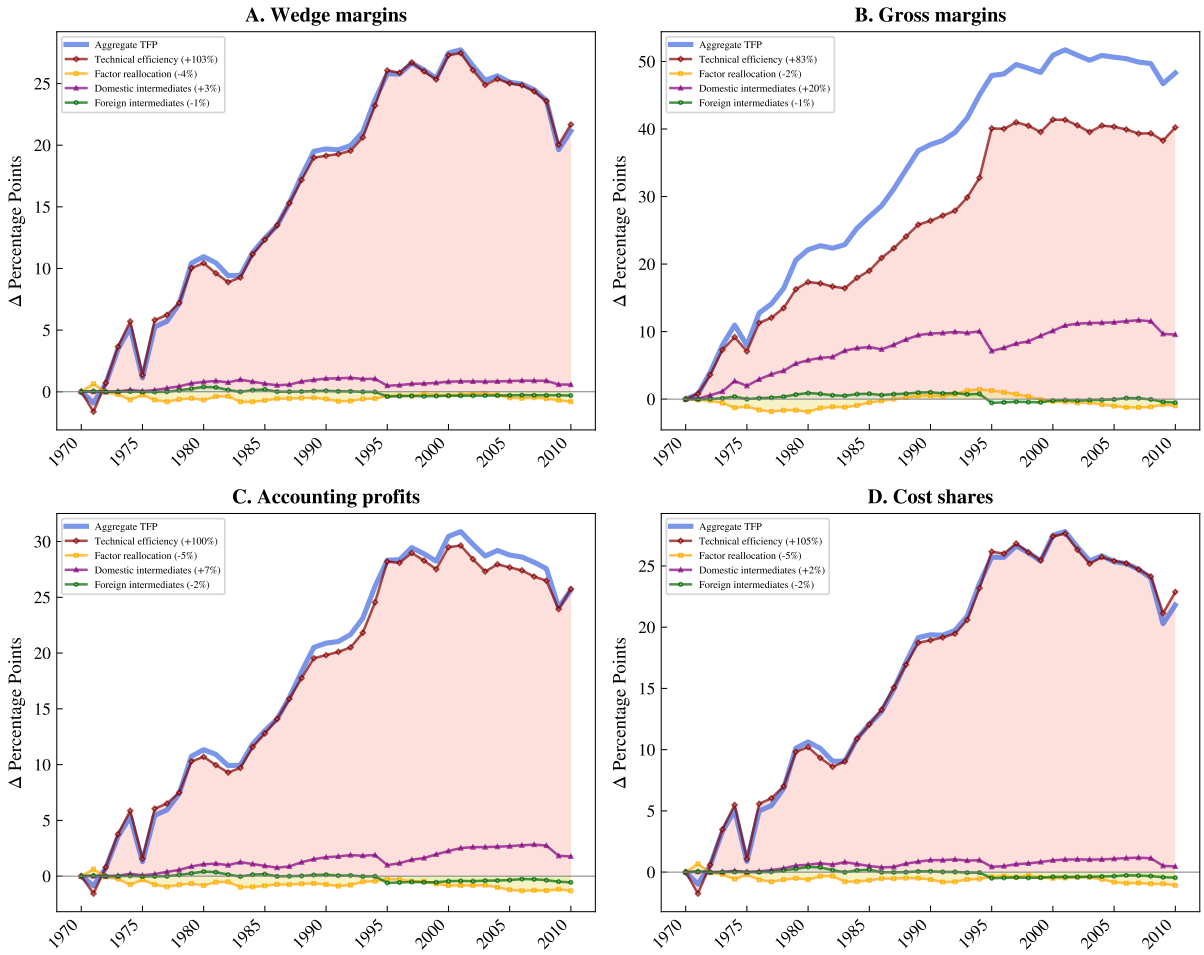


Figure 23 repeats the growth-accounting exercise and the aggregate TFP and welfare decompositions for Italy using base-period weights. As in Spain, productivity and welfare diverge over time: although aggregate TFP declines after 2001, welfare continues to rise. The disconnect is even more pronounced than in Spain. The welfare decomposition shows that these gains are driven primarily by technological progress, whereas reallocation contributes little. This reinforces the point made in the main text: domestic aggregate TFP is not, by itself, a sufficient statistic for the evolution of living standards.

Figure 24 presents a counterfactual exercise in which all weights are held fixed at their 2001 levels. Panel A shows the implied path of aggregate TFP under this fixed-composition scenario relative to its actual path. Under the counterfactual, Italy's TFP would have remained roughly unchanged at its 2001 level. Panel B compares actual and counterfactual welfare. By the end of the sample, welfare would have been approximately 12 percentage points higher had the sectoral composition remained fixed. Again, these interpretation are subject to the caveats emphasized for Spain, and these numbers should be thought as upper bounds.

Figure 23: Growth Accounting and Welfare in Italy, 1970–2010.

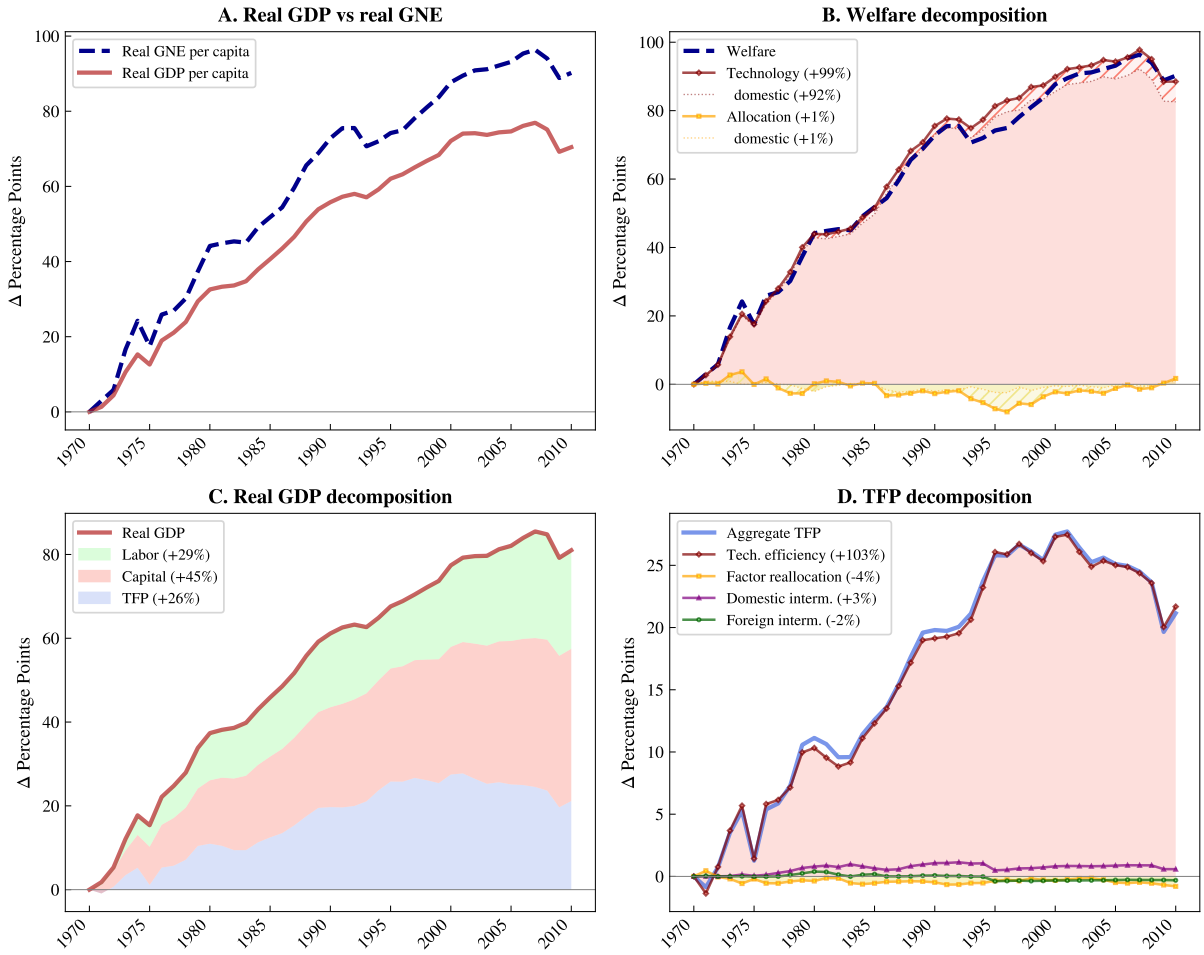
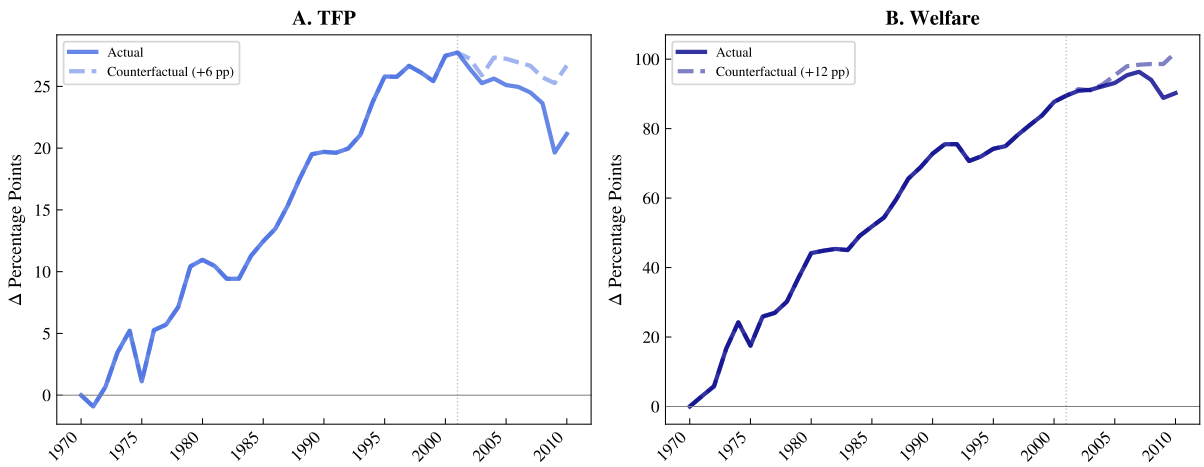


Figure 24: Counterfactual TFP and Welfare in Italy, 1970–2010.



Finally, Figure 25 displays the evolution of income shares in Italy. Unlike Spain, where the income share accruing to wedges rises steadily over time, Italy's wedge income share remains broadly stable at about 5% of GDP throughout the sample. Over the same period, the labor share declines from roughly 70 percent in 1970 to 66 percent

in 2010, while the capital share rises from about 25 percent to 31 percent. The stability in the aggregate wedge explains the close alignment between the traditional and distorted TFP indices documents above.

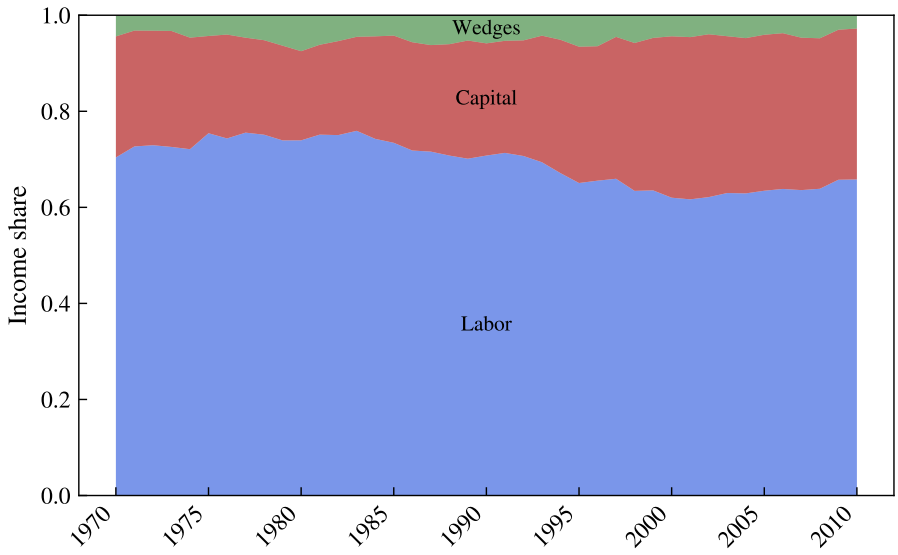



Figure 25: Income Shares in Italy, 1970–2010.